Mathematics

Class 6 (Sindhi Medium)

Solved Exercises — STBB

14 lessons — 1190 questions

1. Sets

Question: Practical Work 1: Fill in the blanks with the symbols ∈ or ∉ for A = {2, 4, 6, 8, 10}. (i) 4 ___ A

Answer: The members of A are 2, 4, 6, 8, 10. 4 is a member of A, so 4 ∈ A. ✓

Question: Practical Work 1: (ii) 1 ___ A

Answer: The members of A are 2, 4, 6, 8, 10. 1 is not a member of A, so 1 ∉ A. ✓

Question: Practical Work 1: (iii) 6 ___ A

Answer: The members of A are 2, 4, 6, 8, 10. 6 is a member of A, so 6 ∈ A. ✓

Question: Practical Work 1: (iv) 9 ___ A

Answer: The members of A are 2, 4, 6, 8, 10. 9 is not a member of A, so 9 ∉ A. ✓

Question: Practical Work 1: (v) 2 ___ A

Answer: The members of A are 2, 4, 6, 8, 10. 2 is a member of A, so 2 ∈ A. ✓

Question: Practical Work 1: (vi) 8 ___ A

Answer: The members of A are 2, 4, 6, 8, 10. 8 is a member of A, so 8 ∈ A. ✓

Question: Practical Work 2: Write the following sets in tabular (roster) form. (i) The set of the first five letters of the English alphabet.

Answer: The first five letters of the English alphabet are a, b, c, d, e. In tabular form: {a, b, c, d, e}.

Question: Practical Work 2: (ii) The set of the 12 months of the year.

Answer: The 12 months of the year are January, February, March, April, May, June, July, August, September, October, November, December. In tabular form: {January, February, March, April, May, June, July, August, September, October, November, December}.

Question: Practical Work 2: (iii) The set of the names of the days of the week that begin with "J".

Answer: The days of the week are Monday, Tuesday, Wednesday, Thursday, Friday, Saturday, Sunday. Of these, only Friday begins with "J".

In tabular form: {Friday}.

Question: Practical Work 2: (iv) The set of the first five natural numbers.

Answer: The first five natural numbers are 1, 2, 3, 4, 5. In tabular form: {1, 2, 3, 4, 5}.

Question: Exercise 1.1, Question 1: Which of the following are sets? (i) The collection of the names of the provinces of Pakistan.

Answer: The four provinces of Pakistan are clearly known: Sindh, Punjab, Khyber Pakhtunkhwa, Balochistan. This is a well-defined collection, so it is a set. ✓

Question: Exercise 1.1, Question 1: (ii) The collection of the names of the players of the Pakistani hockey team.

Answer: The players of the Pakistani hockey team are clearly known, so this is a well-defined collection and a set. ✓

Question: Exercise 1.1, Question 1: (iii) The collection of tasty foods.

Answer: "Tasty" differs from person to person — one food seems tasty to one person and not to another. It is not well-defined, so it is not a set. ✗

Question: Exercise 1.1, Question 1: (iv) The collection of the intelligent students of your class.

Answer: There is no clear definition of "intelligent" — opinions differ on who is intelligent. It is not well-defined, so it is not a set. ✗

Question: Exercise 1.1, Question 1: (v) The collection of trees.

Answer: Everything that is a tree belongs to this collection, and everything that is not a tree does not. This is a well-defined collection, so it is a set. ✓

Question: Exercise 1.1, Question 1: (vi) The collection of the English teachers in your school.

Answer: The English teachers in your school are clearly known, so this is a well-defined collection and a set. ✓

Question: Exercise 1.1, Question 2: If A = {5, 6, 7, 8} and B = {a, b, c, d}, fill in the blanks with the symbol ∈ or ∉. (i) 5 ___ A

Answer: 5 is a member of A, so 5 ∈ A. ✓

Question: Exercise 1.1, Question 2: (ii) 9 ___ A

Answer: 9 is not a member of A, so 9 ∉ A. ✓

Question: Exercise 1.1, Question 2: (iii) m ___ A

Answer: m is not a member of A (A contains only the numbers 5, 6, 7, 8), so m ∉ A. ✓

Question: Exercise 1.1, Question 2: (iv) c ___ B

Answer: c is a member of B, so c ∈ B. ✓

Question: Exercise 1.1, Question 2: (v) b ___ B

Answer: b is a member of B, so b ∈ B. ✓

Question: Exercise 1.1, Question 2: (vi) 7 ___ B

Answer: 7 is not a member of B (B contains only a, b, c, d), so 7 ∉ B. ✓

Question: Exercise 1.1, Question 3: Write in tabular form. (i) The set of the names of the provinces of Pakistan.

Answer: The provinces of Pakistan are Sindh, Punjab, Khyber Pakhtunkhwa, Balochistan. In tabular form: {Sindh, Punjab, Khyber Pakhtunkhwa, Balochistan}.

Question: Exercise 1.1, Question 3: (ii) The set of natural numbers from 50 to 70.

Answer: The natural numbers from 50 to 70 are 50, 51, ..., 70. In tabular form: {50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70}.

Question: Exercise 1.1, Question 3: (iii) The set of the first ten natural numbers that are exactly divisible by 2.

Answer: Of the first ten natural numbers (1 to 10), those exactly divisible by 2 are 2, 4, 6, 8, 10. In tabular form: {2, 4, 6, 8, 10}.

Question: Exercise 1.1, Question 3: (iv) The set of the letters of the word "pakistan".

Answer: The letters of the word pakistan are p, a, k, i, s, t, a, n. In tabular form each member is written only once, so the repeated a is not written again: {p, a, k, i, s, t, n}.

Question: Exercise 1.1, Question 3: (v) The set of the mathematics teachers in your school.

Answer: This set depends on your school. In tabular form the names of those teachers are listed, e.g. {Teacher 1, Teacher 2, ...} — i.e. all the teachers who teach mathematics in your school.

Question: Exercise 1.1, Question 3: (vi) The set of the names of the colours of the national flag.

Answer: The colours of the national flag of Pakistan are green and white. In tabular form: {Green, White}.

Question: Exercise 1.1, Question 4: If P = {a, e, i, o, u} and Q = {1, 2, 3, ..., 10}, state which of the following are true or false. (i) a ∈ P ____

Answer: a is a member of P (a is a vowel of the English alphabet), so a ∈ P

is true. ✓

Question: Exercise 1.1, Question 4: (ii) 1 ∈ Q ____

Answer: 1 is a member of Q (Q contains all natural numbers from 1 to 10), so 1 ∈ Q is true. ✓

Question: Exercise 1.1, Question 4: (iii) I ∈ Q ____

Answer: I is a letter, not a number, so I is not a member of Q. I ∈ Q is false. ✗

Question: Exercise 1.1, Question 4: (iv) v ∈ P ____

Answer: v is not a vowel (P contains only a, e, i, o, u), so v is not a member of P. v ∈ P is false. ✗

Question: Exercise 1.1, Question 4: (v) y ∈ Q ____

Answer: y is a letter, not a number, so y is not a member of Q. y ∈ Q is false. ✗

Question: Exercise 1.1, Question 4: (vi) e ∈ P ____

Answer: e is a member of P (e is a vowel), so e ∈ P is true. ✓

Question: Exercise 1.1, Question 4: (vii) 3 ∈ P ____

Answer: 3 is a number, not a vowel, so 3 is not a member of P. 3 ∈ P is false. ✗

Question: Exercise 1.1, Question 4: (viii) 7 ∈ Q ____

Answer: 7 is a member of Q (Q contains all natural numbers from 1 to 10), so 7 ∈ Q is true. ✓

Question: Practical Work: Put a (✓) mark for a finite set and a (✗) mark for an infinite set. (i) The set of all residents of Pakistan.

Answer: The number of residents of Pakistan is limited (though very large). So this is a finite set: (✓).

Question: Practical Work: (ii) The set of all numbers less than 9999.

Answer: The natural numbers less than 9999 are 1 to 9998 — a limited number. So this is a finite set: (✓).

Question: Practical Work: (iii) The set of the drops of water in a river.

Answer: The number of drops of water in a river is very large, but limited — the drops can in principle be counted. So this is a finite set: (✓).

Question: Practical Work: (iv) The set of all letters of the English alphabet.

Answer: The English alphabet has 26 letters — a limited number. So this is a finite set: (✓).

Question: Practical Work: (v) {1, 2, 3, ...}

Answer: This is the set of natural numbers, which never ends — the number of members is unlimited. So this is an infinite set: (✗).

Question: Practical Work: (vi) {d, o, r}

Answer: This set has only 3 members — a limited number. So this is a finite set: (✓).

Question: Practical Work: (vii) {2, 4, 6, ...}

Answer: This is the set of even numbers, which never ends — the number of members is unlimited. So this is an infinite set: (✗).

Question: Practical Work: Identify and write the following sets from the table below. (i) Two finite sets

Answer: Finite sets are those with a limited number of members, e.g. B = {6} and C = {a, b, c, d} (both are finite sets).

Question: Practical Work: (ii) Two infinite sets

Answer: Infinite sets are those with an unlimited number of members, e.g. A = {1, 3, 5, 7, ...} and E = {5, 10, 15, 20, 25, ...} (both are infinite sets).

Question: Practical Work: (iii) Two empty sets

Answer: Empty sets are those with no members, e.g. the set of natural numbers between 2 and 3 = { } and the set of triangles with five sides = { }.

Question: Practical Work: (iv) Two equal sets

Answer: Equal sets are those with exactly the same members, e.g. D = {2, 4, 6, 8} and F = {4, 6, 2, 8} — both have the same members, so D = F.

Question: Practical Work: (v) Two equivalent sets

Answer: Equivalent sets are those with the same number of members, e.g. C = {a, b, c, d} and D = {2, 4, 6, 8} — both have 4 members, so C ~ D.

Question: Practical Work: (vi) Two singleton sets

Answer: Singleton sets are those with exactly one member, e.g. B = {6} and the set of factors of the number 1 = {1} (each has only one member).

Question: Practical Work: (vii) Two subsets

Answer: Subsets are sets whose members are all contained in another set, e.g. {a} ⊆ C and D ⊆ F (since D = F, every set is a subset of itself).

Question: Practical Work: (viii) Two supersets

Answer: Supersets are sets containing all the members of another set, e.g. C is a superset of {a} ({a} ⊆ C, so C ⊃ {a}), and F is a superset of D (D ⊆

F, so F ⊃ D).

Question: Exercise 1.2, Question 1: Give reasons why the following collections are not sets. (i) {d, o, o, r}

Answer: In this collection the member o appears twice. In a set no member is written more than once, so as written this is not a set. (The correct roster form would be {d, o, r}.)

Question: Exercise 1.2, Question 1: (ii) {★, O, ★, △}

Answer: In this collection the member ★ appears twice. In a set no member is written more than once, so as written this is not a set.

Question: Exercise 1.2, Question 1: (iii) {a, f, d, a}

Answer: In this collection the member a appears twice. In a set no member is written more than once, so as written this is not a set.

Question: Exercise 1.2, Question 1: (iv) {2, 2, 3, 3, 4, 4}

Answer: In this collection every member (2, 3, 4) appears twice. In a set no member is written more than once, so as written this is not a set. (The correct roster form would be {2, 3, 4}.)

Question: Exercise 1.2, Question 1: (v) The set of beautiful birds

Answer: There is no clear definition of "beautiful" — one bird seems beautiful to one person and not to another. It is not well-defined, so it is not a set.

Question: Exercise 1.2, Question 1: (vi) The set of Sindh players

Answer: It is not clear which game, which team, or which level of players is meant. It is not well-defined, so it is not a set.

Question: Exercise 1.2, Question 2: Which of the following sets are finite or infinite? (i) A = {0, 1, 2, ..., 50}

Answer: This set has 51 members, from 0 to 50 — a limited number. So it is a finite set.

Question: Exercise 1.2, Question 2: (ii) B = {100, 200, 300, 400, ...}

Answer: These are the multiples of 100, which never end — the number of members is unlimited. So it is an infinite set.

Question: Exercise 1.2, Question 2: (iii) The set of the hairs on a goat's body.

Answer: The number of hairs on a goat's body is limited. So it is a finite set.

Question: Exercise 1.2, Question 2: (iv) The set of the stars in the sky.

Answer: The number of stars in the sky is very large, but limited — the

stars can in principle be counted. So it is a finite set.

Question: Exercise 1.2, Question 2: (v) The set of a cat's whiskers.

Answer: The number of a cat's whiskers is limited. So it is a finite set.

Question: Exercise 1.2, Question 2: (vi) The set of lines passing through a point.

Answer: Infinitely many lines can pass through one point — there is no limit to the number. So it is an infinite set.

Question: Exercise 1.2, Question 2: (vii) The set of natural numbers greater than 20.

Answer: The natural numbers greater than 20 are 21, 22, 23, ... and never end — the number is unlimited. So it is an infinite set.

Question: Exercise 1.2, Question 2: (viii) The set of all cities of Pakistan.

Answer: The number of cities in Pakistan is limited. So it is a finite set.

Question: Exercise 1.2, Question 2: (ix) The set of all schools of Sindh.

Answer: The number of schools in Sindh is limited. So it is a finite set.

Question: Exercise 1.2, Question 2: (x) The set of all even numbers.

Answer: The even numbers 2, 4, 6, 8, ... never end — the number is unlimited. So it is an infinite set.

Question: Exercise 1.2, Question 3: Which of the following are empty sets? (i) The set of students in your class older than 20 years.

Answer: In Class 6 there is no student older than 20 years. This set has no members, so it is an empty set. ✓

Question: Exercise 1.2, Question 3: (ii) The set of letters in the English alphabet after "z".

Answer: In the English alphabet z is the last letter; there is no letter after it. This set has no members, so it is an empty set. ✓

Question: Exercise 1.2, Question 3: (iii) The set of the names of the children in your neighbourhood whose names begin with "K".

Answer: This set depends on your neighbourhood — it cannot be decided from the book. If there is no such child it will be an empty set, and if there is one it will not be empty.

Question: Exercise 1.2, Question 3: (iv) The set of the months of the solar year whose names begin with "Z".

Answer: The months of the solar year are January, February, March, April, May, June, July, August, September, October, November, December — none begins with "Z". This set has no members, so it is an empty set. ✓

Question: Exercise 1.2, Question 3: (v) The set of fish living in sand.

Answer: No fish lives in sand (desert). This set has no members, so it is an empty set. ✓

Question: Exercise 1.2, Question 3: (vi) The set of even numbers between 4 and 10.

Answer: The even numbers between 4 and 10 are 6 and 8. This set has members, so it is not an empty set. ✗

Question: Exercise 1.2, Question 3: (vii) {0}

Answer: This set has one member, 0, so it is not an empty set. ✗

Question: Exercise 1.2, Question 3: (viii) {Φ}

Answer: This set has one member, Φ, so it is not an empty set — it is a singleton set. ✗

Question: Exercise 1.2, Question 4: Which of the following pairs are equal sets? (i) {1, 2, 3} and {2, 3, 1}

Answer: Both sets have the same members 1, 2, 3 — the order of members in a set does not matter. So these are equal sets. ✓

Question: Exercise 1.2, Question 4: (ii) {p, a, t} and {t, a, p}

Answer: Both sets have the same members p, a, t. So these are equal sets. ✓

Question: Exercise 1.2, Question 4: (iii) {k, i, t, e} and {b, i, t, e}

Answer: The first set has k and the second has b — the members are not the same. So these are not equal sets. ✗

Question: Exercise 1.2, Question 4: (iv) {x, y, z} and the set of the first three letters of the English alphabet

Answer: The first three letters of the English alphabet are a, b, c, i.e. the second set is {a, b, c}. The members of {x, y, z} are different. So these are not equal sets. ✗

Question: Exercise 1.2, Question 4: (v) The set of odd numbers less than 2 and { }

Answer: The odd number less than 2 is 1, i.e. the first set is {1}. The second set { } is empty. The members are not the same, so these are not equal sets. ✗

Question: Exercise 1.2, Question 5: Which of the following pairs are equivalent sets? (i) {m, i, l, e} and {n, i, l, e}

Answer: Both sets have 4 members each — the number of members is the same. So these are equivalent sets. ✓

Question: Exercise 1.2, Question 5: (ii) {1, 2, 3} and {3, 2, 1}

Answer: Both sets have 3 members each — the number of members is the same. So these are equivalent sets (and equal too). ✓

Question: Exercise 1.2, Question 5: (iii) {1, 2, 3, 4, 5} and {D, □, ○, ★, □}

Answer: Both sets have 5 members each — the number of members is the same. So these are equivalent sets. ✓

Question: Exercise 1.2, Question 5: (iv) {6, 66, 666, 6666} and {666, 7777, 77}

Answer: The first set has 4 members and the second has 3 members — the numbers are not the same. So these are not equivalent sets. ✗

Question: Exercise 1.2, Question 6: State which of the following are true or false. (i) If B = {m, o, n}, then B is the set of the letters of the word "moon".

Answer: The letters of the word moon are m, o, o, n. In a set each member is written only once, so {m, o, n} is indeed the set of the letters of moon. This statement is true. ✓

Question: Exercise 1.2, Question 6: (ii) If A = {1, 2} and B = {2, 3, 4, 5}, then A ⊆ B.

Answer: The member 1 of A is not in B. For a subset, all members of A must be in B. So A ⊆ B is false. ✗

Question: Exercise 1.2, Question 6: (iii) If A = {4, 5, 10} and B = {5, 10, 20}, then A ~ B.

Answer: A has 3 members and B also has 3 members — the number is the same. So A ~ B is true. ✓

Question: Exercise 1.2, Question 6: (vi) If A = {0, 1, 2, 3, 5} and B is the set of natural numbers less than 6, then A = B. (In the book, part (vi) is printed before part (v).)

Answer: The natural numbers less than 6 are 1, 2, 3, 4, 5, i.e. B = {1, 2, 3, 4, 5}. A contains 0 and does not contain 4, so the members of A and B are not the same. A = B is false. ✗

Question: Exercise 1.2, Question 6: (v) If x is equal to y then x is an equivalent set or ... (This sentence is printed garbled in the book.)

Answer: This sentence is printed garbled in the book. On its natural reading — if x equals y then {x} and {y} are equivalent (and equal) sets — it is true. ✓

Exercise 1.2, Question 7: Choose the suitable symbol from the given symbols to show the relation between the sets. (=, ⊂, ⊃, ~) (i) {10, 20, 30} _____ {5, 10, 15, 20, 25, 30}

Answer: are in the second set, but the second set is larger — the first set

is a proper subset of the second. So {10, 20, 30} ⊂

Exercise 1.2, Question 7: (ii) {7, 14, 21, 28, 35, 42, 49} _____ the set of multiples of 7 less than 50

Answer: =

Exercise 1.2, Question 7: (iii) {7, 5, 8} _____ {5, 6, 7, 8, ...}

Answer: are in the second set, but the second set is infinite and larger. So {7, 5, 8} ⊂

Exercise 1.2, Question 7: (iv) {0} _____ { }

Answer: has one member, 0, and

Exercise 1.2, Question 7: (v) {t, e, a} _____ {a, e, t}

=

Exercise 1.2, Question 7: (vi) The set of all cities of Sindh _____ {Hyderabad, Karachi, Sukkur}

Answer: ⊃

Exercise 1.2, Question 7: (vii) {a, e, i, o, u} _____ the set of the first five letters of the English alphabet

Answer: are different from these, but both sets have 5 members each. So {a, e, i, o, u} ~

Exercise 1.2, Question 7: (viii) {1, 2, 3, ...} _____ {11, 22, 33}

⊃

Review Exercise 1, Question 1: Fill in the blanks. (i) If A = {a, b, c}, then A set a, b, c are its _____.

Answer: members.

Review Exercise 1, Question 1: (ii) A set which has no member is called a _____.

Answer: an empty set. The blank takes "empty set".

Review Exercise 1, Question 1: (iii) A set in which the number of members is limited is called a _____.

Answer: finite set. The blank takes "finite set".

Review Exercise 1, Question 1: (iv) If two sets A and B have _____ members, they are called equal sets.

Answer: all the same

Question: Review Exercise 1, Question 2: Choose the correct answer. (i) If

set B is the set of vowels of the English alphabet, then: (a) p ∈ B (b) e ∉ B (c) b ∈ B (d) a ∈ B

Answer: The vowels of the English alphabet are a, e, i, o, u. p and b are not vowels, and e is a vowel (so e ∉ B is false). The correct answer is (d) a ∈ B. ✓

Review Exercise 1, Question 2: (ii) {f} is called a _____. (a) empty set (b) infinite set (c) subset (d) singleton set

Answer: singleton set. The correct answer is

Review Exercise 1, Question 2: (iii) If set A is equal to set B, we use the symbol _____. (a) ∈ (b) ⊂ (c) ~ (d) =

Answer: sets. The correct answer is

Question: Review Exercise 1, Question 2: (iv) If A = {a, b, c} and B = {a, b, c, d, e}, then: (a) A = B (b) A ~ B (c) A ⊂ B (d) B ⊂ A

Answer: All members of A are in B, and B has extra members too, so A is a proper subset of B. The correct answer is (c) A ⊂ B. ✓

Question: Review Exercise 1, Question 3: Write two examples of a set.

Answer: Example 1: {1, 2, 3, 4, 5} — the set of the first five natural numbers. Example 2: {a, b, c} — the set of three letters. Both are well-defined collections, so both are sets.

Question: Review Exercise 1, Question 4: Write the following sets in tabular form. (i) A = the set of even numbers between 2 and 10

Answer: The even numbers between 2 and 10 are 4, 6, 8. In tabular form: A = {4, 6, 8}.

Question: Review Exercise 1, Question 4: (ii) B = the set of odd numbers from 1 to 17

Answer: The odd numbers from 1 to 17 are 1, 3, 5, 7, 9, 11, 13, 15, 17. In tabular form: B = {1, 3, 5, 7, 9, 11, 13, 15, 17}.

Question: Review Exercise 1, Question 4: (iii) C = the set of natural numbers less than 30 divisible by 2

Answer: The natural numbers less than 30 divisible by 2 are 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28. In tabular form: C = {2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28}.

Question: Review Exercise 1, Question 4: (iv) D = the set of months of the solar year beginning with "J"

Answer: The months of the solar year beginning with "J" are January, June, July. In tabular form: D = {January, June, July}.

Question: Review Exercise 1, Question 5: Which of the following are finite sets and which are infinite sets? (i) A = the set of the pages of your book

Answer: The number of pages of a book is limited. So A is a finite set.

Question: Review Exercise 1, Question 5: (ii) B = the set of all even numbers

Answer: The even numbers 2, 4, 6, 8, ... never end. So B is an infinite set.

Question: Review Exercise 1, Question 5: (iii) C = the set of prime factors of 60

Answer: The prime factors of 60 are 2, 3, 5 — a limited number. So C is a finite set.

Question: Review Exercise 1, Question 5: (iv) D = the set of all multiples of 7

Answer: The multiples of 7 are 7, 14, 21, 28, ... and never end. So D is an infinite set.

Question: Review Exercise 1, Question 5: (v) E = the set of one-digit natural numbers

Answer: The one-digit natural numbers are 1 to 9 — a limited number. So E is a finite set.

Question: Review Exercise 1, Question 6: Which of the following are empty sets? (i) L = the set of the names of the days of the week beginning with "S"

Answer: The day Monday (Somwaar) begins with "S". This set has a member, so it is not an empty set. ✗

Question: Review Exercise 1, Question 6: (ii) F = the set of the months of the year with 32 days

Answer: No month of the year has 32 days. This set has no members, so it is an empty set. ✓

Question: Review Exercise 1, Question 6: (iii) M = the set of odd numbers exactly divisible by 2

Answer: An odd number is never exactly divisible by 2. This set has no members, so it is an empty set. ✓

Question: Review Exercise 1, Question 6: (iv) N = the set of prime numbers exactly divisible by 2

Answer: The prime number 2 is exactly divisible by 2. This set has the member 2, so it is not an empty set. ✗

Question: Review Exercise 1, Question 7: If A = {d}, B = {c, d}, C = {a, b, c}, D = {a, b}, state which of the following are true or false. (i) D ⊆ C

Answer: All members of D, a and b, are in C. So D ⊆ C is true. ✓

Question: Review Exercise 1, Question 7: (ii) A ⊄ C

Answer: The member d of A is not in C. So A is not a subset of C, i.e. A ⊄ C is true. ✓

Question: Review Exercise 1, Question 7: (iii) B ⊆ D

Answer: The member c of B is not in D. So B is not a subset of D, i.e. B ⊆ D is false. ✗

Question: Review Exercise 1, Question 7: (iv) C ⊄ D

Answer: The member c of C is not in D. So C is not a subset of D, i.e. C ⊄ D is true. ✓

Question: Review Exercise 1, Question 8: State which of the following are true or false. (i) The members of any set are always of the same kind.

Answer: The members of a set can be of different kinds, e.g. {1, a, ★} is a set. So this statement is false. ✗

Question: Review Exercise 1, Question 8: (ii) The order of the members of a set has no importance.

Answer: Members of a set can be written in any order, e.g. {1, 2, 3} = {3, 2, 1}. So this statement is true. ✓

Question: Review Exercise 1, Question 8: (iii) A member can occur more than once in a set.

Answer: In a set no member is written more than once. So this statement is false. ✗

Question: Review Exercise 1, Question 8: (iv) Any collection of things is called a set.

Answer: Only a well-defined collection is called a set — e.g. "the collection of tasty foods" is not a set. So this statement is false. ✗

Question: Review Exercise 1, Question 8: (v) The symbol "∈" is used for a member of any set.

Answer: "∈" means "is a member of". This symbol is used for a member of a set. So this statement is true. ✓

Question: Review Exercise 1, Question 8: (vi) A well-defined collection of distinct things is called a set.

Answer: That is exactly the definition of a set — a well-defined collection of distinct things. So this statement is true. ✓

Question: Review Exercise 1, Question 8: (vii) {a, b, c, d} is not a set.

Answer: {a, b, c, d} has four clear members enclosed in {} — it is a set. So this statement is false. ✗

Question: Review Exercise 1, Question 8: (viii) The symbol "∉" is used for a member of any set.

Answer: "∉" means "is not a member of" — it is used for a non-member, not for a member. So this statement is false. ✗

Question: Review Exercise 1, Question 8: (ix) 1, 2, 3, 4 is a set.

Answer: The members of a set are enclosed in {}. Here there are no {}, so 1, 2, 3, 4 as written is not a set. This statement is false. ✗

2. Whole Numbers

Question: Exercise 2.1, Question 1: Write the number if possible. (i) The smallest natural number

Answer: Natural numbers start from 1. The smallest natural number is 1.

Question: Exercise 2.1, Question 1: (ii) The smallest whole number

Answer: Whole numbers start from 0. The smallest whole number is 0.

Question: Exercise 2.1, Question 1: (iii) The largest natural number

Answer: Natural numbers never end — after every natural number there is a larger one. So there is no largest natural number.

Question: Exercise 2.1, Question 1: (iv) The largest whole number

Answer: Whole numbers never end — after every whole number there is a larger one. So there is no largest whole number.

Question: Exercise 2.1, Question 2: Write the first ten natural numbers.

Answer: The first ten natural numbers are 1 to 10: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10.

Question: Exercise 2.1, Question 3: Write the first ten whole numbers.

Answer: The first ten whole numbers are 0 to 9: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9.

Question: Exercise 2.1, Question 4: Show on the number line. (i) Whole numbers greater than 8

Answer: On the number line, all points to the right of 8 (9, 10, 11, 12, ...) are shown — the whole numbers greater than 8 are 9, 10, 11, ...

Question: Exercise 2.1, Question 4: (ii) Whole numbers less than 8

Answer: On the number line, the points to the left of 8, from 0 to 7, are shown — the whole numbers less than 8 are 0, 1, 2, 3, 4, 5, 6, 7.

Question: Exercise 2.1, Question 4: (iii) Whole numbers greater than or equal to 8

Answer: On the number line, 8 together with all points to its right (8, 9, 10, 11, ...) are shown — the whole numbers greater than or equal to 8 are 8, 9, 10, ...

Question: Exercise 2.1, Question 4: (iv) Whole numbers less than or equal to 8

Answer: On the number line, 8 together with the points to its left, from 0 to 8, are shown — the whole numbers less than or equal to 8 are 0, 1, 2, 3, 4, 5, 6, 7, 8.

Question: Exercise 2.1, Question 4: (v) Whole numbers greater than 3 and less than 15

Answer: On the number line the points between 3 and 15 are shown (excluding 3 and 15) — those whole numbers are 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14.

Question: Exercise 2.1, Question 4: (vi) Whole numbers greater than or equal to 5 and less than or equal to 12

Answer: On the number line all points from 5 to 12 (including 5 and 12) are shown — those whole numbers are 5, 6, 7, 8, 9, 10, 11, 12.

Question: Exercise 2.1, Question 4: (vii) Whole numbers greater than or equal to 4 and less than or equal to 11

Answer: On the number line all points from 4 to 11 (including 4 and 11) are shown — those whole numbers are 4, 5, 6, 7, 8, 9, 10, 11.

Question: Exercise 2.1, Question 4: (viii) Whole numbers greater than or equal to 3 and less than 9

Answer: On the number line all points from 3 to 8 (including 3, excluding 9) are shown — those whole numbers are 3, 4, 5, 6, 7, 8.

Question: Exercise 2.1, Question 4: (ix) Odd whole numbers greater than 3 and less than 9

Answer: The odd whole numbers greater than 3 and less than 9 are 5 and 7. On the number line only the points 5 and 7 are shown.

Question: Exercise 2.1, Question 4: (x) Even whole numbers greater than or equal to 8 and less than 16

Answer: The even whole numbers greater than or equal to 8 and less than 16 are 8, 10, 12, 14. On the number line only these points are shown.

Question: Exercise 2.1, Question 5: Using the number line, find the sum of the following whole numbers. (i) 1 and 5

Answer: On the number line, start at 0 and move 1 step right (reach 1), then move 5 steps further (reach 6). So 1 + 5 = 6.

Question: Exercise 2.1, Question 5: (ii) 6 and 3

Answer: On the number line, start at 0 and move 6 steps right (reach 6), then move 3 steps further (reach 9). So 6 + 3 = 9.

Question: Exercise 2.1, Question 5: (iii) 10 and 2

Answer: On the number line, start at 0 and move 10 steps right (reach 10), then move 2 steps further (reach 12). So 10 + 2 = 12.

Question: Exercise 2.1, Question 5: (iv) 8 and 4

Answer: On the number line, start at 0 and move 8 steps right (reach 8), then move 4 steps further (reach 12). So 8 + 4 = 12.

Question: Exercise 2.1, Question 5: (v) 2, 3 and 5

Answer: On the number line, start at 0 and move 2 steps right (reach 2), then 3 steps further (reach 5), then 5 steps further (reach 10). So 2 + 3 + 5 = 10.

Question: Exercise 2.1, Question 5: (vi) 2, 3 and 4

Answer: On the number line, start at 0 and move 2 steps right (reach 2), then 3 steps further (reach 5), then 4 steps further (reach 9). So 2 + 3 + 4 = 9.

Question: Practical Work: Using the commutative and associative properties of addition, fill in the blanks. (i) 56,123 + 71,045 = 71,045 + 56,123

Answer: This is the commutative property — changing the order does not change the sum. This blank is already filled in the book: 56,123 + 71,045 = 71,045 + 56,123. The sum on both sides is 127,168.

Practical Work: (ii) 24,125 + (41,625 + 7,123) = (24,125 + _____) + 7,123

Answer: 41,625)

Practical Work: (iii) 47,813 + _____ = 51,623 + _____

Answer: 51,623, 47,813 = 99,436.

Practical Work: (iv) 567 + (_____ + 1,784) = (_____ + 962) + _____

Answer: (962, (567, 1,784 = 3,313. The blanks take 962, 567, 1,784 in order.

Question: Exercise 2.2, Question 1: Find the sum of the largest 6-digit number and the smallest 7-digit number.

Answer: The largest 6-digit number is 999,999 and the smallest 7-digit number is 1,000,000. Sum: 999,999 + 1,000,000 = 1,999,999.

Question: Exercise 2.2, Question 2: The population of a village is 2,700. If there are 1,070 men and 915 women, find the number of children.

Answer: Given: total population = 2,700, men = 1,070, women = 915. Children = 2,700 − 1,070 − 915. First 2,700 − 1,070 = 1,630, then 1,630 − 915 = 715. So the number of children is 715.

Question: Exercise 2.2, Question 3: Arif deposited Rs 45,800 in his bank account. He withdrew Rs 3,500 from his bank account. How much money is now left in his account?

Answer: Given: deposited amount = Rs 45,800, withdrawn amount = Rs 3,500. Remaining amount = 45,800 − 3,500 = Rs 42,300.

Question: Exercise 2.2, Question 4: Kashif had Rs 82,000. He gave Rs 6,500 to his wife, Rs 10,550 to his son and Rs 15,335 to his daughter. Tell how much money in total is left with him?

Answer: Given: total money = Rs 82,000. Money given = 6,500 + 10,550 + 15,335 = Rs 32,385. Remaining money = 82,000 − 32,385 = Rs 49,615.

Question: Exercise 2.2, Question 5: For the following numbers, check the commutative property of addition. (i) 7,628 and 39,780

Answer: Left side: 7,628 + 39,780 = 47,408. Right side: 39,780 + 7,628 = 47,408. Both sides are equal, so the commutative property is verified: a + b = b + a.

Question: Exercise 2.2, Question 5: (ii) 924,981 and 228

Answer: Left side: 924,981 + 228 = 925,209. Right side: 228 + 924,981 = 925,209. Both sides are equal, so the commutative property is verified.

Question: Exercise 2.2, Question 5: (iii) 29,000 and 106,99 (the book prints "106,99", which appears to be a misprint for 10,699.)

Answer: The book prints the second number as "106,99", read here as 10,699. Left side: 29,000 + 10,699 = 39,699. Right side: 10,699 + 29,000 = 39,699. Both sides are equal, so the commutative property is verified.

Question: Exercise 2.2, Question 5: (iv) 50,102 and 9,019,854

Answer: Left side: 50,102 + 9,019,854 = 9,069,956. Right side: 9,019,854 + 50,102 = 9,069,956. Both sides are equal, so the commutative property is verified.

Question: Exercise 2.2, Question 6: For the following numbers, check the associative property of addition. (i) 44,380, 2,389 and 34,006 (the numbers are printed run together in the book: "44,380 and 2,389,34,006".)

Answer: Left side: (44,380 + 2,389) + 34,006 = 46,769 + 34,006 = 80,775. Right side: 44,380 + (2,389 + 34,006) = 44,380 + 36,395 = 80,775. Both sides are equal, so the associative property is verified: (a + b) + c = a + (b + c).

Question: Exercise 2.2, Question 6: (ii) 127, 583,031 and 3,405 (the numbers are printed run together in the book: "127,583,031 and 3,405".)

Answer: Left side: (127 + 583,031) + 3,405 = 583,158 + 3,405 = 586,563. Right side: 127 + (583,031 + 3,405) = 127 + 586,436 = 586,563. Both sides are equal, so the associative property is verified.

Question: Exercise 2.2, Question 6: (iii) 25,996, 6,090 and 231 (the numbers are printed run together in the book: "25,996 and 6,090,231".)

Answer: Left side: (25,996 + 6,090) + 231 = 32,086 + 231 = 32,317.

Right side: 25,996 + (6,090 + 231) = 25,996 + 6,321 = 32,317. Both sides are equal, so the associative property is verified.

Question: Exercise 2.2, Question 6: (iv) 102,341, 3,007 and 412 (the numbers are printed run together in the book: "102,341 and 3,007,412".)

Answer: Left side: (102,341 + 3,007) + 412 = 105,348 + 412 = 105,760. Right side: 102,341 + (3,007 + 412) = 102,341 + 3,419 = 105,760. Both sides are equal, so the associative property is verified.

Exercise 2.2, Question 7: Fill in the blanks to make the following correct according to the laws of addition. (i) 5,020 + 849 = _____ + 5,020

Answer: 849

Exercise 2.2, Question 7: (ii) 97864 + 0 = _____

Answer: 97864. The blank takes 97864.

Exercise 2.2, Question 7: (iii) 74,9 + 0 = 0 + _____ = _____ (the book prints "74,9", which appears to be a misprint for 749.)

Answer: 749, 749. The blanks take 749, 749.

Exercise 2.2, Question 7: (iv) 3,971 + (430 + 300) = (3,971 + _____) + 300

Answer: 430)

Exercise 2.2, Question 7: (v) 4,853 + 93 = 93 + 4,853 = _____

Answer: 4,946. The blank takes 4,946.

Question: Exercise 2.2, Question 8: Tell your friend to think of a number, double the number, add 9 to it. Then add to it the number he started with. After that divide the sum by 3, and subtract 3 more from the quotient. Which number did you get?

Answer: Suppose the thought number is n. Steps: double = 2n; add 9 = 2n + 9; add the original number = 2n + 9 + n = 3n + 9; divide by 3 = (3n + 9) ÷ 3 = n + 3; subtract 3 = n + 3 − 3 = n. So you always get back the very number that was thought of. E.g. for n = 5: 10, 19, 24, 8, 5.

Question: Exercise 2.3, Question 1: Find the product of the following whole numbers. (i) 854 x 96

Answer: 854 x 96: first 854 x 6 = 5,124. Then 854 x 90 = 76,860. Sum: 5,124 + 76,860 = 81,984. So 854 x 96 = 81,984.

Question: Exercise 2.3, Question 1: (ii) 736 x 103

Answer: 736 x 103: first 736 x 3 = 2,208. Then 736 x 100 = 73,600. Sum: 2,208 + 73,600 = 75,808. So 736 x 103 = 75,808.

Question: Exercise 2.3, Question 1: (iii) 256 x 1,008

Answer: 256 x 1,008: first 256 x 8 = 2,048. Then 256 x 1,000 = 256,000. Sum: 2,048 + 256,000 = 258,048. So 256 x 1,008 = 258,048.

Question: Exercise 2.3, Question 1: (iv) 995 x 158

Answer: 995 x 158: first 995 x 8 = 7,960. Then 995 x 50 = 49,750. Then 995 x 100 = 99,500. Sum: 7,960 + 49,750 + 99,500 = 157,210. So 995 x 158 = 157,210.

Question: Exercise 2.3, Question 2: Divide and find the quotient and remainder. (i) 7,772 ÷ 58

Answer: Long division: 58 x 1 = 58 (58 from 77 leaves 19); 58 x 3 = 174 from 197 leaves 23; 58 x 4 = 232 from 232 leaves 0. So quotient = 134, remainder = 0.

Question: Exercise 2.3, Question 2: (ii) 96,324 ÷ 245

Answer: Long division: 245 x 3 = 735 (735 from 963 leaves 228); 245 x 9 = 2,205 from 2282 leaves 77; 245 x 3 = 735 from 774 leaves 39. So quotient = 393, remainder = 39.

Question: Exercise 2.3, Question 2: (iii) 16,025 ÷ 1000

Answer: Dividing by 1,000, the last three digits become the remainder: 16,025 = 1,000 x 16 + 25. So quotient = 16, remainder = 25.

Question: Exercise 2.3, Question 2: (iv) 92,845 ÷ 300

Answer: Long division: 300 x 3 = 900 (900 from 928 leaves 28); 300 does not go into 284, so write 0; 300 x 9 = 2,700 from 2845 leaves 145. So quotient = 309, remainder = 145.

Question: Exercise 2.3, Question 3: A gardener plans to plant 570 trees in 19 rows. Each row must have an equal number of trees. Tell how many trees will be in each row?

Answer: Given: total trees = 570, rows = 19. Trees per row = 570 ÷ 19. Since 19 x 30 = 570, there will be 30 trees in each row.

Question: Exercise 2.3, Question 4: A shopkeeper bought 125 television sets. If the price of one television set is Rs 9,820, find the total price of all the sets.

Answer: Given: number of sets = 125, price of one set = Rs 9,820. Total price = 125 x 9,820. 9,820 x 100 = 982,000; 9,820 x 25 = 245,500; sum = Rs 1,227,500.

Question: Exercise 2.3, Question 5: Find the largest 5-digit number that is exactly divisible by 75.

Answer: The largest 5-digit number is 99,999. 99,999 ÷ 75: 75 x 1,333 = 99,975, remainder = 99,999 − 99,975 = 24. So the required number = 99,999 − 24 = 99,975. Check: 75 x 1,333 = 99,975.

Question: Exercise 2.3, Question 6: Multiply the largest 4-digit number by

the smallest 3-digit number.

Answer: The largest 4-digit number is 9,999 and the smallest 3-digit number is 100. Product: 9,999 x 100 = 999,900.

Exercise 2.4, Question 1: Fill in the blanks. (i) 5 x 0 = 0.

Exercise 2.4, Question 1: (ii) 6 x 1 = 6.

Exercise 2.4, Question 1: (iii) 2 x 5 = 5 x _____

Answer: 2 = 10. The blank takes 2.

Exercise 2.4, Question 1: (iv) 3 x (1 x 4) = (3 x 1) x _____

Answer: 4 = 12. The blank takes 4.

Exercise 2.4, Question 1: (v) 5 x ((7 x 8) = ((5 x 7) x 8

Exercise 2.4, Question 1: (vi) 541 x (645 + _____) = (541 x 645) + (541 x 964)

Answer: 964)

Exercise 2.4, Question 1: (vii) 345 x (650 − 125) = (345 x 650) _____ (345 x _____)

Answer: −, 125). The blanks take − and 125.

Exercise 2.4, Question 2: Write true or false. (i) 545 x 248 = 545 + 248 _____

Answer: = 793. They are not equal, so this statement is false. ✗

Exercise 2.4, Question 2: (ii) 67 − (125 − 12) = (67 − 125) − 12 _____

Answer: = (−58) − 12 = −70. They are not equal, so this statement is false. ✗

Exercise 2.4, Question 2: (iii) 64 x (245 − 10) = (64 x 245) − (64 x 10) _____

Answer: 15,040. Both are equal, so this statement is true. ✓

Exercise 2.4, Question 2: (iv) 95 x (25 x 14) = (95 x 25) x 14 _____

Answer: = 33,250. Both are equal, so this statement is true. ✓

Question: Exercise 2.4, Question 3: Name the properties used in the following and verify them. (i) 9 x 7 = 7 x 9

Answer: This is the commutative property of multiplication. Check: 9 x 7 = 63 and 7 x 9 = 63 — both are equal, so the property is verified.

Question: Exercise 2.4, Question 3: (ii) 4 x (6 x 3) = (4 x 6) x 3

Answer: This is the associative property of multiplication. Check: 4 x 18 = 72 and 24 x 3 = 72 — both are equal, so the property is verified.

Question: Exercise 2.4, Question 3: (iii) 5 x (7 − 1) = (5 x 7) − (5 x 1)

Answer: This is the distributive property of multiplication over subtraction. Check: 5 x 6 = 30 and 35 − 5 = 30 — both are equal, so the property is verified.

Question: Exercise 2.4, Question 3: (iv) 2 x (8 + 3) = (2 x 8) + (2 x 3)

Answer: This is the distributive property of multiplication over addition. Check: 2 x 11 = 22 and 16 + 6 = 22 — both are equal, so the property is verified.

Question: Exercise 2.4, Question 4: Check the commutative property of addition for 530 and 750.

Answer: Left side: 530 + 750 = 1,280. Right side: 750 + 530 = 1,280. Both sides are equal, so the commutative property of addition is verified.

Question: Exercise 2.4, Question 5: Check the associative property of multiplication for 240, 425, 35.

Answer: Left side: (240 x 425) x 35 = 102,000 x 35 = 3,570,000. Right side: 240 x (425 x 35) = 240 x 14,875 = 3,570,000. Both sides are equal, so the associative property of multiplication is verified.

Question: Exercise 2.4, Question 6: Check the distributive property of multiplication over addition for 300, 615, 975.

Answer: Left side: 300 x (615 + 975) = 300 x 1,590 = 477,000. Right side: (300 x 615) + (300 x 975) = 184,500 + 292,500 = 477,000. Both sides are equal, so the distributive property of multiplication over addition is verified.

Review Exercise 2, Question 1: Using the laws of multiplication of whole numbers, fill in the blanks. (i) 2 x _____ = 3 x _____

Answer: 3, 2 = 6. The blanks take 3 and 2.

Review Exercise 2, Question 1: (ii) 112 x 528 = _____ x _____

Answer: 528, 112 = 59,136. The blanks take 528 and 112.

Review Exercise 2, Question 1: (iii) 5 x (3 x 8) = (5 x 3) x _____

Answer: 8 = 120. The blank takes 8.

Review Exercise 2, Question 1: (iv) _____ x 7 = 7 x _____ = 7

Answer: 1 is the identity element of multiplication: 1, 1

Review Exercise 2, Question 1: (v) 9 x (2 + 16) = (_____ x 2) + (_____ x 16)

Answer: (9, (9

Review Exercise 2, Question 1: (vi) (20 + 4) x 1 = (_____ x 1) + (4 x _____)

Answer: (20, 1) = 24. The blanks take 20 and 1.

Review Exercise 2, Question 1: (vii) 17 x (8 − 3) = (17 x _____) − (_____ x 3)

Answer: 8), (17

Review Exercise 2, Question 1: (viii) (12 − 6) x 32 = (12 x _____) − (6 x _____)

Answer: 32), 32) = 192. The blanks take 32 and 32.

Question: Review Exercise 2, Question 2: Fill in the empty boxes. (i) 5 x 4 = □ x 5

Answer: By the commutative property of multiplication: 5 x 4 = 4 x 5 = 20. The box takes 4.

Question: Review Exercise 2, Question 2: (ii) 15 x (10 x 6) = (□ x 10) x 6

Answer: By the associative property of multiplication: 15 x (10 x 6) = (15 x 10) x 6 = 900. The box takes 15.

Question: Review Exercise 2, Question 2: (iii) 10 x (15 + 6) = (10 x 15) + (□ x 6)

Answer: By the distributive property of multiplication over addition: 10 x (15 + 6) = (10 x 15) + (10 x 6) = 210. The box takes 10.

Question: Review Exercise 2, Question 2: (iv) 10 x (100 + 60) = (□ x 100) + (10 x □)

Answer: By the distributive property of multiplication over addition: 10 x (100 + 60) = (10 x 100) + (10 x 60) = 1,600. The boxes take 10 and 60.

Question: Review Exercise 2, Question 3: Write true or false. (i) 5 − 3 = 3 − 5

Answer: Left side: 5 − 3 = 2. Right side: 3 − 5 = −2. They are not equal, so this statement is false. ✗

Question: Review Exercise 2, Question 3: (ii) 3 x 1 + 7 = 3 x 8

Answer: Left side: 3 x 1 + 7 = 3 + 7 = 10. Right side: 3 x 8 = 24. They are not equal, so this statement is false. ✗

Question: Review Exercise 2, Question 3: (iii) 2 + (0 − 8) = (2 + 0) + 8

Answer: Left side: 2 + (0 − 8) = 2 + (−8) = −6. Right side: (2 + 0) + 8 = 10. They are not equal, so this statement is false. ✗

Question: Review Exercise 2, Question 3: (iv) 9 + (7 + 5) = (9 + 7) + 5

Answer: Left side: 9 + 12 = 21. Right side: 16 + 5 = 21. Both are equal — this is the associative property of addition. So this statement is true. ✓

Question: Review Exercise 2, Question 3: (v) 4 x (35 x 2) = (4 x 35) x 2

Answer: Left side: 4 x 70 = 280. Right side: 140 x 2 = 280. Both are equal — this is the associative property of multiplication. So this statement is true. ✓

Question: Review Exercise 2, Question 3: (vi) 24 − (50 − 6) = (24 − 50) − 6

Answer: Left side: 24 − 44 = −20. Right side: (−26) − 6 = −32. They are not equal, so this statement is false. ✗

Question: Review Exercise 2, Question 3: (vii) 0 ÷ 14 = 14

Answer: Dividing 0 by any number gives 0. 0 ÷ 14 = 0, which is not equal to 14. So this statement is false. ✗

Question: Review Exercise 2, Question 3: (viii) 0 ÷ 125 = 0

Answer: Dividing 0 by any number gives 0. 0 ÷ 125 = 0. So this statement is true. ✓

Question: Review Exercise 2, Question 3: (ix) 18 ÷ 18 = 0

Answer: Dividing any number by itself gives 1. 18 ÷ 18 = 1, which is not equal to 0. So this statement is false. ✗

Question: Review Exercise 2, Question 3: (x) 75 ÷ 75 = 1

Answer: Dividing any number by itself gives 1. 75 ÷ 75 = 1. So this statement is true. ✓

Question: Review Exercise 2, Question 4: Write the predecessor of each of the following numbers. (i) 671

Answer: The predecessor of a number is 1 less than it. The predecessor of 671 = 671 − 1 = 670.

Question: Review Exercise 2, Question 4: (ii) 245

Answer: The predecessor of 245 = 245 − 1 = 244.

Question: Review Exercise 2, Question 4: (iii) 99

Answer: The predecessor of 99 = 99 − 1 = 98.

Question: Review Exercise 2, Question 4: (iv) 999

Answer: The predecessor of 999 = 999 − 1 = 998.

Review Exercise 2, Question 5: Choose the correct answer. (i) The smallest natural number is _____. (a) 0 (b) 1 (c) 2 (d) 100

Answer: 1. The correct answer is

Review Exercise 2, Question 5: (ii) In the set of whole numbers, the predecessor of 1 is _____. (a) 0 (b) 2 (c) 3 (d) 1

Answer: 1 − 1 = 0. The correct answer is

Review Exercise 2, Question 5: (iii) The smallest 7-digit number is _____. (a) 123,456 (b) 9,999,999 (c) 1,111,111 (d) 1000,000

Answer: 1,000,000 (written in the book as 1000,000). The correct answer is

Review Exercise 2, Question 5: (iv) The largest 6-digit number is _____. (a) 876,543 (b) 999,999 (c) 111,111 (d) 100,000

Answer: 999,999. The correct answer is

Review Exercise 2, Question 5: (v) Numbers exactly divisible by 2 are called _____ numbers. (a) prime (b) even (c) odd (d) straight

Answer: even numbers — they have 0, 2, 4, 6 or 8 in the units place. The correct answer is

Question: Review Exercise 2, Question 6: Show the following whole numbers on the number line. (i) 9, 3, 1, 0

Answer: On the number line the points 0, 1, 3 and 9 are shown.

Question: Review Exercise 2, Question 6: (ii) Whole numbers greater than 3

Answer: On the number line all points to the right of 3 (4, 5, 6, ...) are shown — the whole numbers greater than 3 are 4, 5, 6, ...

Question: Review Exercise 2, Question 6: (iii) Whole numbers less than or equal to 8

Answer: On the number line all points from 0 to 8 (including 8) are shown — the whole numbers less than or equal to 8 are 0, 1, 2, 3, 4, 5, 6, 7, 8.

Question: Review Exercise 2, Question 6: (iv) Whole numbers greater than 5 but less than 10

Answer: On the number line the points between 5 and 10 (excluding 5 and 10) are shown — those whole numbers are 6, 7, 8, 9.

Question: Review Exercise 2, Question 6: (v) Whole numbers greater than or equal to 1 but less than or equal to 8

Answer: On the number line all points from 1 to 8 (including 1 and 8) are shown — those whole numbers are 1, 2, 3, 4, 5, 6, 7, 8.

Question: Review Exercise 2, Question 7: Find the sum of 4, 2, 3, 5 on the number line.

Answer: On the number line, start at 0 and move 4 steps right (reach 4), then 2 steps further (reach 6), then 3 steps further (reach 9), then 5 steps further (reach 14). So 4 + 2 + 3 + 5 = 14.

Question: Review Exercise 2, Question 8: Check the commutative property of multiplication for 190 and 330.

Answer: Left side: 190 x 330 = 62,700. Right side: 330 x 190 = 62,700. Both sides are equal, so the commutative property of multiplication is verified.

Question: Review Exercise 2, Question 9: Check the associative property of multiplication for 20, 30, 60.

Answer: Left side: (20 x 30) x 60 = 600 x 60 = 36,000. Right side: 20 x (30 x 60) = 20 x 1,800 = 36,000. Both sides are equal, so the associative property of multiplication is verified.

Question: Review Exercise 2, Question 10: Check the distributive property of multiplication over addition for 700, 500, 100.

Answer: Left side: 700 x (500 + 100) = 700 x 600 = 420,000. Right side: (700 x 500) + (700 x 100) = 350,000 + 70,000 = 420,000. Both sides are equal, so the distributive property of multiplication over addition is verified.

Question: Review Exercise 2, Question 11: What will be the price of 640 photocopies? At the rate of Rs 1.50 per copy

Answer: Given: number of copies = 640, price per copy = Rs 1.50. Total price = 640 x 1.50 = Rs 960.

Question: Review Exercise 2, Question 12: Write the largest 4-digit number that is exactly divisible by 44.

Answer: The largest 4-digit number is 9,999. 9,999 ÷ 44: 44 x 227 = 9,988, remainder = 9,999 − 9,988 = 11. So the required number = 9,999 − 11 = 9,988. Check: 44 x 227 = 9,988.

3. Factors & Multiples

Question: Exercise 3.1, Question 1: Which of the following are factors of 20? (a) 2 (b) 8 (c) 5 (d) 3

Answer: A factor is a number that divides 20 exactly. 20 ÷ 2 = 10 (remainder 0), so 2 is a factor. 20 ÷ 8 = 2, remainder 4, so 8 is not a factor. 20 ÷ 5 = 4 (remainder 0), so 5 is a factor. 20 ÷ 3 = 6, remainder 2, so 3 is not a factor. Correct answers: (a) 2 and (c) 5. ✓

Question: Exercise 3.1, Question 2: Find all the factors of each of the following numbers. (a) 15

Answer: Numbers dividing 15: 15 = 1 x 15 = 3 x 5. So the factors of 15 are 1, 3, 5, 15.

Question: Exercise 3.1, Question 2: (b) 30

Answer: 30 = 1 x 30 = 2 x 15 = 3 x 10 = 5 x 6. So the factors of 30 are 1, 2, 3, 5, 6, 10, 15, 30.

Question: Exercise 3.1, Question 2: (c) 50

Answer: 50 = 1 x 50 = 2 x 25 = 5 x 10. So the factors of 50 are 1, 2, 5, 10, 25, 50.

Question: Exercise 3.1, Question 2: (d) 125

Answer: 125 = 1 x 125 = 5 x 25. So the factors of 125 are 1, 5, 25, 125.

Question: Exercise 3.1, Question 2: (e) 150

Answer: 150 = 1 x 150 = 2 x 75 = 3 x 50 = 5 x 30 = 6 x 25 = 10 x 15. So the factors of 150 are 1, 2, 3, 5, 6, 10, 15, 25, 30, 50, 75, 150.

Question: Exercise 3.1, Question 3: Which of the following are multiples of 6? (a) 10 (b) 18 (c) 96 (d) 200

Answer: A multiple is a number exactly divisible by 6. 10 ÷ 6 = 1, remainder 4 (no). 18 ÷ 6 = 3 (remainder 0), so 18 is a multiple. 96 ÷ 6 = 16 (remainder 0), so 96 is a multiple. 200 ÷ 6 = 33, remainder 2 (no). Correct answers: (b) 18 and (c) 96. ✓

Question: Exercise 3.1, Question 4: Write the first seven multiples of the following. (a) 4

Answer: Multiples of 4: 4 x 1 = 4, 4 x 2 = 8, 4 x 3 = 12, 4 x 4 = 16, 4 x 5 = 20, 4 x 6 = 24, 4 x 7 = 28. First seven multiples: 4, 8, 12, 16, 20, 24, 28.

Question: Exercise 3.1, Question 4: (b) 7

Answer: Multiples of 7: 7 x 1 = 7, 7 x 2 = 14, 7 x 3 = 21, 7 x 4 = 28, 7 x 5 = 35, 7 x 6 = 42, 7 x 7 = 49. First seven multiples: 7, 14, 21, 28, 35, 42, 49.

Question: Exercise 3.1, Question 4: (c) 12

Answer: Multiples of 12: 12 x 1 = 12, 12 x 2 = 24, 12 x 3 = 36, 12 x 4 = 48, 12 x 5 = 60, 12 x 6 = 72, 12 x 7 = 84. First seven multiples: 12, 24, 36, 48, 60, 72, 84.

Question: Exercise 3.1, Question 4: (d) 15

Answer: Multiples of 15: 15 x 1 = 15, 15 x 2 = 30, 15 x 3 = 45, 15 x 4 = 60, 15 x 5 = 75, 15 x 6 = 90, 15 x 7 = 105. First seven multiples: 15, 30, 45, 60, 75, 90, 105.

Question: Exercise 3.1, Question 4: (e) 20

Answer: Multiples of 20: 20 x 1 = 20, 20 x 2 = 40, 20 x 3 = 60, 20 x 4 = 80, 20 x 5 = 100, 20 x 6 = 120, 20 x 7 = 140. First seven multiples: 20, 40, 60, 80, 100, 120, 140.

Question: Exercise 3.1, Question 5: Write the prime numbers between the given numbers. (a) 6 and 15

Answer: Numbers between 6 and 15: 7, 8, 9, 10, 11, 12, 13, 14. Of these, the primes (only 1 and itself as factors) are 7, 11, 13. 8, 9, 10, 12, 14 are composite.

Question: Exercise 3.1, Question 5: (b) 20 and 40

Answer: The prime numbers between 20 and 40 are 23, 29, 31, 37. (21, 22, 24, 25, 26, 27, 28, 30, 32, 33, 34, 35, 36, 38, 39 are composite.)

Question: Exercise 3.1, Question 5: (c) 60 and 95

Answer: The prime numbers between 60 and 95 are 61, 67, 71, 73, 79, 83, 89.

Question: Exercise 3.1, Question 6: Write the even prime number.

Answer: Even numbers are exactly divisible by 2. Of the prime numbers, only 2 is even (its factors are 1 and 2). All other primes are odd. So the even prime number is 2.

Question: Exercise 3.1, Question 7: Separate the even and odd numbers from the following numbers. (a) 10

Answer: 10 has 0 in the units place, so 10 is exactly divisible by 2. 10 is an even number.

Question: Exercise 3.1, Question 7: (b) 47

Answer: 47 has 7 in the units place, so 47 is not exactly divisible by 2. 47 is an odd number.

Question: Exercise 3.1, Question 7: (c) 75

Answer: 75 has 5 in the units place, so 75 is not exactly divisible by 2. 75 is an odd number.

Question: Exercise 3.1, Question 7: (d) 91

Answer: 91 has 1 in the units place, so 91 is not exactly divisible by 2. 91 is an odd number.

Question: Exercise 3.1, Question 7: (e) 100

Answer: 100 has 0 in the units place, so 100 is exactly divisible by 2. 100 is an even number.

Question: Exercise 3.1, Question 7: (f) 117

Answer: 117 has 7 in the units place, so 117 is not exactly divisible by 2. 117 is an odd number.

Question: Exercise 3.1, Question 8: Write all the even numbers between 21 and 51.

Answer: The even numbers between 21 and 51 are 22, 24, 26, 28, 30, 32, 34, 36, 38, 40, 42, 44, 46, 48, 50.

Question: Exercise 3.1, Question 9: Write all the odd numbers between 10 and 40.

Answer: The odd numbers between 10 and 40 are 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31, 33, 35, 37, 39.

Question: Exercise 3.1, Question 10: Write all the composite numbers less than 40.

Answer: The composite numbers less than 40 (those with more than 2 factors) are 4, 6, 8, 9, 10, 12, 14, 15, 16, 18, 20, 21, 22, 24, 25, 26, 27, 28, 30, 32, 33, 34, 35, 36, 38, 39.

Question: Exercise 3.1, Question 11: Write all the composite numbers between 71 and 101.

Answer: The composite numbers between 71 and 101 are 72, 74, 75, 76, 77, 78, 80, 81, 82, 84, 85, 86, 87, 88, 90, 91, 92, 93, 94, 95, 96, 98, 99, 100.

Question: Exercise 3.1, Question 12: Write all the prime numbers between 20 and 60.

Answer: The prime numbers between 20 and 60 are 23, 29, 31, 37, 41, 43, 47, 53, 59.

Question: Exercise 3.1, Question 13: The number 1 is neither prime nor composite. Why? Give the reason.

Answer: A prime number has exactly two factors (1 and the number itself)

and a composite number has more than two factors. 1 has only one factor — 1 itself. So 1 is neither prime nor composite.

Question: Exercise 3.2, Question 1: Check. (i) Can the number 251,061 be exactly divided by 3?

Answer: Divisibility test for 3: sum of digits = 2 + 5 + 1 + 0 + 6 + 1 = 15. 15 is exactly divisible by 3, so 251,061 can be exactly divided by 3. ✓

Question: Exercise 3.2, Question 1: (ii) Can the number 92,348 be exactly divided by 4?

Answer: Divisibility test for 4: the last two digits are 48. 48 is exactly divisible by 4 (48 ÷ 4 = 12), so 92,348 can be exactly divided by 4. ✓

Question: Exercise 3.2, Question 1: (iii) Can the number 49,230 be exactly divided by 9?

Answer: Divisibility test for 9: sum of digits = 4 + 9 + 2 + 3 + 0 = 18. 18 is exactly divisible by 9, so 49,230 can be exactly divided by 9. ✓

Question: Exercise 3.2, Question 1: (iv) Can the number 421,50 be exactly divided by 8? (The book prints "421,50", read here as 42,150.)

Answer: The book prints "421,50", read here as 42,150. Divisibility test for 8: the last three digits are 150. 150 ÷ 8 = 18, remainder 6 — 150 is not exactly divisible by 8, so 42,150 cannot be exactly divided by 8. ✗

Question: Exercise 3.2, Question 1: (v) Can the number 1000 be exactly divided by 2?

Answer: Divisibility test for 2: the units digit is 0, so 1000 can be exactly divided by 2. ✓

Question: Exercise 3.2, Question 1: (vi) Can the number 241,566 be exactly divided by 3?

Answer: Divisibility test for 3: sum of digits = 2 + 4 + 1 + 5 + 6 + 6 = 24. 24 is exactly divisible by 3, so 241,566 can be exactly divided by 3. ✓

Question: Exercise 3.2, Question 1: (vii) Can the number 24,268 be exactly divided by 2?

Answer: Divisibility test for 2: the units digit is 8, so 24,268 can be exactly divided by 2. ✓

Question: Exercise 3.2, Question 1: (viii) Can the number 241,361 be exactly divided by 4?

Answer: Divisibility test for 4: the last two digits are 61. 61 is not exactly divisible by 4 (61 ÷ 4 = 15, remainder 1), so 241,361 cannot be exactly divided by 4. ✗

Question: Exercise 3.2, Question 1: (ix) Can the number 123,864 be exactly divided by 8?

Answer: Divisibility test for 8: the last three digits are 864. 864 ÷ 8 = 108 (remainder 0), so 123,864 can be exactly divided by 8. ✓

Question: Exercise 3.2, Question 1: (x) Can the number 4,158,720 be exactly divided by 9?

Answer: Divisibility test for 9: sum of digits = 4 + 1 + 5 + 8 + 7 + 2 + 0 = 27. 27 is exactly divisible by 9, so 4,158,720 can be exactly divided by 9. ✓

Question: Exercise 3.2, Question 1: (xi) Can the number 210,006 be exactly divided by 2?

Answer: Divisibility test for 2: the units digit is 6, so 210,006 can be exactly divided by 2. ✓

Question: Exercise 3.2, Question 1: (xii) Can the number 1,234,562 be exactly divided by 3?

Answer: Divisibility test for 3: sum of digits = 1 + 2 + 3 + 4 + 5 + 6 + 2 = 23. 23 is not exactly divisible by 3, so 1,234,562 cannot be exactly divided by 3. ✗

Question: Exercise 3.2, Question 2: Which of the following numbers can be exactly divided by 6? (i) 2,456

Answer: Test for 6: the number must be exactly divisible by both 2 and 3. The units digit is 6, so it is divisible by 2. Sum of digits = 2 + 4 + 5 + 6 = 17, which is not divisible by 3. So 2,456 cannot be exactly divided by 6. ✗

Question: Exercise 3.2, Question 2: (ii) 7,121,700

Answer: Test for 6: the units digit is 0, so it is divisible by 2. Sum of digits = 7 + 1 + 2 + 1 + 7 + 0 + 0 = 18, which is divisible by 3. So 7,121,700 can be exactly divided by 6. ✓

Question: Exercise 3.2, Question 2: (iii) 19,206

Answer: Test for 6: the units digit is 6, so it is divisible by 2. Sum of digits = 1 + 9 + 2 + 0 + 6 = 18, which is divisible by 3. So 19,206 can be exactly divided by 6. ✓

Question: Exercise 3.2, Question 3: Which of the following numbers can be exactly divided by 5? (i) 24,567

Answer: Test for 5: the units digit must be 0 or 5. Here the units digit is 7, so 24,567 cannot be exactly divided by 5. ✗

Question: Exercise 3.2, Question 3: (ii) 230,590

Answer: Test for 5: the units digit is 0, so 230,590 can be exactly divided by 5. ✓

Question: Exercise 3.2, Question 3: (iii) 111,165

Answer: Test for 5: the units digit is 5, so 111,165 can be exactly divided

by 5. ✓

Question: Exercise 3.2, Question 4: Which of the following numbers can be exactly divided by 10? (i) 25,670

Answer: Test for 10: the units digit must be 0. Here the units digit is 0, so 25,670 can be exactly divided by 10. ✓

Question: Exercise 3.2, Question 4: (ii) 123,600

Answer: Test for 10: the units digit is 0, so 123,600 can be exactly divided by 10. ✓

Question: Exercise 3.2, Question 4: (iii) 14,675

Answer: Test for 10: the units digit must be 0. Here the units digit is 5, so 14,675 cannot be exactly divided by 10. ✗

Question: Exercise 3.2, Question 5: Which of the following numbers can be exactly divided by 11? (i) 215,762

Answer: Test for 11: alternate sums of digits (from the right): odd places: 2 + 7 + 1 = 10; even places: 6 + 5 + 2 = 13. Difference = 13 − 10 = 3, which is not divisible by 11. So 215,762 cannot be exactly divided by 11. ✗

Question: Exercise 3.2, Question 5: (ii) 52,958,400

Answer: Test for 11: odd places: 0 + 4 + 5 + 2 = 11; even places: 0 + 8 + 9 + 5 = 22. Difference = 22 − 11 = 11, which is divisible by 11. So 52,958,400 can be exactly divided by 11. ✓

Question: Exercise 3.2, Question 5: (iii) 94,857,290

Answer: Test for 11: odd places: 0 + 2 + 5 + 4 = 11; even places: 9 + 7 + 8 + 9 = 33. Difference = 33 − 11 = 22, which is divisible by 11. So 94,857,290 can be exactly divided by 11. ✓

Question: Exercise 3.2, Question 6: Which of the following numbers can be exactly divided by 12? (i) 245,168

Answer: Test for 12: the number must be exactly divisible by both 3 and 4. Sum of digits = 2 + 4 + 5 + 1 + 6 + 8 = 26, which is not divisible by 3. So 245,168 cannot be exactly divided by 12. ✗

Question: Exercise 3.2, Question 6: (ii) 512,100

Answer: Test for 12: sum of digits = 5 + 1 + 2 + 1 + 0 + 0 = 9, which is divisible by 3. The last two digits 00 are divisible by 4. So 512,100 can be exactly divided by 12. ✓

Question: Exercise 3.2, Question 6: (iii) 2,561,742

Answer: Test for 12: sum of digits = 2 + 5 + 6 + 1 + 7 + 4 + 2 = 27, which is divisible by 3. The last two digits 42 are not divisible by 4 (42 ÷ 4 = 10, remainder 2). So 2,561,742 cannot be exactly divided by 12. ✗

Question: Exercise 3.2, Question 7: Which of the following numbers can be exactly divided by 15? (i) 51,300

Answer: Test for 15: the number must be exactly divisible by both 3 and 5. Sum of digits = 5 + 1 + 3 + 0 + 0 = 9, which is divisible by 3. The units digit is 0, so it is divisible by 5. So 51,300 can be exactly divided by 15. ✓

Question: Exercise 3.2, Question 7: (ii) 523,449

Answer: Test for 15: sum of digits = 5 + 2 + 3 + 4 + 4 + 9 = 27, which is divisible by 3. The units digit is 9, so it is not divisible by 5. So 523,449 cannot be exactly divided by 15. ✗

Question: Exercise 3.2, Question 7: (iii) 236,790

Answer: Test for 15: sum of digits = 2 + 3 + 6 + 7 + 9 + 0 = 27, which is divisible by 3. The units digit is 0, so it is divisible by 5. So 236,790 can be exactly divided by 15. ✓

Question: Exercise 3.2, Question 8: Which of the following numbers can be exactly divided by 25? (i) 2,563,400

Answer: Test for 25: the last two digits are 00. 00 is divisible by 25, so 2,563,400 can be exactly divided by 25. ✓

Question: Exercise 3.2, Question 8: (ii) 341,236

Answer: Test for 25: the last two digits are 36. 36 is not divisible by 25, so 341,236 cannot be exactly divided by 25. ✗

Question: Exercise 3.2, Question 8: (iii) 4,102,375

Answer: Test for 25: the last two digits are 75. 75 ÷ 25 = 3 (remainder 0), so 4,102,375 can be exactly divided by 25. ✓

Question: Exercise 3.2, Question 9: Which of the following numbers can be exactly divided by 2 and 3? (i) 14,562

Answer: The units digit is 2, so it is divisible by 2. Sum of digits = 1 + 4 + 5 + 6 + 2 = 18, which is divisible by 3. So 14,562 can be exactly divided by both 2 and 3. ✓

Question: Exercise 3.2, Question 9: (ii) 1,101,361

Answer: The units digit is 1, so it is not divisible by 2. So 1,101,361 cannot be exactly divided by both 2 and 3. ✗

Question: Exercise 3.2, Question 9: (iii) 518,214

Answer: The units digit is 4, so it is divisible by 2. Sum of digits = 5 + 1 + 8 + 2 + 1 + 4 = 21, which is divisible by 3. So 518,214 can be exactly divided by both 2 and 3. ✓

Question: Exercise 3.2, Question 10: Which of the following numbers can be exactly divided by 4 and 8? (i) 513,864

Answer: Test for 4: the last two digits 64 are divisible by 4. Test for 8: the last three digits 864, 864 ÷ 8 = 108 (remainder 0). So 513,864 can be exactly divided by both 4 and 8. ✓

Question: Exercise 3.2, Question 10: (ii) 617,116

Answer: Test for 4: the last two digits 16 are divisible by 4. Test for 8: the last three digits 116, 116 ÷ 8 = 14, remainder 4 — not divisible by 8. So 617,116 cannot be exactly divided by both 4 and 8. ✗

Question: Exercise 3.2, Question 10: (iii) 2,415,848

Answer: Test for 4: the last two digits 48 are divisible by 4. Test for 8: the last three digits 848, 848 ÷ 8 = 106 (remainder 0). So 2,415,848 can be exactly divided by both 4 and 8. ✓

Question: Exercise 3.2, Question 11: Which of the following numbers can be exactly divided by 2, 5, 10? (i) 120,340

Answer: The units digit is 0, so it is exactly divisible by 2, 5 and 10 all three. So 120,340 can be exactly divided by 2, 5, 10. ✓

Question: Exercise 3.2, Question 11: (ii) 210,785

Answer: The units digit is 5: it is divisible by 5, but not by 2 and 10. So 210,785 cannot be exactly divided by 2, 5, 10 all three — it is divisible only by 5. ✗

Question: Exercise 3.2, Question 11: (iii) 412,360

Answer: The units digit is 0, so it is exactly divisible by 2, 5 and 10 all three. So 412,360 can be exactly divided by 2, 5, 10. ✓

Question: Exercise 3.3, Question 1: Write the following factors in power form. (I) 2 x 2 x 2 x 3 x 5 x 3

Answer: Count the repeated factors: 2 three times, 3 two times, 5 once. So in power form: $2^3 \times 3^2 \times 5$.

Question: Exercise 3.3, Question 1: (ii) 3 x 5 x 5 x 5 x 7 x 7

Answer: Count the repeated factors: 3 once, 5 three times, 7 two times. So in power form: $3 \times 5^3 \times 7^2$.

Question: Exercise 3.3, Question 1: (iii) 2 x 3 x 3 x 7 x 7 x 3 x 3

Answer: Count the repeated factors: 2 once, 3 four times, 7 two times. So in power form: $2 \times 3^4 \times 7^2$.

Question: Exercise 3.3, Question 2: Using the division method, find the prime factors of each of the following numbers and write them in power form. (i) 24

Answer: Division method: 24 ÷ 2 = 12, 12 ÷ 2 = 6, 6 ÷ 2 = 3, 3 ÷ 3 = 1. Prime factors: 2 x 2 x 2 x 3 = $2^3 \times 3$. Check: $2^3 \times 3 = 8 \times 3 = 24$.

Question: Exercise 3.3, Question 2: (ii) 48

Answer: Division method: 48 ÷ 2 = 24, 24 ÷ 2 = 12, 12 ÷ 2 = 6, 6 ÷ 2 = 3, 3 ÷ 3 = 1. Prime factors: 2 x 2 x 2 x 2 x 3 = 2^4 x 3. Check: 2^4 x 3 = 16 x 3 = 48.

Question: Exercise 3.3, Question 2: (iii) 216

Answer: Division method: 216 ÷ 2 = 108, 108 ÷ 2 = 54, 54 ÷ 2 = 27, 27 ÷ 3 = 9, 9 ÷ 3 = 3, 3 ÷ 3 = 1. Prime factors: 2 x 2 x 2 x 3 x 3 x 3 = 2^3 x 3^3. Check: 8 x 27 = 216.

Question: Exercise 3.3, Question 2: (iv) 250

Answer: Division method: 250 ÷ 2 = 125, 125 ÷ 5 = 25, 25 ÷ 5 = 5, 5 ÷ 5 = 1. Prime factors: 2 x 5 x 5 x 5 = 2 x 5^3. Check: 2 x 125 = 250.

Question: Exercise 3.3, Question 2: (v) 468

Answer: Division method: 468 ÷ 2 = 234, 234 ÷ 2 = 117, 117 ÷ 3 = 39, 39 ÷ 3 = 13, 13 ÷ 13 = 1. Prime factors: 2 x 2 x 3 x 3 x 13 = 2^2 x 3^2 x 13. Check: 4 x 9 x 13 = 468.

Question: Exercise 3.3, Question 2: (vi) 540

Answer: Division method: 540 ÷ 2 = 270, 270 ÷ 2 = 135, 135 ÷ 3 = 45, 45 ÷ 3 = 15, 15 ÷ 3 = 5, 5 ÷ 5 = 1. Prime factors: 2 x 2 x 3 x 3 x 3 x 5 = 2^2 x 3^3 x 5. Check: 4 x 27 x 5 = 540.

Question: Exercise 3.3, Question 2: (vii) 1024

Answer: Division method: 1024 ÷ 2 = 512, 512 ÷ 2 = 256, 256 ÷ 2 = 128, 128 ÷ 2 = 64, 64 ÷ 2 = 32, 32 ÷ 2 = 16, 16 ÷ 2 = 8, 8 ÷ 2 = 4, 4 ÷ 2 = 2, 2 ÷ 2 = 1. Prime factors: 2 ten times = 2^{10}. Check: 2^{10} = 1024.

Question: Exercise 3.3, Question 2: (viii) 5000

Answer: Division method: 5000 ÷ 2 = 2500, 2500 ÷ 2 = 1250, 1250 ÷ 2 = 625, 625 ÷ 5 = 125, 125 ÷ 5 = 25, 25 ÷ 5 = 5, 5 ÷ 5 = 1. Prime factors: 2 x 2 x 2 x 5 x 5 x 5 x 5 = 2^3 x 5^4. Check: 8 x 625 = 5000.

Question: Exercise 3.3, Question 3: Can every prime factorization be written in power form?

Answer: Yes. In a prime factorization each prime factor may be repeated some number of times, and repeated factors can be written with a power. Example: 12 = 2 x 2 x 3 = 2^2 x 3.

Question: Exercise 3.4, Question 1: Find the HCF by the prime factorization method. (i) 50, 75

Answer: Prime factors of 50: 2 x 5^2. Prime factors of 75: 3 x 5^2. Common prime factors in both: 5^2 = 25. So HCF (50, 75) = 25.

Question: Exercise 3.4, Question 1: (ii) 98, 196

Answer: Prime factors of 98: 2 x 7^2. Prime factors of 196: 2^2 x 7^2. Common prime factors in both (with the lower powers): 2 x 7^2 = 98. So HCF (98, 196) = 98.

Question: Exercise 3.4, Question 1: (iii) 144, 198

Answer: Prime factors of 144: 2^4 x 3^2. Prime factors of 198: 2 x 3^2 x 11. Common prime factors in both (with the lower powers): 2 x 3^2 = 18. So HCF (144, 198) = 18.

Question: Exercise 3.4, Question 1: (iv) 120, 144, 204

Answer: 120 = 2^3 x 3 x 5. 144 = 2^4 x 3^2. 204 = 2^2 x 3 x 17. Common prime factors in all three (with the lower powers): 2^2 x 3 = 12. So HCF (120, 144, 204) = 12.

Question: Exercise 3.4, Question 1: (v) 106, 159, 265

Answer: 106 = 2 x 53. 159 = 3 x 53. 265 = 5 x 53. The common prime factor in all three is 53. So HCF (106, 159, 265) = 53.

Question: Exercise 3.4, Question 1: (vi) 12, 48, 36, 24

Answer: 12 = 2^2 x 3. 48 = 2^4 x 3. 36 = 2^2 x 3^2. 24 = 2^3 x 3. Common prime factors in all four (with the lower powers): 2^2 x 3 = 12. So HCF (12, 48, 36, 24) = 12.

Question: Exercise 3.4, Question 1: (vii) 60, 70, 420, 480

Answer: 60 = 2^2 x 3 x 5. 70 = 2 x 5 x 7. 420 = 2^2 x 3 x 5 x 7. 480 = 2^5 x 3 x 5. Common prime factors in all four (with the lower powers): 2 x 5 = 10. So HCF (60, 70, 420, 480) = 10.

Question: Exercise 3.4, Question 2: Find the HCF by the division method. (i) 12, 20

Answer: Division method: divide 20 by 12: quotient 1, remainder 8. Divide 12 by 8: quotient 1, remainder 4. Divide 8 by 4: quotient 2, remainder 0. The last divisor is 4. So HCF (12, 20) = 4.

Question: Exercise 3.4, Question 2: (ii) 81, 117

Answer: Division method: divide 117 by 81: quotient 1, remainder 36. Divide 81 by 36: quotient 2, remainder 9. Divide 36 by 9: quotient 4, remainder 0. The last divisor is 9. So HCF (81, 117) = 9.

Question: Exercise 3.4, Question 2: (iii) 935, 1320

Answer: Division method: divide 1320 by 935: quotient 1, remainder 385. Divide 935 by 385: quotient 2, remainder 165. Divide 385 by 165: quotient 2, remainder 55. Divide 165 by 55: quotient 3, remainder 0. The last divisor is 55. So HCF (935, 1320) = 55.

Question: Exercise 3.4, Question 2: (iv) 252, 576

Answer: Division method: divide 576 by 252: quotient 2, remainder 72.

Divide 252 by 72: quotient 3, remainder 36. Divide 72 by 36: quotient 2, remainder 0. The last divisor is 36. So HCF (252, 576) = 36.

Question: Exercise 3.4, Question 2: (v) 2241, 8217, 747

Answer: First the HCF of 747 and 2241: divide 2241 by 747: quotient 3, remainder 0, so HCF = 747. Now the HCF of 747 and 8217: divide 8217 by 747: 747 x 11 = 8217, remainder 0, so HCF = 747. So HCF (2241, 8217, 747) = 747.

Question: Exercise 3.4, Question 2: (vi) 30, 120, 90, 210

Answer: By the division method, two at a time: HCF (30, 120) = 30 (divide 120 by 30: remainder 0). HCF (30, 90) = 30 (divide 90 by 30: remainder 0). HCF (30, 210) = 30 (divide 210 by 30: remainder 0). So HCF (30, 120, 90, 210) = 30.

Question: Exercise 3.4, Question 3: What will be the HCF of two prime numbers?

Answer: Two prime numbers have only one common factor, 1, because the factors of each prime number are only 1 and the number itself, and the two primes are different from each other. So the HCF of two prime numbers = 1.

Question: Exercise 3.5, Question 1: Find the LCM by the prime factorization method. (i) 12, 25, 40

Answer: $12 = 2^2 \times 3$. $25 = 5^2$. $40 = 2^3 \times 5$. For the LCM take the highest power of each prime factor: $2^3 \times 3 \times 5^2 = 8 \times 3 \times 25 = 600$. So LCM (12, 25, 40) = 600.

Question: Exercise 3.5, Question 1: (ii) 21, 49, 63

Answer: $21 = 3 \times 7$. $49 = 7^2$. $63 = 3^2 \times 7$. For the LCM take the highest power of each prime factor: $3^2 \times 7^2 = 9 \times 49 = 441$. So LCM (21, 49, 63) = 441.

Question: Exercise 3.5, Question 1: (iii) 144, 180, 384

Answer: $144 = 2^4 \times 3^2$. $180 = 2^2 \times 3^2 \times 5$. $384 = 2^7 \times 3$. For the LCM take the highest power of each prime factor: $2^7 \times 3^2 \times 5 = 128 \times 9 \times 5 = 5{,}760$. So LCM (144, 180, 384) = 5,760.

Question: Exercise 3.5, Question 1: (iv) 108, 135, 162

Answer: $108 = 2^2 \times 3^3$. $135 = 3^3 \times 5$. $162 = 2 \times 3^4$. For the LCM take the highest power of each prime factor: $2^2 \times 3^4 \times 5 = 4 \times 81 \times 5 = 1{,}620$. So LCM (108, 135, 162) = 1,620.

Question: Exercise 3.5, Question 1: (v) 35, 65, 75

Answer: $35 = 5 \times 7$. $65 = 5 \times 13$. $75 = 3 \times 5^2$. For the LCM take the highest power of each prime factor: $3 \times 5^2 \times 7 \times 13 = 3 \times 25 \times 7 \times 13 = 6{,}825$. So LCM (35, 65, 75) = 6,825.

Question: Exercise 3.5, Question 1: (vi) 24, 36, 48, 72

Answer: 24 = 2^3 x 3. 36 = 2^2 x 3^2. 48 = 2^4 x 3. 72 = 2^3 x 3^2. For the LCM take the highest power of each prime factor: 2^4 x 3^2 = 16 x 9 = 144. So LCM (24, 36, 48, 72) = 144.

Question: Exercise 3.5, Question 2: Find the LCM by the division method. (i) 45, 55

Answer: Division method: divide 45 and 55 by 5: 9, 11. Now 9 and 11 have no common divisor. LCM = 5 x 9 x 11 = 495.

Question: Exercise 3.5, Question 2: (ii) 21, 35, 70

Answer: Division method: divide 21, 35, 70 by 7: 3, 5, 10. Then divide 3, 5, 10 by 5: 3, 1, 2. Now there is no common divisor. LCM = 7 x 5 x 3 x 1 x 2 = 210.

Question: Exercise 3.5, Question 2: (iii) 24, 40, 60

Answer: Division method: divide 24, 40, 60 by 2: 12, 20, 30. Again by 2: 6, 10, 15. Then by 3: 2, 10, 5. Then by 2: 1, 5, 5. Then by 5: 1, 1, 1. LCM = 2 x 2 x 3 x 2 x 5 = 120.

Question: Exercise 3.5, Question 2: (iv) 72, 108, 120

Answer: Division method: divide 72, 108, 120 by 2: 36, 54, 60. Again by 2: 18, 27, 30. Then by 3: 6, 9, 10. Then by 2: 3, 9, 5. Then by 3: 1, 3, 5. Then by 3: 1, 1, 5. Then by 5: 1, 1, 1. LCM = 2 x 2 x 3 x 2 x 3 x 3 x 5 = 1,080.

Question: Exercise 3.5, Question 3: The product of two numbers is 360 and their HCF is 16. Find their LCM.

Answer: Rule: product of two numbers = HCF x LCM, so LCM = product ÷ HCF = 360 ÷ 16 = 22.5. This is not a whole number, and 16 does not divide 360 exactly. So as printed in the book, no valid whole-number LCM exists — the numbers given in the book contain an error. (Note: if the HCF is 16, the product of the two numbers must be exactly divisible by 16.)

Question: Exercise 3.6, Question 1: Find the longest tape that can measure both 540 cm and 360 cm exactly.

Answer: Given: lengths = 540 cm and 360 cm. The longest measure = HCF (540, 360). 540 = 2^2 x 3^3 x 5. 360 = 2^3 x 3^2 x 5. Common prime factors (with lower powers): 2^2 x 3^2 x 5 = 4 x 9 x 5 = 180. So the longest tape is 180 cm.

Question: Exercise 3.6, Question 2: Find the largest number that leaves remainder 7 when dividing 232 and remainder 5 when dividing 305.

Answer: Given: dividing 232 leaves remainder 7, dividing 305 leaves remainder 5. After removing the remainders: 232 − 7 = 225, 305 − 5 = 300. The required number = HCF (225, 300). 225 = 3^2 x 5^2. 300 = 2^2 x 3 x 5^2. Common prime factors (with lower powers): 3 x 5^2 = 75. Check: 232 ÷ 75 = 3, remainder 7; 305 ÷ 75 = 4, remainder 5. So the required number is 75.

Question: Exercise 3.6, Question 3: Find the largest number that leaves remainder 5 each time when dividing 245 and 1029.

Answer: Given: dividing both leaves remainder 5. After removing the remainder: 245 − 5 = 240, 1029 − 5 = 1024. The required number = HCF (240, 1024). 240 = 2^4 x 3 x 5. 1024 = 2^{10}. Common prime factors (with lower powers): 2^4 = 16. Check: 245 ÷ 16 = 15, remainder 5; 1029 ÷ 16 = 64, remainder 5. So the required number is 16.

Question: Exercise 3.6, Question 4: Two petrol tanks have capacities of 600 litres and 570 litres. Find the capacity of a measure that can measure both tanks exactly.

Answer: Given: capacities = 600 litres and 570 litres. The largest measure = HCF (600, 570). 600 = 2^3 x 3 x 5^2. 570 = 2 x 3 x 5 x 19. Common prime factors: 2 x 3 x 5 = 30. So the capacity of the measure is 30 litres.

Question: Exercise 3.6, Question 5: A room is 8 m long, 6 m wide and 4 m high. Find the length of the longest tape that can measure all three dimensions of the room exactly.

Answer: Given: dimensions = 8 m, 6 m, 4 m. The longest tape = HCF (8, 6, 4). 8 = 2^3. 6 = 2 x 3. 4 = 2^2. Common prime factor (with lower power): 2. So the longest tape is 2 m.

Question: Exercise 3.6, Question 6: Three bells ring every 5 minutes, 10 minutes and 15 minutes. If they ring together at 8:00 a.m., at what time will they ring together again?

Answer: Given: intervals = 5 min, 10 min, 15 min; time of ringing together = 8:00 a.m. The interval after which they ring together again = LCM (5, 10, 15). 5 = 5. 10 = 2 x 5. 15 = 3 x 5. LCM = 2 x 3 x 5 = 30 minutes. So they will ring together again at 8:30 a.m.

Question: Exercise 3.6, Question 7: Find the smallest number that leaves remainder 9 when divided by 30, 45, 60.

Answer: Given: divisors = 30, 45, 60; remainder = 9. First find LCM (30, 45, 60): 30 = 2 x 3 x 5. 45 = 3^2 x 5. 60 = 2^2 x 3 x 5. LCM = 2^2 x 3^2 x 5 = 180. The required number = 180 + 9 = 189. Check: 189 ÷ 30 = 6, remainder 9; 189 ÷ 45 = 4, remainder 9; 189 ÷ 60 = 3, remainder 9.

Question: Exercise 3.6, Question 8: Find the smallest quantity that can be measured exactly with buckets of 12 litres, 16 litres and 24 litres.

Answer: Given: bucket capacities = 12 litres, 16 litres, 24 litres. The required quantity = LCM (12, 16, 24). 12 = 2^2 x 3. 16 = 2^4. 24 = 2^3 x 3. LCM = 2^4 x 3 = 48. So the smallest quantity is 48 litres.

Question: Exercise 3.6, Question 9: Three traffic lights turn green every 48 seconds, 72 seconds and 108 seconds. If they turn green together at 7:00 a.m., at what time will they turn green together again?

Answer: Given: intervals = 48 sec, 72 sec, 108 sec; time of turning green together = 7:00 a.m. The interval after which they turn green together again = LCM (48, 72, 108). 48 = 2^4 x 3. 72 = 2^3 x 3^2. 108 = 2^2 x 3^3. LCM =

$2^4 \times 3^3 = 16 \times 27 = 432$ seconds = 7 minutes 12 seconds. So they will turn green together again at 7:07:12 a.m.

Question: Review Exercise 3, Question 1: Find all the factors of each of the following numbers. (i) 60

Answer: 60 = 1 x 60 = 2 x 30 = 3 x 20 = 4 x 15 = 5 x 12 = 6 x 10. So the factors of 60 are 1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60.

Question: Review Exercise 3, Question 1: (ii) 250

Answer: 250 = 1 x 250 = 2 x 125 = 5 x 50 = 10 x 25. So the factors of 250 are 1, 2, 5, 10, 25, 50, 125, 250.

Question: Review Exercise 3, Question 2: Write the first five multiples of the following. (i) 13

Answer: Multiples of 13: 13 x 1 = 13, 13 x 2 = 26, 13 x 3 = 39, 13 x 4 = 52, 13 x 5 = 65. First five multiples: 13, 26, 39, 52, 65.

Question: Review Exercise 3, Question 2: (ii) 20

Answer: Multiples of 20: 20 x 1 = 20, 20 x 2 = 40, 20 x 3 = 60, 20 x 4 = 80, 20 x 5 = 100. First five multiples: 20, 40, 60, 80, 100.

Question: Review Exercise 3, Question 3: Write all the prime numbers from 1 to 50.

Answer: The prime numbers from 1 to 50 are 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47.

Question: Review Exercise 3, Question 4: Write all the composite numbers from 30 to 60.

Answer: The composite numbers from 30 to 60 (including the endpoints) are 30, 32, 33, 34, 35, 36, 38, 39, 40, 42, 44, 45, 46, 48, 49, 50, 51, 52, 54, 55, 56, 57, 58, 60. (31, 37, 41, 43, 47, 53, 59 are prime.)

Question: Review Exercise 3, Question 5: Check whether the following numbers are exactly divisible by 2 or not. (i) 31621

Answer: The units digit is 1, which is not even. So 31621 cannot be exactly divided by 2. ✗

Question: Review Exercise 3, Question 5: (ii) 7008

Answer: The units digit is 8, which is even. So 7008 can be exactly divided by 2. ✓

Question: Review Exercise 3, Question 5: (iii) 91130

Answer: The units digit is 0, which is even. So 91130 can be exactly divided by 2. ✓

Question: Review Exercise 3, Question 5: (iv) 5178

Answer: The units digit is 8, which is even. So 5178 can be exactly divided by 2. ✓

Question: Review Exercise 3, Question 6: Check whether the following numbers are exactly divisible by 3 or not. (i) 51237

Answer: Sum of digits = 5 + 1 + 2 + 3 + 7 = 18. 18 is exactly divisible by 3, so 51237 can be exactly divided by 3. ✓

Question: Review Exercise 3, Question 6: (ii) 30001

Answer: Sum of digits = 3 + 0 + 0 + 0 + 1 = 4. 4 is not exactly divisible by 3, so 30001 cannot be exactly divided by 3. ✗

Question: Review Exercise 3, Question 6: (iii) 1001001

Answer: Sum of digits = 1 + 0 + 0 + 1 + 0 + 0 + 1 = 3. 3 is exactly divisible by 3, so 1001001 can be exactly divided by 3. ✓

Question: Review Exercise 3, Question 6: (iv) 56712

Answer: Sum of digits = 5 + 6 + 7 + 1 + 2 = 21. 21 is exactly divisible by 3, so 56712 can be exactly divided by 3. ✓

Question: Review Exercise 3, Question 7: Check whether the following numbers are exactly divisible by 4 or not. (i) 2173

Answer: The last two digits are 73. 73 is not exactly divisible by 4 (73 ÷ 4 = 18, remainder 1), so 2173 cannot be exactly divided by 4. ✗

Question: Review Exercise 3, Question 7: (ii) 41524

Answer: The last two digits are 24. 24 ÷ 4 = 6 (remainder 0), so 41524 can be exactly divided by 4. ✓

Question: Review Exercise 3, Question 7: (iii) 71611

Answer: The last two digits are 11. 11 is not exactly divisible by 4, so 71611 cannot be exactly divided by 4. ✗

Question: Review Exercise 3, Question 7: (iv) 40048

Answer: The last two digits are 48. 48 ÷ 4 = 12 (remainder 0), so 40048 can be exactly divided by 4. ✓

Question: Review Exercise 3, Question 8: Check whether the following numbers are exactly divisible by 5 or not. (i) 2010

Answer: The units digit is 0, so 2010 can be exactly divided by 5. ✓

Question: Review Exercise 3, Question 8: (ii) 31625

Answer: The units digit is 5, so 31625 can be exactly divided by 5. ✓

Question: Review Exercise 3, Question 8: (iii) 7128

Answer: The units digit is 8, which is neither 0 nor 5. So 7128 cannot be exactly divided by 5. ✗

Question: Review Exercise 3, Question 8: (iv) 1001

Answer: The units digit is 1, which is neither 0 nor 5. So 1001 cannot be exactly divided by 5. ✗

Question: Review Exercise 3, Question 9: Check whether the following numbers are exactly divisible by 10 or not. (i) 2165

Answer: The units digit is 5, not 0. So 2165 cannot be exactly divided by 10. ✗

Question: Review Exercise 3, Question 9: (ii) 71230

Answer: The units digit is 0, so 71230 can be exactly divided by 10. ✓

Question: Review Exercise 3, Question 9: (iii) 10000

Answer: The units digit is 0, so 10000 can be exactly divided by 10. ✓

Question: Review Exercise 3, Question 9: (iv) 25618

Answer: The units digit is 8, not 0. So 25618 cannot be exactly divided by 10. ✗

Question: Review Exercise 3, Question 10: Using the division method and the factor tree, find the prime factors of the following numbers and write them in power form. (i) 450

Answer: Division method: 450 ÷ 2 = 225, 225 ÷ 3 = 75, 75 ÷ 3 = 25, 25 ÷ 5 = 5, 5 ÷ 5 = 1. Factor tree: 450 = 45 x 10 = (9 x 5) x (2 x 5) = (3 x 3 x 5) x (2 x 5). Prime factors: 2 x 3 x 3 x 5 x 5 = 2 x 3^2 x 5^2. Check: 2 x 9 x 25 = 450.

Question: Review Exercise 3, Question 10: (ii) 720

Answer: Division method: 720 ÷ 2 = 360, 360 ÷ 2 = 180, 180 ÷ 2 = 90, 90 ÷ 2 = 45, 45 ÷ 3 = 15, 15 ÷ 3 = 5, 5 ÷ 5 = 1. Factor tree: 720 = 72 x 10 = (8 x 9) x (2 x 5) = (2 x 2 x 2 x 3 x 3) x (2 x 5). Prime factors: 2 x 2 x 2 x 2 x 3 x 3 x 5 = 2^4 x 3^2 x 5. Check: 16 x 9 x 5 = 720.

Question: Review Exercise 3, Question 11: Find the HCF and LCM by the prime factorization method. (i) 18, 24

Answer: 18 = 2 x 3^2. 24 = 2^3 x 3. HCF (common prime factors, lower powers): 2 x 3 = 6. LCM (highest power of each prime factor): 2^3 x 3^2 = 8 x 9 = 72. So HCF = 6, LCM = 72.

Question: Review Exercise 3, Question 11: (ii) 12, 15, 40

Answer: 12 = 2^2 x 3. 15 = 3 x 5. 40 = 2^3 x 5. HCF (prime factors common to all three): none — so HCF = 1. LCM (highest power of each prime factor): 2^3 x 3 x 5 = 8 x 3 x 5 = 120. So HCF = 1, LCM = 120.

Question: Review Exercise 3, Question 12: Find the HCF and LCM by the division method. (i) 65, 80

Answer: HCF (division method): divide 80 by 65: quotient 1, remainder 15. Divide 65 by 15: quotient 4, remainder 5. Divide 15 by 5: quotient 3, remainder 0. The last divisor is 5, so HCF = 5. LCM = (65 x 80) ÷ HCF = 5,200 ÷ 5 = 1,040. Check: 1,040 = 65 x 16 = 80 x 13.

Question: Review Exercise 3, Question 12: (ii) 26, 65, 169

Answer: HCF: HCF (26, 65): divide 65 by 26: quotient 2, remainder 13. Divide 26 by 13: quotient 2, remainder 0, so HCF = 13. HCF (13, 169): divide 169 by 13: 13 x 13 = 169, remainder 0, so HCF = 13. LCM: LCM (26, 65) = (26 x 65) ÷ 13 = 1,690 ÷ 13 = 130. LCM (130, 169) = (130 x 169) ÷ 13 = 21,970 ÷ 13 = 1,690. So HCF = 13, LCM = 1,690.

Question: Review Exercise 3, Question 13: Find the largest number that leaves remainder 3 when dividing 43, remainder 5 when dividing 55 and remainder 7 when dividing 67.

Answer: Given: dividing 43 leaves remainder 3, dividing 55 leaves remainder 5, dividing 67 leaves remainder 7. After removing the remainders: 43 − 3 = 40, 55 − 5 = 50, 67 − 7 = 60. The required number = HCF (40, 50, 60). $40 = 2^3 \times 5$. $50 = 2 \times 5^2$. $60 = 2^2 \times 3 \times 5$. Common prime factors (with lower powers): 2 x 5 = 10. Check: 43 ÷ 10 = 4, remainder 3; 55 ÷ 10 = 5, remainder 5; 67 ÷ 10 = 6, remainder 7. So the required number is 10.

Question: Review Exercise 3, Question 14: Find the smallest number that leaves remainder 6 when divided by 20, 30, 45.

Answer: Given: divisors = 20, 30, 45; remainder = 6. First find LCM (20, 30, 45): $20 = 2^2 \times 5$. $30 = 2 \times 3 \times 5$. $45 = 3^2 \times 5$. LCM = $2^2 \times 3^2 \times 5 = 180$. The required number = 180 + 6 = 186. Check: 186 ÷ 20 = 9, remainder 6; 186 ÷ 30 = 6, remainder 6; 186 ÷ 45 = 4, remainder 6.

4. Integers

Question: Exercise 4.1, Question 1 (i): Show the following integers on the number line: −2, −1, 0, +1, +2

Answer: Mark 0 in the middle of the number line. Mark the points +1 and +2 at equal distances to the right of 0, and mark the points −1 and −2 at equal distances to the left of 0.

Question: Exercise 4.1, Question 1 (ii): Show the following integers on the number line: −3, −2, −1, 0, +1, +2, +3, +4

Answer: Mark 0 in the middle of the number line. Mark the points +1, +2, +3 and +4 at equal distances to the right of 0, and mark the points −1, −2 and −3 at equal distances to the left of 0.

Question: Exercise 4.1, Question 1 (iii): Show the following integers on the number line: −4, −3, −2, −1, 0, +1, +2

Answer: Mark 0 in the middle of the number line. Mark the points +1 and +2 at equal distances to the right of 0, and mark the points −1, −2, −3 and −4 at equal distances to the left of 0.

Question: Exercise 4.1, Question 1 (iv): Show the following integers on the number line: +5, −5, −4, +3, +1, −2

Answer: Mark 0 in the middle of the number line. Mark the points +1, +3 and +5 at equal distances to the right of 0, and mark the points −2, −4 and −5 at equal distances to the left of 0.

Question: Exercise 4.1, Question 2 (i): Tell whether the first number is greater or smaller than the second: +15, −6

Answer: On the number line +15 lies to the right of −6, so +15 is greater than −6. That is, +15 > −6.

Question: Exercise 4.1, Question 2 (ii): Tell whether the first number is greater or smaller than the second: −8, 0

Answer: On the number line −8 lies to the left of 0, so −8 is smaller than 0. That is, −8 < 0.

Question: Exercise 4.1, Question 2 (iii): Tell whether the first number is greater or smaller than the second: +16, 0

Answer: On the number line +16 lies to the right of 0, so +16 is greater than 0. That is, +16 > 0.

Question: Exercise 4.1, Question 2 (iv): Tell whether the first number is greater or smaller than the second: −2, −8

Answer: On the number line −2 lies to the right of −8, so −2 is greater than −8. That is, −2 > −8.

Question: Exercise 4.1, Question 2 (v): Tell whether the first number is greater or smaller than the second: +7, +9

Answer: On the number line +7 lies to the left of +9, so +7 is smaller than +9. That is, +7 < +9.

Question: Exercise 4.1, Question 2 (vi): Tell whether the first number is greater or smaller than the second: −4, −1

Answer: On the number line −4 lies to the left of −1, so −4 is smaller than −1. That is, −4 < −1.

Exercise 4.1, Question 3 (i): Fill in the blank by choosing the correct answer: −5 is on the _____ side of +6 (left, right)

Answer: left

Exercise 4.1, Question 3 (ii): Fill in the blank by choosing the correct answer: +6 is on the _____ side of −7 (left, right)

Answer: right

Exercise 4.1, Question 3 (iii): Fill in the blank by choosing the correct answer: 0 is on the _____ side of −15 (left, right)

Answer: right

Exercise 4.1, Question 3 (iv): Fill in the blank by choosing the correct answer: 0 is on the _____ side of 20 (left, right)

Answer: left

Exercise 4.1, Question 4 (i): Fill in the blank by choosing the correct answer: +10 _____ −20 (<, >)

Answer: is greater than −20. Completed: +10 > −20.

Exercise 4.1, Question 4 (ii): Fill in the blank by choosing the correct answer: −16 _____ −4 (<, >)

Answer: is smaller than −4. Completed: −16 < −4.

Exercise 4.1, Question 4 (iii): Fill in the blank by choosing the correct answer: +25 _____ −100 (<, >)

Answer: is greater than −100. Completed: +25 > −100.

Exercise 4.1, Question 4 (iv): Fill in the blank by choosing the correct answer: +30 _____ +50 (<, >)

Answer: is smaller than +50. Completed: +30 < +50.

Exercise 4.1, Question 4 (v): Fill in the blank by choosing the correct

answer: −17 _____ +17 (<, >)

Answer: is smaller than +17. Completed: −17 < +17.

Exercise 4.1, Question 4 (vi): Fill in the blank by choosing the correct answer: 0 _____ −5 (<, >)

Answer: is greater than −5. Completed: 0 > −5.

Question: Exercise 4.1, Question 5 (i): Write the following integers in ascending order: +5, −7, +1, 0, −3, −1

Answer: On the number line numbers increase as we go from left to right. Ascending order: −7, −3, −1, 0, +1, +5.

Question: Exercise 4.1, Question 5 (ii): Write the following integers in ascending order: −3, +4, 0, −1, +2, +5

Answer: On the number line numbers increase as we go from left to right. Ascending order: −3, −1, 0, +2, +4, +5.

Question: Exercise 4.1, Question 5 (iii): Write the following integers in ascending order: 0, −4, +4, −5, +5

Answer: On the number line numbers increase as we go from left to right. Ascending order: −5, −4, 0, +4, +5.

Question: Exercise 4.1, Question 5 (iv): Write the following integers in ascending order: −4, −1, −7, −2, −8

Answer: On the number line numbers increase as we go from left to right. Ascending order: −8, −7, −4, −2, −1.

Question: Exercise 4.1, Question 6 (i): Write the absolute values of the following integers: −5

Answer: The absolute value of a number equals its distance from 0, so |−5| = 5.

Question: Exercise 4.1, Question 6 (ii): Write the absolute values of the following integers: +20

Answer: The absolute value of a number equals its distance from 0, so |+20| = 20.

Question: Exercise 4.1, Question 6 (iii): Write the absolute values of the following integers: 0

Answer: The absolute value of a number equals its distance from 0, so |0| = 0.

Question: Exercise 4.1, Question 6 (iv): Write the absolute values of the following integers: −18

Answer: The absolute value of a number equals its distance from 0, so |−18| = 18.

Question: Exercise 4.1, Question 6 (v): Write the absolute values of the following integers: +50

Answer: The absolute value of a number equals its distance from 0, so |+50| = 50.

Question: Exercise 4.1, Question 7 (i): Write the absolute values of the following integers in ascending and descending order: −4, +1, −6, +3, 0, +5

Answer: First find the absolute values: |−4| = 4, |+1| = 1, |−6| = 6, |+3| = 3, |0| = 0, |+5| = 5. Ascending order: 0, +1, +3, −4, +5, −6. Descending order: −6, +5, −4, +3, +1, 0.

Question: Exercise 4.1, Question 7 (ii): Write the absolute values of the following integers in ascending and descending order: −25, 0, +17, −10, +30, −60

Answer: First find the absolute values: |−25| = 25, |0| = 0, |+17| = 17, |−10| = 10, |+30| = 30, |−60| = 60. Ascending order: 0, −10, +17, −25, +30, −60. Descending order: −60, +30, −25, +17, −10, 0.

Question: Exercise 4.1, Question 7 (iii): Write the absolute values of the following integers in ascending and descending order: −20, −10, +11, +7, 0, −4

Answer: First find the absolute values: |−20| = 20, |−10| = 10, |+11| = 11, |+7| = 7, |0| = 0, |−4| = 4. Ascending order: 0, −4, +7, −10, +11, −20. Descending order: −20, +11, −10, +7, −4, 0.

Question: Exercise 4.1, Question 7 (iv): Write the absolute values of the following integers in ascending and descending order: +8, −5, +13, −9, −12, +3

Answer: First find the absolute values: |+8| = 8, |−5| = 5, |+13| = 13, |−9| = 9, |−12| = 12, |+3| = 3. Ascending order: +3, −5, +8, −9, −12, +13. Descending order: +13, −12, −9, +8, −5, +3.

✓ Exercise 4.1, Question 8 (i): Consider the following statements and write true or false: A positive integer is always greater than a negative integer

✗ Exercise 4.1, Question 8 (ii): Consider the following statements and write true or false: Zero is a positive integer

✓ Exercise 4.1, Question 8 (iii): Consider the following statements and write true or false: Every negative integer is smaller than zero

Exercise 4.1, Question 8 (iv): Consider the following statements and write true or false: The absolute value of −5 is smaller than the

absolute value of +4

Exercise 4.1, Question 8 (v): Consider the following statements and write true or false: −5 is smaller than −10

Exercise 4.1, Question 8 (vi): Consider the following statements and write true or false: −25 is greater than −100

Exercise 4.1, Question 8 (vii): Consider the following statements and write true or false: The absolute value of any number can never be negative

Question: Exercise 4.2, Question 1 (i): Add the following integers: (+2) + (+7)

Answer: Both numbers have the same (positive) sign, so add their absolute values and attach the common sign: (+2) + (+7) = +(2 + 7) = +9.

Question: Exercise 4.2, Question 1 (ii): Add the following integers: (−5) + (−6)

Answer: Both numbers have the same (negative) sign, so add their absolute values and attach the common sign: (−5) + (−6) = −(5 + 6) = −11.

Question: Exercise 4.2, Question 1 (iii): Add the following integers: (+7) + (−4)

Answer: The two numbers have different signs, so subtract their absolute values and attach the sign of the number with the greater absolute value: |+7| = 7, |−4| = 4, difference = 7 − 4 = 3, and the number with the greater absolute value (+7) has the + sign. (+7) + (−4) = +(7 − 4) = +3.

Question: Exercise 4.2, Question 1 (iv): Add the following integers: (−8) + (+2)

Answer: The two numbers have different signs, so subtract their absolute values and attach the sign of the number with the greater absolute value: |−8| = 8, |+2| = 2, difference = 8 − 2 = 6, and the number with the greater absolute value (−8) has the − sign. (−8) + (+2) = −(8 − 2) = −6.

Question: Exercise 4.2, Question 1 (v): Add the following integers: (+5) + (−8)

Answer: The two numbers have different signs: |+5| = 5, |−8| = 8, difference = 8 − 5 = 3, and the number with the greater absolute value (−8) has the − sign. (+5) + (−8) = −(8 − 5) = −3.

Question: Exercise 4.2, Question 1 (vi): Add the following integers: (+9) + (−9)

Answer: The two numbers have different signs and equal absolute values: |+9| = 9, |−9| = 9, difference = 9 − 9 = 0. (+9) + (−9) = 0.

Question: Exercise 4.2, Question 2 (i): Add the following integers: (+5) + (+2) + (+3)

Answer: All three numbers are positive: (+5) + (+2) + (+3) = +(5 + 2 + 3) = +10.

Question: Exercise 4.2, Question 2 (ii): Add the following integers: (−5) + (−3) + (−4)

Answer: All three numbers are negative: (−5) + (−3) + (−4) = −(5 + 3 + 4) = −12.

Question: Exercise 4.2, Question 2 (iii): Add the following integers: (+4) + (+5) + (+1)

Answer: All three numbers are positive: (+4) + (+5) + (+1) = +(4 + 5 + 1) = +10.

Question: Exercise 4.2, Question 2 (iv): Add the following integers: (−2) + (−6) + (−5)

Answer: All three numbers are negative: (−2) + (−6) + (−5) = −(2 + 6 + 5) = −13.

Question: Exercise 4.2, Question 3 (i): Solve the following: (+10) + (+20)

Answer: Both numbers have the same (positive) sign: (+10) + (+20) = +(10 + 20) = +30.

Question: Exercise 4.2, Question 3 (ii): Solve the following: (−15) + (−25)

Answer: Both numbers have the same (negative) sign: (−15) + (−25) = −(15 + 25) = −40.

Question: Exercise 4.2, Question 3 (iii): Solve the following: (−20) + (−7)

Answer: Both numbers have the same (negative) sign: (−20) + (−7) = −(20 + 7) = −27.

Question: Exercise 4.2, Question 3 (iv): Solve the following: (+5) + (−14)

Answer: The signs are different: |+5| = 5, |−14| = 14, difference = 14 − 5 = 9, and the number with the greater absolute value (−14) has the − sign. (+5) + (−14) = −(14 − 5) = −9.

Question: Exercise 4.2, Question 3 (v): Solve the following: (−14) + (+8)

Answer: The signs are different: |−14| = 14, |+8| = 8, difference = 14 − 8 = 6, and the number with the greater absolute value (−14) has the − sign. (−14) + (+8) = −(14 − 8) = −6.

Question: Exercise 4.2, Question 3 (vi): Solve the following: (−30) + (+30)

Answer: The signs are different and the absolute values are equal: |−30| = 30, |+30| = 30, difference = 30 − 30 = 0. (−30) + (+30) = 0.

Question: Exercise 4.3, Question 1 (i): Find the difference in the following, using the number line: (+6) − (+4)

Answer: In subtraction, change the sign of the second number and add: (+6) − (+4) = (+6) + (−4) = +(6 − 4) = +2.

Question: Exercise 4.3, Question 1 (ii): Find the difference in the following, using the number line: (−8) − (−3)

Answer: In subtraction, change the sign of the second number and add: (−8) − (−3) = (−8) + (+3) = −(8 − 3) = −5.

Question: Exercise 4.3, Question 1 (iii): Find the difference in the following, using the number line: (−9) − (−3)

Answer: In subtraction, change the sign of the second number and add: (−9) − (−3) = (−9) + (+3) = −(9 − 3) = −6.

Question: Exercise 4.3, Question 1 (iv): Find the difference in the following, using the number line: (−9) − (+5)

Answer: In subtraction, change the sign of the second number and add: (−9) − (+5) = (−9) + (−5) = −(9 + 5) = −14.

Question: Exercise 4.3, Question 1 (v): Find the difference in the following, using the number line: (+9) − (−2)

Answer: In subtraction, change the sign of the second number and add: (+9) − (−2) = (+9) + (+2) = +(9 + 2) = +11.

Question: Exercise 4.3, Question 1 (vi): Find the difference in the following, using the number line: (−7) − (+4)

Answer: In subtraction, change the sign of the second number and add: (−7) − (+4) = (−7) + (−4) = −(7 + 4) = −11.

Question: Exercise 4.3, Question 2 (i): Subtract: Subtract +6 from −10

Answer: Subtracting +6 from −10 means (−10) − (+6). Change the sign of the second number and add: (−10) − (+6) = (−10) + (−6) = −(10 + 6) = −16.

Question: Exercise 4.3, Question 2 (ii): Subtract: Subtract −20 from −10

Answer: Subtracting −20 from −10 means (−10) − (−20). Change the sign of the second number and add: (−10) − (−20) = (−10) + (+20) = +(20 − 10) = +10.

Question: Exercise 4.3, Question 2 (iii): Subtract: Subtract −8 from −7

Answer: Subtracting −8 from −7 means (−7) − (−8). Change the sign of the second number and add: (−7) − (−8) = (−7) + (+8) = +(8 − 7) = +1.

Question: Exercise 4.3, Question 2 (iv): Subtract: Subtract −15 from −20

Answer: Subtracting −15 from −20 means (−20) − (−15). Change the sign of the second number and add: (−20) − (−15) = (−20) + (+15) = −(20 − 15) = −5.

Question: Exercise 4.3, Question 2 (v): Subtract: Subtract −14 from −6

Answer: Subtracting −14 from −6 means (−6) − (−14). Change the sign of the second number and add: (−6) − (−14) = (−6) + (+14) = +(14 − 6) = +8.

Question: Exercise 4.3, Question 2 (vi): Subtract: Subtract −6 from −9

Answer: Subtracting −6 from −9 means (−9) − (−6). Change the sign of the second number and add: (−9) − (−6) = (−9) + (+6) = −(9 − 6) = −3.

Question: Exercise 4.3, Question 3 (i): Solve the following: (+25) − (−15)

Answer: In subtraction, change the sign of the second number and add: (+25) − (−15) = (+25) + (+15) = +(25 + 15) = +40.

Question: Exercise 4.3, Question 3 (ii): Solve the following: (−30) − (−25)

Answer: In subtraction, change the sign of the second number and add: (−30) − (−25) = (−30) + (+25) = −(30 − 25) = −5.

Question: Exercise 4.3, Question 3 (iii): Solve the following: (−11) − (+5)

Answer: In subtraction, change the sign of the second number and add: (−11) − (+5) = (−11) + (−5) = −(11 + 5) = −16.

Question: Exercise 4.3, Question 3 (iv): Solve the following: (+40) − (+30)

Answer: In subtraction, change the sign of the second number and add: (+40) − (+30) = (+40) + (−30) = +(40 − 30) = +10.

Question: Exercise 4.3, Question 3 (v): Solve the following: (−16) − (−18)

Answer: In subtraction, change the sign of the second number and add: (−16) − (−18) = (−16) + (+18) = +(18 − 16) = +2.

Question: Exercise 4.3, Question 3 (vi): Solve the following: (+15) − (−20)

Answer: In subtraction, change the sign of the second number and add: (+15) − (−20) = (+15) + (+20) = +(15 + 20) = +35.

Question: Exercise 4.3, Question 3 (vii): Solve the following: (−29) − (+17)

Answer: In subtraction, change the sign of the second number and add: (−29) − (+17) = (−29) + (−17) = −(29 + 17) = −46.

Question: Exercise 4.3, Question 3 (viii): Solve the following: (−20) + (−30)

Answer: Both numbers have the same (negative) sign: (−20) + (−30) =

−(20 + 30) = −50.

Question: Exercise 4.3, Question 3 (ix): Solve the following: (−27) + (+17)

Answer: The signs are different: |−27| = 27, |+17| = 17, difference = 27 − 17 = 10, and the number with the greater absolute value (−27) has the − sign. (−27) + (+17) = −(27 − 17) = −10.

Question: Exercise 4.4, Question 1 (i): Multiply the following integers: (+15) x (−4)

Answer: The signs are different, so the product is negative: (+15) x (−4) = −(15 x 4) = −60.

Question: Exercise 4.4, Question 1 (ii): Multiply the following integers: (+20) x (+17)

Answer: The signs are the same (positive), so the product is positive: (+20) x (+17) = +(20 x 17) = +340.

Question: Exercise 4.4, Question 1 (iii): Multiply the following integers: (−16) x (−25)

Answer: The signs are the same (negative), so the product is positive: (−16) x (−25) = +(16 x 25) = +400.

Question: Exercise 4.4, Question 1 (iv): Multiply the following integers: (−36) x (+12)

Answer: The signs are different, so the product is negative: (−36) x (+12) = −(36 x 12) = −432.

Question: Exercise 4.4, Question 1 (v): Multiply the following integers: (+35) x (−14)

Answer: The signs are different, so the product is negative: (+35) x (−14) = −(35 x 14) = −490.

Question: Exercise 4.4, Question 1 (vi): Multiply the following integers: (−43) x (−16)

Answer: The signs are the same (negative), so the product is positive: (−43) x (−16) = +(43 x 16) = +688.

Question: Exercise 4.4, Question 2 (i): Find the quotient: (+25) ÷ (−5)

Answer: The signs are different, so the quotient is negative: (+25) ÷ (−5) = −(25 ÷ 5) = −5.

Question: Exercise 4.4, Question 2 (ii): Find the quotient: (−35) ÷ (−7)

Answer: The signs are the same (negative), so the quotient is positive: (−35) ÷ (−7) = +(35 ÷ 7) = +5.

Question: Exercise 4.4, Question 2 (iii): Find the quotient: (+100) ÷ (+4)

Answer: The signs are the same (positive), so the quotient is positive: (+100) ÷ (+4) = +(100 ÷ 4) = +25.

Question: Exercise 4.4, Question 2 (iv): Find the quotient: (−252) ÷ (+4)

Answer: The signs are different, so the quotient is negative: (−252) ÷ (+4) = −(252 ÷ 4) = −63.

Question: Exercise 4.4, Question 2 (v): Find the quotient: (−234) ÷ (+3)

Answer: The signs are different, so the quotient is negative: (−234) ÷ (+3) = −(234 ÷ 3) = −78.

Question: Exercise 4.4, Question 2 (vi): Find the quotient: (−126) ÷ (−6)

Answer: The signs are the same (negative), so the quotient is positive: (−126) ÷ (−6) = +(126 ÷ 6) = +21.

✗ Exercise 4.4, Question 3 (i): Consider the following operations and choose true or false: 0 ÷ 5 = 5

✓ Exercise 4.4, Question 3 (ii): Consider the following operations and choose true or false: 0 ÷ +6 = 0

✗ Exercise 4.4, Question 3 (iii): Consider the following operations and choose true or false: +9 x 0 = +9

✓ Exercise 4.4, Question 3 (iv): Consider the following operations and choose true or false: +6 x 0 = 0

✗ Exercise 4.4, Question 3 (v): Consider the following operations and choose true or false: (−10) ÷ (−10) = −1

✓ Exercise 4.4, Question 3 (vi): Consider the following operations and choose true or false: (−20) x (−1) = +20

✗ Exercise 4.4, Question 3 (vii): Consider the following operations and choose true or false: (+5) ÷ (−5) = 1

✓ Exercise 4.4, Question 3 (viii): Consider the following operations and choose true or false: (−18) ÷ (−3) = (+6)

Question: Review Exercise 4, Question 1: Write all the integers that come between the given integers −5 and +5

Answer: The integers between −5 and +5 are: −4, −3, −2, −1, 0, +1, +2, +3, +4.

Question: Review Exercise 4, Question 2 (i): Show the following integers on the number line: −5, −2, 0, +3, +5

Answer: Mark 0 in the middle of the number line. Mark the points +3 and +5 at equal distances to the right of 0, and mark the points −2 and −5 at equal distances to the left of 0.

Question: Review Exercise 4, Question 2 (ii): Show the following integers on the number line: −6, −3, −1, +1, +4, +6

Answer: Mark 0 in the middle of the number line. Mark the points +1, +4 and +6 at equal distances to the right of 0, and mark the points −1, −3 and −6 at equal distances to the left of 0.

Question: Review Exercise 4, Question 3 (i): Write the absolute value of each of the following integers: −3, +4, −4, +6, 0

Answer: |−3| = 3, |+4| = 4, |−4| = 4, |+6| = 6, |0| = 0.

Question: Review Exercise 4, Question 3 (ii): Write the absolute value of each of the following integers: +2, −7, +1, 0, +4, −2

Answer: |+2| = 2, |−7| = 7, |+1| = 1, |0| = 0, |+4| = 4, |−2| = 2.

Question: Review Exercise 4, Question 4 (i): Solve the addition of the following integers with the help of the number line: (+6) + (−9)

Answer: On the number line, walk 6 steps right from 0 to reach +6, then walk 9 steps left to reach −3. (+6) + (−9) = −3.

Question: Review Exercise 4, Question 4 (ii): Solve the addition of the following integers with the help of the number line: (−9) + (−6)

Answer: On the number line, walk 9 steps left from 0 to reach −9, then walk 6 more steps left to reach −15. (−9) + (−6) = −15.

Question: Review Exercise 4, Question 4 (iii): Solve the addition of the following integers with the help of the number line: (+9) + (+7) + (−5)

Answer: On the number line, walk 9 steps right from 0 to reach +9, then 7 more steps right to reach +16, then 5 steps left to reach +11. (+9) + (+7) + (−5) = +11.

Question: Review Exercise 4, Question 4 (iv): Solve the addition of the following integers with the help of the number line: (−1) + (−2) + (3)

Answer: On the number line, walk 1 step left from 0 to reach −1, then 2 more steps left to reach −3, then 3 steps right to reach 0. (−1) + (−2) + (3) = 0.

Question: Review Exercise 4, Question 5 (i): Solve the subtraction of the following integers with the help of the number line: (+5) − (+3)

Answer: In subtraction, change the sign of the second number and add: (+5) − (+3) = (+5) + (−3) = +(5 − 3) = +2.

Question: Review Exercise 4, Question 5 (ii): Solve the subtraction of the following integers with the help of the number line: (−7) − (−2)

Answer: In subtraction, change the sign of the second number and add: (−7) − (−2) = (−7) + (+2) = −(7 − 2) = −5.

Question: Review Exercise 4, Question 5 (iii): Solve the subtraction of the following integers with the help of the number line: (+4) − (−3)

Answer: In subtraction, change the sign of the second number and add: (+4) − (−3) = (+4) + (+3) = +(4 + 3) = +7.

Question: Review Exercise 4, Question 6 (i): Solve the addition and subtraction operations of the following integers: (+6) + (−10)

Answer: The signs are different: |+6| = 6, |−10| = 10, difference = 10 − 6 = 4, and the number with the greater absolute value (−10) has the − sign. (+6) + (−10) = −(10 − 6) = −4.

Question: Review Exercise 4, Question 6 (ii): Solve the addition and subtraction operations of the following integers: (−10) + (−4)

Answer: Both numbers have the same (negative) sign: (−10) + (−4) = −(10 + 4) = −14.

Question: Review Exercise 4, Question 6 (iii): Solve the addition and subtraction operations of the following integers: (−8) + (+5)

Answer: The signs are different: |−8| = 8, |+5| = 5, difference = 8 − 5 = 3, and the number with the greater absolute value (−8) has the − sign. (−8) + (+5) = −(8 − 5) = −3.

Question: Review Exercise 4, Question 6 (iv): Solve the addition and subtraction operations of the following integers: (+20) − (−10)

Answer: In subtraction, change the sign of the second number and add: (+20) − (−10) = (+20) + (+10) = +(20 + 10) = +30.

Question: Review Exercise 4, Question 6 (v): Solve the addition and subtraction operations of the following integers: (−20) − (−15)

Answer: In subtraction, change the sign of the second number and add: (−20) − (−15) = (−20) + (+15) = −(20 − 15) = −5.

Question: Review Exercise 4, Question 6 (vi): Solve the addition and subtraction operations of the following integers: (+40) − (+25)

Answer: In subtraction, change the sign of the second number and add: (+40) − (+25) = (+40) + (−25) = +(40 − 25) = +15.

Question: Review Exercise 4, Question 7 (i): Solve the multiplication operations of the following integers: (+10) x (−5)

Answer: The signs are different, so the product is negative: (+10) x (−5) = −(10 x 5) = −50.

Question: Review Exercise 4, Question 7 (ii): Solve the multiplication operations of the following integers: (−6) x (−9)

Answer: The signs are the same (negative), so the product is positive: (−6) x (−9) = +(6 x 9) = +54.

Question: Review Exercise 4, Question 7 (iii): Solve the multiplication operations of the following integers: (+20) x (+6)

Answer: The signs are the same (positive), so the product is positive: (+20) x (+6) = +(20 x 6) = +120.

Question: Review Exercise 4, Question 7 (iv): Solve the multiplication operations of the following integers: (−15) x (+3)

Answer: The signs are different, so the product is negative: (−15) x (+3) = −(15 x 3) = −45.

Question: Review Exercise 4, Question 7 (v): Solve the multiplication operations of the following integers: (+6) x 0

Answer: Multiplying any number by zero gives zero: (+6) x 0 = 0.

Question: Review Exercise 4, Question 7 (vi): Solve the multiplication operations of the following integers: (−10) x (−8)

Answer: The signs are the same (negative), so the product is positive: (−10) x (−8) = +(10 x 8) = +80.

Question: Review Exercise 4, Question 8 (i): Solve the division operations of the following integers: (+20) ÷ (−4)

Answer: The signs are different, so the quotient is negative: (+20) ÷ (−4) = −(20 ÷ 4) = −5.

Question: Review Exercise 4, Question 8 (ii): Solve the division operations of the following integers: (−15) ÷ (−3)

Answer: The signs are the same (negative), so the quotient is positive: (−15) ÷ (−3) = +(15 ÷ 3) = +5.

Question: Review Exercise 4, Question 8 (iii): Solve the division operations of the following integers: (−36) ÷ (+6)

Answer: The signs are different, so the quotient is negative: (−36) ÷ (+6) = −(36 ÷ 6) = −6.

Question: Review Exercise 4, Question 8 (iv): Solve the division operations of the following integers: (+40) ÷ (+10)

Answer: The signs are the same (positive), so the quotient is positive: (+40) ÷ (+10) = +(40 ÷ 10) = +4.

Question: Practical Work (i): If -10 metres show a distance of 10 metres towards the west, then a distance of 10 metres towards the east will be shown by +10 metres.

Answer: This is the book's worked example: the opposite direction of west is east, so the opposite sign is used: the opposite of -10 is +10. So a distance of 10 metres towards the east is shown by +10 metres.

Question: Practical Work (ii): If +6 million show an increase of 6 million in income, then a decrease in income will be shown by ___.

Answer: The opposite of an increase in income is a decrease, so the opposite sign is used: the opposite of +6 is -6. So the blank takes -6 million: a decrease in income will be shown by -6 million.

Question: Practical Work (iii): If +20 metres show a distance of 20 metres above sea level, then a distance below sea level will be shown by ___.

Answer: The opposite of above sea level is below sea level, so the opposite sign is used: the opposite of +20 is -20. So the blank takes -20 metres: a distance below sea level will be shown by -20 metres.

Question: Practical Work (iv): If -20 rupees show a loss of 20 rupees in some business, then a profit in the business will be shown by ___.

Answer: The opposite of a loss is a profit, so the opposite sign is used: the opposite of -20 is +20. So the blank takes +20 rupees: a profit in the business will be shown by +20 rupees.

Question: Practical Work (v): If -50 metres show a distance of 50 metres towards the south, then a distance of 60 metres towards the north will be shown by ___.

Answer: The opposite direction of south is north, so the opposite sign is used: the opposite of -50 is +50, but here the distance is 60 metres, so the blank takes +60 metres: a distance of 60 metres towards the north will be shown by +60 metres.

Question: Practical Work 1 (i): -2 is on the left side of +3. (left, right)

Answer: This is the book's worked example: on the number line, -2 lies to the left of +3, so the answer is the left side.

Question: Practical Work 1 (ii): +5 is on the ___ side of -10. (left, right)

Answer: On the number line, +5 lies to the right of -10, because +5 is greater than -10 (+5 > -10). So the blank takes "right": +5 is on the right side of -10.

Question: Practical Work 1 (iii): 0 is on the ___ side of -6. (left, right)

Answer: On the number line, 0 lies to the right of -6, because 0 is greater than -6 (0 > -6). So the blank takes "right": 0 is on the right side of -6.

Question: Practical Work 2 (i): +5 > -5 (<, >)

Answer: This is the book's worked example: a positive number is always greater than a negative number, so +5 > -5.

Question: Practical Work 2 (ii): +7 ___ +11 (<, >)

Answer: +7 is less than +11, because on the number line +7 lies to the left of +11. So the blank takes <: +7 < +11.

Question: Practical Work 2 (iii): -9 ___ -4 (<, >)

Answer: -9 is less than -4, because on the number line -9 lies to the left of -4. So the blank takes <: -9 < -4.

Question: Practical Work 2 (iv): +2 ___ -20 (<, >)

Answer: A positive number is always greater than a negative number. So the blank takes >: +2 > -20.

5. Simplest Form

Question: Exercise 5.1, Question 1: Simplify: 1/5 − {2/5 ÷ 4/15 x 1/2 − (1/2 + 2/5)}

Answer: By BODMAS, first work inside the brackets. (1/2 + 2/5) = 9/10. Then 2/5 ÷ 4/15 = 2/5 x 15/4 = 3/2, and 3/2 x 1/2 = 3/4. Now inside the braces: 3/4 − 9/10 = 15/20 − 18/20 = −3/20. Then 1/5 − (−3/20) = 4/20 + 3/20 = 7/20.

Question: Exercise 5.1, Question 2: Simplify: 3/4 + (4/3 − (1/2 + 2/3) [vinculum over 1/2 + 2/3] + 4/5)

Answer: First solve the vinculum part: 1/2 + 2/3 = 3/6 + 4/6 = 7/6. Then inside the brackets: 4/3 − 7/6 + 4/5 = 8/6 − 7/6 + 4/5 = 1/6 + 4/5 = 5/30 + 24/30 = 29/30. Then 3/4 + 29/30 = 45/60 + 58/60 = 103/60 = 1 43/60.

Question: Exercise 5.1, Question 3: Simplify: 7/8 + {5/7 + (1/2 + 1/3) ÷ 5/6}

Answer: First inside the small brackets: (1/2 + 1/3) = 5/6. Then 5/6 ÷ 5/6 = 1. Inside the braces: 5/7 + 1 = 12/7. Then 7/8 + 12/7 = 49/56 + 96/56 = 145/56 = 2 33/56.

Question: Exercise 5.1, Question 4: Simplify: 5/2 x (5/6 − 2/3) ÷ 2/5 x (3/5 + 6/25)

Answer: First inside the brackets: (5/6 − 2/3) = 5/6 − 4/6 = 1/6, and (3/5 + 6/25) = 15/25 + 6/25 = 21/25. Then multiply and divide from left to right: 5/2 x 1/6 = 5/12, 5/12 ÷ 2/5 = 5/12 x 5/2 = 25/24, 25/24 x 21/25 = 21/24 = 7/8.

Question: Exercise 5.1, Question 5: Simplify: (2/3 − (3/4 x 1/8) [vinculum]) ÷ (2/3 − (3/4 + 5/8) [vinculum])

Answer: First solve the vinculum parts: 3/4 x 1/8 = 3/32, and 3/4 + 5/8 = 6/8 + 5/8 = 11/8. Then the first bracket: 2/3 − 3/32 = 64/96 − 9/96 = 55/96. The second bracket: 2/3 − 11/8 = 16/24 − 33/24 = −17/24. Then divide: 55/96 ÷ (−17/24) = 55/96 x (−24/17) = −55/68.

Question: Exercise 5.1, Question 6: Simplify: 1 1/2 + [1/3 ÷ {2/5 ÷ (4/5 − (1/10 + 3/5) [vinculum])}]

Answer: First the vinculum part: 1/10 + 3/5 = 1/10 + 6/10 = 7/10. Then the small bracket: 4/5 − 7/10 = 8/10 − 7/10 = 1/10. Then the braces: 2/5 ÷ 1/10 = 2/5 x 10 = 4. Then the square brackets: 1/3 ÷ 4 = 1/12. Then 1 1/2 + 1/12 = 3/2 + 1/12 = 18/12 + 1/12 = 19/12 = 1 7/12.

Question: Exercise 5.1, Question 7: Simplify: [1/2 + {(3/4 ÷ 9/16) x 1 1/2} − 2 1/4]

Answer: First the small brackets: 3/4 ÷ 9/16 = 3/4 x 16/9 = 4/3. Then 4/3 x 1 1/2 = 4/3 x 3/2 = 2. Then inside the square brackets: 1/2 + 2 − 2 1/4 = 5/2 − 9/4 = 10/4 − 9/4 = 1/4.

Question: Exercise 5.1, Question 8: Simplify: 3 1/2 + {(10 2/5 − 5 1/3) ÷ 3 2/3} − 1 1/5

Answer: First the small brackets: 10 2/5 − 5 1/3 = 52/5 − 16/3 = 156/15 − 80/15 = 76/15. Then 76/15 ÷ 3 2/3 = 76/15 ÷ 11/3 = 76/15 x 3/11 = 76/55. Then 3 1/2 + 76/55 − 1 1/5 = 7/2 + 76/55 − 6/5 = 385/110 + 152/110 − 132/110 = 405/110 = 81/22 = 3 15/22.

Question: Exercise 5.1, Question 9: Simplify: [(2 3/4 + 1 2/6) x (1 1/5 ÷ 2/5)]

Answer: First inside the brackets: 2 3/4 + 1 2/6 = 11/4 + 8/6 = 11/4 + 4/3 = 33/12 + 16/12 = 49/12. Second bracket: 1 1/5 ÷ 2/5 = 6/5 x 5/2 = 3. Then 49/12 x 3 = 49/4 = 12 1/4.

Question: Exercise 5.1, Question 10: Simplify: {(1/2 − 1/3) + (1 2/3 ÷ 1/9)} − {(5/3 − 1/6) ÷ 9/6}

Answer: First large braces: (1/2 − 1/3) = 1/6, and (1 2/3 ÷ 1/9) = 5/3 x 9 = 15, sum = 1/6 + 15 = 91/6. Second large braces: (5/3 − 1/6) = 9/6, and 9/6 ÷ 9/6 = 1. Then 91/6 − 1 = 85/6 = 14 1/6.

Question: Exercise 5.1, Question 11: Simplify: [2 1/4 x {3 1/3 − 1/2 + (5/3 − 1/6)}] + 5/6

Answer: First the small brackets: (5/3 − 1/6) = 10/6 − 1/6 = 9/6 = 3/2. Then the braces: 3 1/3 − 1/2 + 3/2 = 10/3 + 1 = 13/3. Then the square brackets: 2 1/4 x 13/3 = 9/4 x 13/3 = 39/4. Then 39/4 + 5/6 = 117/12 + 10/12 = 127/12 = 10 7/12.

Question: Exercise 5.1, Question 12: Simplify: (3 1/2 x 5 1/3 − 10 2/5) x (3 1/8 − 6 1/2 ÷ 2 1/6)

Answer: First bracket: 3 1/2 x 5 1/3 = 7/2 x 16/3 = 56/3, then 56/3 − 10 2/5 = 56/3 − 52/5 = 280/15 − 156/15 = 124/15. Second bracket: 6 1/2 ÷ 2 1/6 = 13/2 ÷ 13/6 = 13/2 x 6/13 = 3, then 3 1/8 − 3 = 25/8 − 3 = 1/8. Then 124/15 x 1/8 = 124/120 = 31/30 = 1 1/30.

Question: Exercise 5.1, Question 13: Simplify: (2 1/5 ÷ 3 2/25) x 4 2/5 ÷ (3 − 1/5)

Answer: First inside the brackets: (2 1/5 ÷ 3 2/25) = 11/5 ÷ 77/25 = 11/5 x 25/77 = 5/7, and (3 − 1/5) = 14/5. Then from left to right: 5/7 x 4 2/5 = 5/7 x 22/5 = 22/7, then 22/7 ÷ 14/5 = 22/7 x 5/14 = 110/98 = 55/49 = 1 6/49.

Question: Exercise 5.1, Question 14: Simplify: 4.02 x {1.2 x (2.11 + 3.89) [vinculum]}

Answer: First the vinculum part: 2.11 + 3.89 = 6.00. Then the braces: 1.2 x 6.00 = 7.2. Then 4.02 x 7.2 = 28.944.

Question: Exercise 5.1, Question 15: Simplify: 4.093 + {5.2 x (1.5 + 2.4) [vinculum]}

Answer: First the vinculum part: 1.5 + 2.4 = 3.9. Then the braces: 5.2 x 3.9 = 20.28. Then 4.093 + 20.28 = 24.373.

Question: Exercise 5.1, Question 16: Simplify: 5.7 x {6.7 + (1.3 x 7.2 ÷ (1.5 x 2.4) [vinculum])}

Answer: First the vinculum part: 1.5 x 2.4 = 3.6. Then the small brackets: 1.3 x 7.2 = 9.36, and 9.36 ÷ 3.6 = 2.6. Then the braces: 6.7 + 2.6 = 9.3. Then 5.7 x 9.3 = 53.01.

Question: Exercise 5.1, Question 17: Simplify: 0.3 x 2.6 + {3.7 − (1.2 x (4.5 ÷ 1.5) [vinculum])}

Answer: First the vinculum part: 4.5 ÷ 1.5 = 3. Then the small brackets: 1.2 x 3 = 3.6. Then the braces: 3.7 − 3.6 = 0.1. Then 0.3 x 2.6 = 0.78, and 0.78 + 0.1 = 0.88.

Question: Exercise 5.1, Question 18: Simplify: {(3.41 − (1.05 + 1.5) [vinculum]) x 3.5} ÷ 1.25

Answer: First the vinculum part: 1.05 + 1.5 = 2.55. Then the small brackets: 3.41 − 2.55 = 0.86. Then the braces: 0.86 x 3.5 = 3.01. Then 3.01 ÷ 1.25 = 2.408.

Question: Exercise 5.1, Question 19: Simplify: 30.45 + [6.35 x {8.95 − (3.5 ÷ 0.7 x 0.01)}]

Answer: First the small brackets: 3.5 ÷ 0.7 = 5, and 5 x 0.01 = 0.05. Then the braces: 8.95 − 0.05 = 8.90. Then the square brackets: 6.35 x 8.90 = 56.515. Then 30.45 + 56.515 = 86.965.

Question: Exercise 5.1, Question 20: Simplify: 2.297 + [0.2 x {9 ÷ 0.3} − 0.09]

Answer: First the braces: 9 ÷ 0.3 = 30, and 0.2 x 30 = 6. Then inside the square brackets: 6 − 0.09 = 5.91. Then 2.297 + 5.91 = 8.207.

Question: Exercise 5.1, Question 21: Simplify: {(2.5 − 2) + 1.5} x 0.5 − 0.05

Answer: First the small brackets: (2.5 − 2) = 0.5. Then the braces: 0.5 + 1.5 = 2. Then 2 x 0.5 = 1, and 1 − 0.05 = 0.95.

Question: Exercise 5.1, Question 22: Simplify: 2.04 + [2.82 ÷ {2.4 − (1.3 x (0.3 + 0.6) [vinculum])}]

Answer: First the vinculum part: 0.3 + 0.6 = 0.9. Then the small brackets: 1.3 x 0.9 = 1.17. Then the braces: 2.4 − 1.17 = 1.23. Then the square brackets: 2.82 ÷ 1.23 = 94/41 ≈ 2.2927. Then 2.04 + 94/41 = 4441/1025 ≈ 4.3327. (With the book's given numbers the answer is a non-terminating decimal.)

Question: Exercise 5.1, Question 23: Simplify: [{7.45 − 1.35 + 1.5} x 1.25

÷ 2.5]

Answer: First the braces: 7.45 − 1.35 + 1.5 = 7.60. Then inside the square brackets from left to right: 7.60 x 1.25 = 9.5, and 9.5 ÷ 2.5 = 3.8.

Question: Exercise 5.2, Question 1: There are 640 students in a school. 3/10 of the students use their own vehicles to come to school. 5/8 of the students come in public transport. The remaining students walk to school. Tell how many students walk?

Answer: Given data: total students = 640, own vehicles = 3/10 part, public transport = 5/8 part. Fraction walking = 1 − (3/10 + 5/8) = 1 − (12/40 + 25/40) = 1 − 37/40 = 3/40. Students walking = 3/40 x 640 = 48. Answer: 48 students walk.

Question: Exercise 5.2, Question 2: A crate has 140 oranges in all. Of these, 3/7 of the oranges were sold. 1/14 of the oranges went bad. Tell how many oranges are left in the crate?

Answer: Given data: total oranges = 140, sold = 3/7 part, bad = 1/14 part. Remaining fraction = 1 − (3/7 + 1/14) = 1 − (6/14 + 1/14) = 1 − 7/14 = 1/2. Oranges left = 1/2 x 140 = 70. Answer: 70 oranges are left.

Question: Exercise 5.2, Question 3: A large crowd gathered at the stadium to watch a cricket match. 1/8 of the crowd came by train. 2/7 came by ordinary bus. 1/4 came in their own vehicles. The rest came on foot. Tell what fraction of the people came on foot?

Answer: Given data: by train = 1/8 part, by bus = 2/7 part, own vehicles = 1/4 part. Fraction on foot = 1 − (1/8 + 2/7 + 1/4) = 1 − (7/56 + 16/56 + 14/56) = 1 − 37/56 = 19/56. Answer: 19/56 of the people came on foot.

Question: Exercise 5.2, Question 4: Maria's total property is Rs. 58000. She gave 16/29 of her property to her mother. After that she deposited 1/4 of the remaining property in the Zakat fund. Tell how much money is left with Maria?

Answer: Given data: total property = Rs. 58000. Given to mother = 16/29 x 58000 = 16 x 2000 = Rs. 32000. Remaining property = 58000 − 32000 = Rs. 26000. Zakat fund = 1/4 x 26000 = Rs. 6500. Left with Maria = 26000 − 6500 = Rs. 19500. Answer: Rs. 19500 is left.

Question: Exercise 5.2, Question 5: Amir planted 425 tomato plants in his field. After a few days 25 plants dried up. Tell what fraction of the plants dried up?

Answer: Given data: total plants = 425, dried plants = 25. Fraction dried = 25/425 = 1/17. Answer: 1/17 of the plants dried up.

Question: Exercise 5.2, Question 6: Ijaz gave 2/5 of his property to his son, and 1/7 to his son. He gave the remaining amount to his wife. Tell what fraction he gave to his wife?

Answer: Given data (the book prints "son" in both places): to son = 2/5 part, to son = 1/7 part. Share given to wife = 1 − (2/5 + 1/7) = 1 − (14/35 + 5/35) = 1 − 19/35 = 16/35. Answer: He gave 16/35 to his wife.

Question: Exercise 5.2, Question 7: Adil had 26.50 metres of cloth. He used 5.80 metres of his cloth to make his suit. He gave 11.50 metres of his cloth to his brother. Tell how much cloth is left with Adil?

Answer: Given data: total cloth = 26.50 m, for suit = 5.80 m, given to brother = 11.50 m. Cloth left = 26.50 − 5.80 − 11.50 = 20.70 − 11.50 = 9.20 m. Answer: 9.20 m of cloth is left.

Question: Exercise 5.2, Question 8: Abish bought 5 packets of cooking oil at Rs. 121.80 per packet. She bought 3 bags of rice at Rs. 2350.50 per bag. She bought 40 kg of flour at Rs. 42.70 per kg. Tell how much total money she spent?

Answer: Given data: oil = 5 x 121.80 = Rs. 609.00, rice = 3 x 2350.50 = Rs. 7051.50, flour = 40 x 42.70 = Rs. 1708.00. Total spent = 609.00 + 7051.50 + 1708.00 = Rs. 9368.50. Answer: She spent Rs. 9368.50.

Question: Exercise 5.2, Question 9: Umair spent 0.06 of his pocket money on Monday, 0.25 on Tuesday and 0.50 on Wednesday. He spent the remaining amount on Thursday. Tell what fraction he spent on Thursday?

Answer: Given data: Monday = 0.06 part, Tuesday = 0.25 part, Wednesday = 0.50 part. Fraction spent on Thursday = 1 − (0.06 + 0.25 + 0.50) = 1 − 0.81 = 0.19. Answer: He spent 0.19 on Thursday.

Question: Exercise 5.2, Question 10 (i): Sobia bought the following things for her home. Tell how much total Sobia spent? 2 electric bulbs, Rs. 80.95 per bulb

Answer: Given data: 2 bulbs, Rs. 80.95 per bulb. Cost = 2 x 80.95 = Rs. 161.90.

Question: Exercise 5.2, Question 10 (ii): Sobia bought the following things for her home. Tell how much total Sobia spent? 4 small plastic stools, Rs. 105.50 per stool

Answer: Given data: 4 stools, Rs. 105.50 per stool. Cost = 4 x 105.50 = Rs. 422.00.

Question: Exercise 5.2, Question 10 (iii): Sobia bought the following things for her home. Tell how much total Sobia spent? 3 plastic tables, Rs. 530.95 per table

Answer: Given data: 3 tables, Rs. 530.95 per table. Cost = 3 x 530.95 = Rs. 1592.85.

Question: Exercise 5.2, Question 10 (iv): Sobia bought the following things for her home. Tell how much total Sobia spent? 6 plastic chairs, Rs. 458.30 per chair

Answer: Given data: 6 chairs, Rs. 458.30 per chair. Cost = 6 x 458.30 = Rs. 2749.80. Total cost of all items = 161.90 + 422.00 + 1592.85 + 2749.80 = Rs. 4926.55. Answer: Sobia spent Rs. 4926.55 in total.

Question: Review Exercise 5, Question 1 (i): Simplify: 1 2/5 ÷ {2 3/5 − (1 1/4 − (2/3 x 1 1/7) [vinculum])}

Answer: First the vinculum part: 2/3 x 1 1/7 = 2/3 x 8/7 = 16/21. Then the small brackets: 1 1/4 − 16/21 = 5/4 − 16/21 = 105/84 − 64/84 = 41/84. Then the braces: 2 3/5 − 41/84 = 13/5 − 41/84 = 1092/420 − 205/420 = 887/420. Then 1 2/5 ÷ 887/420 = 7/5 x 420/887 = 588/887.

Question: Review Exercise 5, Question 1 (ii): Simplify: 2 1/4 ÷ [2/3 ÷ {1/3 − ((1/2 − 1/5) [vinculum] ÷ 1 1/2)}]

Answer: First the vinculum part: 1/2 − 1/5 = 3/10. Then the small brackets: 3/10 ÷ 1 1/2 = 3/10 ÷ 3/2 = 3/10 x 2/3 = 1/5. Then the braces: 1/3 − 1/5 = 5/15 − 3/15 = 2/15. Then the square brackets: 2/3 ÷ 2/15 = 2/3 x 15/2 = 5. Then 2 1/4 ÷ 5 = 9/4 ÷ 5 = 9/20.

Question: Review Exercise 5, Question 1 (iii): Simplify: 4.3 x [3.9 x 0.8 − 1.3 + {4.8 ÷ 1.2 − ((1.3 − 1.2 x 2.5) [vinculum])}]

Answer: First the vinculum part: 1.2 x 2.5 = 3.0, and 1.3 − 3.0 = −1.7. Then the braces: 4.8 ÷ 1.2 = 4, and 4 − (−1.7) = 5.7. Then inside the square brackets: 3.9 x 0.8 = 3.12, and 3.12 − 1.3 + 5.7 = 7.52. Then 4.3 x 7.52 = 32.336.

Question: Review Exercise 5, Question 1 (iv): Simplify: 2.9 − [1.05 x 1.17 + (1.04 − (2.5 + 1.9) [vinculum])]

Answer: First the vinculum part: 2.5 + 1.9 = 4.4. Then the small brackets: 1.04 − 4.4 = −3.36. Then inside the square brackets: 1.05 x 1.17 = 1.2285, and 1.2285 + (−3.36) = −2.1315. Then 2.9 − (−2.1315) = 5.0315.

Question: Review Exercise 5, Question 2 (i): Solve the following question: Bilal sacrificed a goat. From it he got 13 5/7 kg of meat. He gave 1/3 of the meat in charity. Tell how much meat is left with him?

Answer: Given data: total meat = 13 5/7 kg, given in charity = 1/3 part. Remaining fraction = 1 − 1/3 = 2/3. Meat left = 2/3 x 13 5/7 = 2/3 x 96/7 = 192/21 = 64/7 = 9 1/7 kg. Answer: 9 1/7 kg of meat is left.

Question: Review Exercise 5, Question 2 (ii): Solve the following question: Azhar solved 15 of 25 mathematics questions. Tell what fraction of questions he solved and what fraction of questions remain to be solved?

Answer: Given data: total questions = 25, solved = 15. Fraction solved = 15/25 = 3/5. Fraction remaining = (25 − 15)/25 = 10/25 = 2/5. Answer: He solved 3/5 of the questions and 2/5 remain to be solved.

Question: Review Exercise 5, Question 3: Farheen read 2/5 of a book in one day. She read the remaining part of the book on the second day. Tell what fraction of the book she read on the second day? If the book has 150 pages in all, tell how many pages she read on the first day and then on the second day?

Answer: Given data: first day = 2/5 part, total pages = 150. Fraction read on second day = 1 − 2/5 = 3/5. Pages read on first day = 2/5 x 150 = 60 pages. Pages read on second day = 3/5 x 150 = 90 pages. Answer: On the second day she read 3/5; she read 60 pages on the first day and 90 pages on the second day.

Question: Review Exercise 5, Question 4 (i): A contractor bought different kinds of tiles for the roof of a hut at the following rates. Tell how much total the contractor spent on the tiles? Local tiles 240.34 square metres, price Rs. 327.34 per square metre

Answer: Given data: local tiles = 240.34 sq m, price = Rs. 327.34 per sq m. Cost of local tiles = 240.34 x 327.34 = 78672.8956 ≈ Rs. 78672.90.

Question: Review Exercise 5, Question 4 (ii): A contractor bought different kinds of tiles for the roof of a hut at the following rates. Tell how much total the contractor spent on the tiles? Imported tiles 105.28 square metres, price Rs. 539.99 per square metre

Answer: Given data: imported tiles = 105.28 sq m, price = Rs. 539.99 per sq m. Cost of imported tiles = 105.28 x 539.99 = 56850.1472 ≈ Rs. 56850.15. Total cost of both kinds = 78672.90 + 56850.15 = 135523.05 (exact sum 135523.0428 ≈ Rs. 135523.04). Answer: Total spent approx. Rs. 135523.04.

✗ Review Exercise 5, Question 5 (i): Write (S) for true and (G) for false in the following: The vinculum is the [ ] sign

✗ Review Exercise 5, Question 5 (ii): Write (S) for true and (G) for false in the following: To simplify, if there is a minus sign outside the brackets then the signs inside the brackets are not changed when the brackets are removed

✓ Review Exercise 5, Question 5 (iii): Write (S) for true and (G) for false in the following: To simplify the expression 35 + 2 x 8, first we will multiply 2 by 8 and then add 35 to it

Review Exercise 5, Question 6 (i): Fill in the blank in the following: In BODMAS, M stands for _____

Answer: Multiplication. Completed sentence: In BODMAS, M stands for Multiplication.

Review Exercise 5, Question 6 (ii): Fill in the blank in the following: The name of the brackets { } is _____

Answer: Braces. Completed sentence: The name of the brackets { } is Braces.

Review Exercise 5, Question 6 (iii): Fill in the blank in the following: To simplify brackets, the order of removing the brackets is: vinculum, _____, _____, _____

Answer: ( ), { }, [ ]. That, is, first the vinculum, then the parentheses,, then the braces, then the square brackets.

6. Ratio & Proportion

Question: Exercise 6.1, Question 1 (i): Bring the following ratio to its simplest form: 4 : 50

Answer: Divide both terms by their common factor: 4 : 50 = (4 ÷ 2) : (50 ÷ 2) = 2 : 25.

Question: Exercise 6.1, Question 1 (ii): Bring the following ratio to its simplest form: 0.8 : 72

Answer: First remove the decimal by multiplying both terms by 10: 0.8 : 72 = 8 : 720. Then divide: 8 : 720 = (8 ÷ 8) : (720 ÷ 8) = 1 : 90.

Question: Exercise 6.1, Question 1 (iii): Bring the following ratio to its simplest form: 3.5 : 4.9

Answer: First remove the decimals by multiplying both terms by 10: 3.5 : 4.9 = 35 : 49. Then divide: 35 : 49 = (35 ÷ 7) : (49 ÷ 7) = 5 : 7.

Question: Exercise 6.1, Question 1 (iv): Bring the following ratio to its simplest form: 13/60 : 7/15

Answer: Give both fractions the same denominator: 7/15 = 28/60. So 13/60 : 7/15 = 13/60 : 28/60 = 13 : 28.

Question: Exercise 6.1, Question 1 (v): Bring the following ratio to its simplest form: 5/6 : 3/10

Answer: Give both fractions the same denominator: 5/6 = 25/30, 3/10 = 9/30. So 5/6 : 3/10 = 25/30 : 9/30 = 25 : 9.

Question: Exercise 6.1, Question 1 (vi): Bring the following ratio to its simplest form: 2/3 : 5

Answer: Write the second term as a fraction: 2/3 : 5 = 2/3 : 15/3 = 2 : 15.

Question: Exercise 6.1, Question 1 (vii): Bring the following ratio to its simplest form: 2 1/2 : 4

Answer: Convert the mixed fraction to an improper fraction: 2 1/2 = 5/2. So 2 1/2 : 4 = 5/2 : 4 = 5 : 8 (multiplying both terms by 2).

Question: Exercise 6.1, Question 1 (viii): Bring the following ratio to its simplest form: 1/3 : 1/6 : 1/9

Answer: Give all three fractions the same denominator (multiply by 18): 1/3 : 1/6 : 1/9 = 6/18 : 3/18 : 2/18 = 6 : 3 : 2.

Question: Exercise 6.1, Question 1 (ix): Bring the following ratio to its simplest form: 1.5 : 5 : 5.8

Answer: First remove the decimals by multiplying all three terms by 10: 1.5 : 5 : 5.8 = 15 : 50 : 58. 15, 50 and 58 have no common factor greater than 1, so the simplest form is 15 : 50 : 58.

Question: Exercise 6.1, Question 1 (x): Bring the following ratio to its simplest form: 3 2/5 : 0.6 : 3.5

Answer: 3 2/5 = 17/5, 0.6 = 3/5, 3.5 = 7/2. Multiply all three terms by 10: 17/5 : 3/5 : 7/2 = 34 : 6 : 35. 34, 6 and 35 have no common factor greater than 1, so the simplest form is 34 : 6 : 35.

Question: Exercise 6.1, Question 1 (xi): Bring the following ratio to its simplest form: 1/5 : 1/10 : 1/15

Answer: Give all three fractions the same denominator (multiply by 30): 1/5 : 1/10 : 1/15 = 6/30 : 3/30 : 2/30 = 6 : 3 : 2.

Question: Exercise 6.1, Question 1 (xii): Bring the following ratio to its simplest form: 1/7 : 1/14 : 1/21

Answer: Give all three fractions the same denominator (multiply by 42): 1/7 : 1/14 : 1/21 = 6/42 : 3/42 : 2/42 = 6 : 3 : 2.

Question: Exercise 6.1, Question 2 (i): Write the ratio between the following amounts and bring it to simplest form: Rs. 150 and Rs. 180

Answer: Before writing a ratio both quantities must have the same unit; both are in rupees. Ratio = 150 : 180 = (150 ÷ 30) : (180 ÷ 30) = 5 : 6.

Question: Exercise 6.1, Question 2 (ii): Write the ratio between the following amounts and bring it to simplest form: 250 cm and 1 metre

Answer: First convert both quantities to the same unit: 1 metre = 100 cm. Ratio = 250 : 100 = (250 ÷ 50) : (100 ÷ 50) = 5 : 2.

Question: Exercise 6.1, Question 2 (iii): Write the ratio between the following amounts and bring it to simplest form: 700 grams and 2 kilograms

Answer: First convert both quantities to the same unit: 1 kg = 1000 g, so 2 kg = 2000 g. Ratio = 700 : 2000 = (700 ÷ 100) : (2000 ÷ 100) = 7 : 20.

Question: Exercise 6.1, Question 2 (iv): Write the ratio between the following amounts and bring it to simplest form: 3 hours and 210 minutes

Answer: First convert both quantities to the same unit: 1 hour = 60 minutes, so 3 hours = 180 minutes. Ratio = 180 : 210 = (180 ÷ 30) : (210 ÷ 30) = 6 : 7.

Question: Exercise 6.1, Question 2 (v): Write the ratio between the following amounts and bring it to simplest form: 5 years and 3 months

Answer: First convert both quantities to the same unit: 1 year = 12 months, so 5 years = 60 months. Ratio = 60 : 3 = (60 ÷ 3) : (3 ÷ 3) = 20 : 1.

Question: Exercise 6.1, Question 2 (vi): Write the ratio between the following amounts and bring it to simplest form: 11 days, two weeks and 1 month

Answer: First convert all three quantities to the same unit (days): two weeks = 2 x 7 = 14 days, 1 month = 30 days. Ratio = 11 : 14 : 30. 11, 14 and 30 have no common factor greater than 1, so the simplest form is 11 : 14 : 30.

Question: Exercise 6.1, Question 3 (i): Convert the following fraction into a ratio: 2/9

Answer: To convert a fraction into a ratio, write the numerator as the first term and the denominator as the second term: 2/9 = 2 : 9.

Question: Exercise 6.1, Question 3 (ii): Convert the following fraction into a ratio: 5/6

Answer: To convert a fraction into a ratio, write the numerator as the first term and the denominator as the second term: 5/6 = 5 : 6.

Question: Exercise 6.1, Question 3 (iii): Convert the following fraction into a ratio: 1/75

Answer: To convert a fraction into a ratio, write the numerator as the first term and the denominator as the second term: 1/75 = 1 : 75.

Question: Exercise 6.1, Question 3 (iv): Convert the following fraction into a ratio: 3

Answer: Write the whole number as a fraction: 3 = 3/1, then write the numerator as the first term and the denominator as the second term: 3 = 3 : 1.

Question: Exercise 6.1, Question 3 (v): Convert the following fraction into a ratio: p/q

Answer: To convert a fraction into a ratio, write the numerator as the first term and the denominator as the second term: p/q = p : q.

Question: Exercise 6.1, Question 4 (i): Convert the following ratio into a common fraction: 1 : 5

Answer: To convert a ratio into a fraction, write the first term as the numerator and the second term as the denominator: 1 : 5 = 1/5.

Question: Exercise 6.1, Question 4 (ii): Convert the following ratio into a common fraction: 2 : 19

Answer: To convert a ratio into a fraction, write the first term as the numerator and the second term as the denominator: 2 : 19 = 2/19.

Question: Exercise 6.1, Question 4 (iii): Convert the following ratio into a common fraction: 8 : 1

Answer: To convert a ratio into a fraction, write the first term as the

numerator and the second term as the denominator: 8 : 1 = 8/1 = 8.

Question: Exercise 6.1, Question 4 (iv): Convert the following ratio into a common fraction: 75 : 76

Answer: To convert a ratio into a fraction, write the first term as the numerator and the second term as the denominator: 75 : 76 = 75/76.

Question: Exercise 6.1, Question 4 (v): Convert the following ratio into a common fraction: x : y

Answer: To convert a ratio into a fraction, write the first term as the numerator and the second term as the denominator: x : y = x/y.

Question: Exercise 6.1, Question 5: In a Class 6 science test paper, only 25 of 45 students passed. Write the ratio between the number of passing students and the total number of students

Answer: Given data: total students = 45, passing = 25. Passing : total = 25 : 45 = (25 ÷ 5) : (45 ÷ 5) = 5 : 9. Answer: 5 : 9.

Question: Exercise 6.1, Question 6: Arshad's one month's income is Rs. 20000. Write the ratio of his monthly income to his expense, when his monthly saving is Rs. 5000

Answer: Given data: monthly income = Rs. 20000, saving = Rs. 5000. Monthly expense = 20000 − 5000 = Rs. 15000. Income : expense = 20000 : 15000 = (20000 ÷ 5000) : (15000 ÷ 5000) = 4 : 3. Answer: 4 : 3.

Question: Exercise 6.1, Question 7: Two squares whose sides are 2 m and 5 m respectively. Write the ratio between the perimeters of their squares

Answer: Given data: first square's side = 2 m, second square's side = 5 m. Perimeter of a square = 4 x side. First square's perimeter = 4 x 2 = 8 m, second square's perimeter = 4 x 5 = 20 m. Ratio of perimeters = 8 : 20 = (8 ÷ 4) : (20 ÷ 4) = 2 : 5. Answer: 2 : 5.

Question: Exercise 6.1, Question 8: The weights of two bags of wheat are 16 kg and 14 kg and 400 g respectively. Write the ratio between their weights

Answer: Given data: first bag's weight = 16 kg = 16000 g, second bag's weight = 14 kg 400 g = 14400 g. Ratio of weights = 16000 : 14400 = (16000 ÷ 1600) : (14400 ÷ 1600) = 10 : 9. Answer: 10 : 9.

Question: Exercise 6.1, Question 9: The measures of the angles of a triangle are 30°, 60° and 90° respectively; write the ratio between the measures of those angles of the triangle

Answer: Given data: measures of the triangle's angles = 30°, 60°, 90°. Ratio of the angle measures = 30 : 60 : 90 = (30 ÷ 30) : (60 ÷ 30) : (90 ÷ 30) = 1 : 2 : 3. Answer: 1 : 2 : 3.

Question: Exercise 6.2, Question 1 (i): Identify the means and extremes in each of the following proportions: 2 : 5 = 8 : 20

Answer: In the proportion 2 : 5 = 8 : 20 the extremes (outer terms) are 2 and 20, and the means (middle terms) are 5 and 8. Check: product of extremes = 2 x 20 = 40, product of means = 5 x 8 = 40, both are equal.

Question: Exercise 6.2, Question 1 (ii): Identify the means and extremes in each of the following proportions: 3 : 4 = 6 : 8

Answer: In the proportion 3 : 4 = 6 : 8 the extremes (outer terms) are 3 and 8, and the means (middle terms) are 4 and 6. Check: product of extremes = 3 x 8 = 24, product of means = 4 x 6 = 24, both are equal.

Question: Exercise 6.2, Question 1 (iii): Identify the means and extremes in each of the following proportions: a : b = c : d

Answer: In the proportion a : b = c : d the extremes (outer terms) are a and d, and the means (middle terms) are b and c. The proportion formula: product of extremes = product of means, that is, a x d = b x c.

Question: Exercise 6.2, Question 2 (i): In each of the following questions four numbers are given. Check the numbers and tell whether they are in proportion or not: 18, 24, 30, 40

Answer: Product of extremes = 18 x 40 = 720, product of means = 24 x 30 = 720. The two products are equal, so the numbers are in proportion. That is, 18 : 24 = 30 : 40.

Question: Exercise 6.2, Question 2 (ii): In each of the following questions four numbers are given. Check the numbers and tell whether they are in proportion or not: 14, 19, 3, 4

Answer: Product of extremes = 14 x 4 = 56, product of means = 19 x 3 = 57. The two products are not equal, so the numbers are not in proportion.

Question: Exercise 6.2, Question 2 (iii): In each of the following questions four numbers are given. Check the numbers and tell whether they are in proportion or not: 6, 8, 12, 16

Answer: Product of extremes = 6 x 16 = 96, product of means = 8 x 12 = 96. The two products are equal, so the numbers are in proportion. That is, 6 : 8 = 12 : 16.

Question: Exercise 6.2, Question 2 (iv): In each of the following questions four numbers are given. Check the numbers and tell whether they are in proportion or not: 15, 20, 16, 21

Answer: Product of extremes = 15 x 21 = 315, product of means = 20 x 16 = 320. The two products are not equal, so the numbers are not in proportion.

Question: Exercise 6.2, Question 2 (v): In each of the following questions four numbers are given. Check the numbers and tell whether they are in proportion or not: 20, 30, 40, 50

Answer: Product of extremes = 20 x 50 = 1000, product of means = 30 x 40 = 1200. The two products are not equal, so the numbers are not in proportion.

Question: Exercise 6.2, Question 2 (vi): In each of the following questions four numbers are given. Check the numbers and tell whether they are in proportion or not: 21, 57, 28, 76

Answer: Product of extremes = 21 x 76 = 1596, product of means = 57 x 28 = 1596. The two products are equal, so the numbers are in proportion. That is, 21 : 57 = 28 : 76.

Question: Exercise 6.2, Question 3: Write the value of the unknown term in the given proportion: 2 : x = 3 : 7

Answer: By the proportion formula, product of extremes = product of means: 2 x 7 = x x 3, that is, 14 = 3x. Divide both sides by 3: x = 14/3 = 4 2/3.

Question: Exercise 6.2, Question 4 (i): Write the fourth proportional in each of the following: 2, 3, 6

Answer: Suppose the fourth proportional is x: 2 : 3 :: 6 : x. Product of extremes = product of means: 2 x x = 3 x 6, that is, 2x = 18. Divide both sides by 2: x = 9. The fourth proportional is 9.

Question: Exercise 6.2, Question 4 (ii): Write the fourth proportional in each of the following: 11/24, 7/15, 5/8

Answer: Suppose the fourth proportional is x: 11/24 : 7/15 :: 5/8 : x. Product of extremes = product of means: 11/24 x x = 7/15 x 5/8 = 35/120. So x = 35/120 ÷ 11/24 = 35/120 x 24/11 = 7/11. The fourth proportional is 7/11.

Question: Exercise 6.2, Question 4 (iii): Write the fourth proportional in each of the following: 16, 12, 8

Answer: Suppose the fourth proportional is x: 16 : 12 :: 8 : x. Product of extremes = product of means: 16 x x = 12 x 8 = 96, that is, 16x = 96. Divide both sides by 16: x = 6. The fourth proportional is 6.

Question: Exercise 6.2, Question 4 (iv): Write the fourth proportional in each of the following: 76, 28, 57

Answer: Suppose the fourth proportional is x: 76 : 28 :: 57 : x. Product of extremes = product of means: 76 x x = 28 x 57 = 1596, that is, 76x = 1596. Divide both sides by 76: x = 21. The fourth proportional is 21.

Question: Exercise 6.2, Question 4 (v): Write the fourth proportional in each of the following: 36, 45, 4

Answer: Suppose the fourth proportional is x: 36 : 45 :: 4 : x. Product of extremes = product of means: 36 x x = 45 x 4 = 180, that is, 36x = 180. Divide both sides by 36: x = 5. The fourth proportional is 5.

Question: Exercise 6.2, Question 4 (vi): Write the fourth proportional in each of the following: 40, 30, 24

Answer: Suppose the fourth proportional is x: 40 : 30 :: 24 : x. Product of extremes = product of means: 40 x x = 30 x 24 = 720, that is, 40x = 720.

Divide both sides by 40: x = 18. The fourth proportional is 18.

Question: Exercise 6.2, Question 5 (i): Write the mean proportional in each of the following parts: 15 and 60

Answer: Suppose the mean proportional is x: 15 : x :: x : 60. Product of extremes = product of means: 15 x 60 = x x x, that is, $x^2 = 900$. So x = 30. The mean proportional is 30.

Question: Exercise 6.2, Question 5 (ii): Write the mean proportional in each of the following parts: 18 and 32

Answer: Suppose the mean proportional is x: 18 : x :: x : 32. Product of extremes = product of means: 18 x 32 = x^2, that is, $x^2 = 576$. So x = 24. The mean proportional is 24.

Question: Exercise 6.2, Question 5 (iii): Write the mean proportional in each of the following parts: 28 and 63

Answer: Suppose the mean proportional is x: 28 : x :: x : 63. Product of extremes = product of means: 28 x 63 = x^2, that is, $x^2 = 1764$. So x = 42. The mean proportional is 42.

Question: Exercise 6.2, Question 5 (iv): Write the mean proportional in each of the following parts: 27 and 12

Answer: Suppose the mean proportional is x: 27 : x :: x : 12. Product of extremes = product of means: 27 x 12 = x^2, that is, $x^2 = 324$. So x = 18. The mean proportional is 18.

Question: Exercise 6.2, Question 5 (v): Write the mean proportional in each of the following parts: 40 and 90

Answer: Suppose the mean proportional is x: 40 : x :: x : 90. Product of extremes = product of means: 40 x 90 = x^2, that is, $x^2 = 3600$. So x = 60. The mean proportional is 60.

Question: Exercise 6.2, Question 5 (vi): Write the mean proportional in each of the following parts: 44 and 94

Answer: Suppose the mean proportional is x: 44 : x :: x : 94. Product of extremes = product of means: 44 x 94 = x^2, that is, $x^2 = 4136$. So x = $\sqrt{4136} \approx 64.31$. (4136 is not a perfect square, so the answer is not an exact whole number.) The mean proportional is $\sqrt{4136} \approx 64.31$.

Question: Exercise 6.3, Question 1: Ahmad obtained 60 marks in a test paper of 100 marks. Tell how many marks he would obtain in a test paper of 75 marks?

Answer: Given data: obtained in 100 marks = 60, total marks of second paper = 75. This is direct proportion: total marks : obtained marks = 100 : 60 = 75 : x. Product of extremes = product of means: 100 x x = 60 x 75 = 4500, so x = 4500 ÷ 100 = 45. Answer: He would obtain 45 marks in the 75-mark paper.

Question: Exercise 6.3, Question 2: The weight of 5/9 of a piece of cloth is

35 kg. Tell what the weight of 2/7 of the same piece of cloth will be?

Answer: Given data: weight of 5/9 part = 35 kg. Whole piece = 35 ÷ 5/9 = 35 x 9/5 = 63 kg. Weight of 2/7 part = 2/7 x 63 = 18 kg. Answer: The weight of the 2/7 part will be 18 kg.

Question: Exercise 6.3, Question 3: On a map, an actual distance of 80 km is shown by a line of 5 cm. Similarly, the distance between two cities on the map is shown by a line of 15 cm. Tell what the total actual distance is?

Answer: Given data: 5 cm on map = actual 80 km. This is direct proportion: map distance : actual distance = 5 : 80 = 15 : x. Product of extremes = product of means: 5 x x = 80 x 15 = 1200, so x = 1200 ÷ 5 = 240. Answer: The actual distance is 240 km.

Question: Exercise 6.3, Question 4: If one dozen eggs are bought for Rs. 60, tell for how many rupees 32 eggs will be bought?

Answer: Given data: 1 dozen = 12 eggs = Rs. 60. This is direct proportion: eggs : price = 12 : 60 = 32 : x. Product of extremes = product of means: 12 x x = 60 x 32 = 1920, so x = 1920 ÷ 12 = 160. Answer: 32 eggs will be bought for Rs. 160.

Question: Exercise 6.3, Question 5: Tabassum's one month's salary is Rs. 42000. She spends Rs. 39000 on household expenses. Write the ratio of her monthly saving to her salary. Also write the ratio of her monthly expense to her salary

Answer: Given data: salary = Rs. 42000, expense = Rs. 39000. Saving = 42000 − 39000 = Rs. 3000. Saving : salary = 3000 : 42000 = (3000 ÷ 3000) : (42000 ÷ 3000) = 1 : 14. Expense : salary = 39000 : 42000 = (39000 ÷ 3000) : (42000 ÷ 3000) = 13 : 14. Answer: The ratio of saving to salary is 1 : 14, and the ratio of expense to salary is 13 : 14.

Question: Exercise 6.3, Question 6: If Naveed can read 45 pages of a book in 75 minutes, how much total time will he take to read 876 pages of the same kind and same size?

Answer: Given data: 45 pages = 75 minutes. This is direct proportion: pages : time = 45 : 75 = 876 : x. Product of extremes = product of means: 45 x x = 75 x 876 = 65700, so x = 65700 ÷ 45 = 1460 minutes. 1460 minutes = 24 hours 20 minutes. Answer: Total 1460 minutes, that is, 24 hours 20 minutes.

Question: Exercise 6.3, Question 7: A book has 130 pages in all and there are 24 lines on each page. If we want to bring this book into 80 pages, tell how many lines will be on each page of this book?

Answer: Given data: total pages = 130, lines per page = 24. Total lines = 130 x 24 = 3120. This is inverse proportion: fewer pages means more lines per page. Lines per page in the new pages = 3120 ÷ 80 = 39. Answer: There will be 39 lines on each page.

Question: Exercise 6.3, Question 8: If 80 workers complete a work in 6 days, tell in how many total days 10 workers will complete the same work?

Answer: Given data: 80 workers = 6 days. This is inverse proportion: fewer workers means more days. Total worker-days = 80 x 6 = 480. Days needed by 10 workers = 480 ÷ 10 = 48. Answer: 10 workers will complete the same work in 48 days.

Question: Review Exercise 6, Question 1 (i): Find the following ratio and bring it to simplest form: The ratio of your copy and the width of the length

Answer: This question depends on the student's own measurement; the book gives no fixed measurement of the copy's length or width, so no single definite answer can be derived from the book. Method: measure your copy's length and width in the same unit, write length : width, and divide both terms by their common factor to bring it to simplest form.

Question: Review Exercise 6, Question 1 (ii): Find the following ratio and bring it to simplest form: The ratio of the number of present and absent students of your class

Answer: This question depends on the actual numbers of the student's own class; the book gives no fixed number of present or absent students, so no single definite answer can be derived from the book. Method: count the present and absent students in your class, write present : absent, and divide both terms by their common factor to bring it to simplest form.

Question: Review Exercise 6, Question 1 (iii): Find the following ratio and bring it to simplest form: The ratio between the weights of two friends of the class

Answer: This question depends on the actual weights of two friends; the book gives no fixed weight measurements, so no single definite answer can be derived from the book. Method: measure both friends' weights in the same unit, write first friend's weight : second friend's weight, and divide both terms by their common factor to bring it to simplest form.

Question: Review Exercise 6, Question 1 (iv): Find the following ratio and bring it to simplest form: The ratio of the daily pocket money of your three friends

Answer: This question depends on the actual pocket money of three friends; the book gives no fixed amount, so no single definite answer can be derived from the book. Method: write the three friends' daily pocket money in the same unit, write first : second : third, and divide all three terms by their common factor to bring it to simplest form.

Question: Review Exercise 6, Question 1 (v): Find the following ratio and bring it to simplest form: The ratio of the pages of the books of your class, Mathematics, English and Social Studies

Answer: This question depends on the actual pages of the student's own books; the book gives no fixed page counts, so no single definite answer can be derived from the book. Method: count the pages of the Mathematics, English and Social Studies books, write Mathematics : English : Social Studies, and divide all three terms by their common factor to bring it to simplest form.

Review Exercise 6, Question 2 (i): Find the equal ratios of the following ratio: 2 : 3 = _____

Answer: (2 x 2) : (3 x 2) = 4 : 6. (Another correct answer is also possible, e.g. 6 : 9.)

Review Exercise 6, Question 2 (ii): Find the equal ratios of the following ratio: 10 : 8 = _____

Answer: (10 ÷ 2) : (8 ÷ 2) = 5 : 4. (Another correct answer is also possible, e.g. 20 : 16.)

Review Exercise 6, Question 2 (iii): Find the equal ratios of the following ratio: 1 : 4 = _____

Answer: (1 x 2) : (4 x 2) = 2 : 8. (Another correct answer is also possible, e.g. 3 : 12.)

Review Exercise 6, Question 2 (iv): Find the equal ratios of the following ratio: 21 : 35 = _____

Answer: (21 ÷ 7) : (35 ÷ 7) = 3 : 5. (Another correct answer is also possible, e.g. 42 : 70.)

Review Exercise 6, Question 2 (v): Find the equal ratios of the following ratio: 13 : 91 = _____

Answer: (13 ÷ 13) : (91 ÷ 13) = 1 : 7. (Another correct answer is also possible, e.g. 26 : 182.)

Review Exercise 6, Question 2 (vi): Find the equal ratios of the following ratio: 33 : 110 = _____

Answer: (33 ÷ 11) : (110 ÷ 11) = 3 : 10. (Another correct answer is also possible, e.g. 6 : 20.)

Question: Review Exercise 6, Question 3 (i): Write the value of x in each of the following proportions: 2 : 5 :: x : 10

Answer: Product of extremes = product of means: 2 x 10 = 5 x x, that is, 20 = 5x. Divide both sides by 5: x = 4.

Question: Review Exercise 6, Question 3 (ii): Write the value of x in each of the following proportions: 5 : x :: 10 : 14

Answer: Product of extremes = product of means: 5 x 14 = x x 10, that is, 70 = 10x. Divide both sides by 10: x = 7.

Question: Review Exercise 6, Question 3 (iii): Write the value of x in each of the following proportions: 15 : 30 :: 1 : x

Answer: Product of extremes = product of means: 15 x x = 30 x 1, that is, 15x = 30. Divide both sides by 15: x = 2.

Question: Review Exercise 6, Question 3 (iv): Write the value of x in each of the following proportions: x : 12 :: 3 : 4

Answer: Product of extremes = product of means: x x 4 = 12 x 3, that is, 4x = 36. Divide both sides by 4: x = 9.

Question: Review Exercise 6, Question 3 (v): Write the value of x in each of the following proportions: x : 200 :: 18 : 25

Answer: Product of extremes = product of means: x x 25 = 200 x 18, that is, 25x = 3600. Divide both sides by 25: x = 144.

Question: Review Exercise 6, Question 3 (vi): Write the value of x in each of the following proportions: 132 : 216 :: 33 : x

Answer: Product of extremes = product of means: 132 x x = 216 x 33, that is, 132x = 7128. Divide both sides by 132: x = 54.

Question: Review Exercise 6, Question 4 (i): Which of the following are inverse proportions? The number of copies you bought and their total price

Answer: This is direct proportion, not inverse, because as the copies increase, their total price also increases.

Question: Review Exercise 6, Question 4 (ii): Which of the following are inverse proportions? The number of people and the time they take to do the work

Answer: This is inverse proportion, because as the people increase, the time taken to do the work decreases.

Question: Review Exercise 6, Question 4 (iii): Which of the following are inverse proportions? The speed of the vehicle and the time the vehicle takes to complete the journey

Answer: This is inverse proportion, because as the speed increases, the time taken to complete the journey decreases.

Question: Review Exercise 6, Question 4 (iv): Which of the following are inverse proportions? The total number of packets and the number of pencils in each packet

Answer: This is inverse proportion (for a fixed total number of pencils), because as the packets increase, the pencils in each packet decrease.

Question: Review Exercise 6, Question 5 (i): In a school there are 670 students in all. Of them 350 are girls, the rest are boys, then write the ratios between the following: The number of girls to the total number of students

Answer: Given data: total students = 670, girls = 350, boys = 670 − 350 = 320. Girls : total students = 350 : 670 = (350 ÷ 10) : (670 ÷ 10) = 35 : 67. Answer: 35 : 67.

Question: Review Exercise 6, Question 5 (ii): In a school there are 670 students in all. Of them 350 are girls, the rest are boys, then write the ratios between the following: The number of boys to the total number of students

Answer: Given data: total students = 670, boys = 320. Boys : total students = 320 : 670 = (320 ÷ 10) : (670 ÷ 10) = 32 : 67. Answer: 32 : 67.

Question: Review Exercise 6, Question 5 (iii): In a school there are 670 students in all. Of them 350 are girls, the rest are boys, then write the ratios between the following: The number of girls to the number of boys

Answer: Given data: girls = 350, boys = 320. Girls : boys = 350 : 320 = (350 ÷ 10) : (320 ÷ 10) = 35 : 32. Answer: 35 : 32.

Question: Review Exercise 6, Question 5 (iv): In a school there are 670 students in all. Of them 350 are girls, the rest are boys, then write the ratios between the following: The number of boys to the number of girls

Answer: Given data: boys = 320, girls = 350. Boys : girls = 320 : 350 = (320 ÷ 10) : (350 ÷ 10) = 32 : 35. Answer: 32 : 35.

Question: Review Exercise 6, Question 6: Rehan writes 26 words in 2 minutes. Tell how much time she will take to write 325 words?

Answer: Given data: 26 words = 2 minutes. This is direct proportion: words : time = 26 : 2 = 325 : x. Product of extremes = product of means: 26 x x = 2 x 325 = 650, so x = 650 ÷ 26 = 25. Answer: She will take 25 minutes to write 325 words.

Question: Review Exercise 6, Question 7: Twelve litres of milk are available for Rs. 1200. Tell for how much 13 1/2 litres of milk will be available?

Answer: Given data: 12 litres of milk = Rs. 1200. This is direct proportion: milk : price = 12 : 1200 = 13 1/2 : x. Product of extremes = product of means: 12 x x = 1200 x 27/2 = 16200, so x = 16200 ÷ 12 = 1350. Answer: 13 1/2 litres of milk will be available for Rs. 1350.

Question: Review Exercise 6, Question 8: A 24-metre-high wall is completed by 16 workers in a given time. Tell how many workers will complete an 18-metre-high wall of the same kind in the same time?

Answer: Given data: 24 m wall = 16 workers. This is direct proportion: as the wall height increases, the workers also increase. Wall : workers = 24 : 16 = 18 : x. Product of extremes = product of means: 24 x x = 16 x 18 = 288, so x = 288 ÷ 24 = 12. Answer: 12 workers will complete the 18-metre-high wall in the same time.

Question: Review Exercise 6, Question 9: Food for 600 men is available for 18 days. Tell how many men will finish the same food in 27 days?

Answer: Given data: 600 men = 18 days. This is inverse proportion: more men means the food lasts fewer days. Total man-days = 600 x 18 = 10800. Men who will finish the food in 27 days = 10800 ÷ 27 = 400. Answer: 400 men will finish the same food in 27 days.

Question: Practical Work: The weights of Sahar and Shabah are 45 kg and 15 kg respectively. To show their weights in relation to each other, consider the following statements: (1) Sahar is 30 kg heavier than Shabah. (2) Sahar's weight is three times Shabah's weight. (3) Shabah is 30 kg lighter than Sahar. (4) In comparison with Sahar's weight, Shabah's weight

is 1/3 part. Which of the above statements are related to addition and subtraction?

Answer: Statements (1) and (3) are related to addition and subtraction: Sahar's weight is 30 kg more than Shabah's weight, that is 15 + 30 = 45, and Shabah's weight is 30 kg less than Sahar's weight, that is 45 - 30 = 15. Both statements give the difference (30 kg) between the two weights, which is the operation of addition and subtraction.

Question: Practical Work: The weights of Sahar and Shabah are 45 kg and 15 kg respectively. To show their weights in relation to each other, consider the following statements: (1) Sahar is 30 kg heavier than Shabah. (2) Sahar's weight is three times Shabah's weight. (3) Shabah is 30 kg lighter than Sahar. (4) In comparison with Sahar's weight, Shabah's weight is 1/3 part. Which of the above statements are related to multiplication and division?

Answer: Statements (2) and (4) are related to multiplication and division: Sahar's weight is three times Shabah's weight, that is 3 x 15 = 45 (multiplication), and Shabah's weight is 1/3 of Sahar's weight, that is 15 ÷ 45 = 1/3 or 45 ÷ 3 = 15 (division). Both statements give one weight as a multiple or a part of the other weight, which is the operation of multiplication and division.

7. Financial Arithmetic

Question: Exercise 7.1, Question 1 (i): What percent of these shapes is shaded? (10 × 10 square with 20 cells shaded)

Answer: Total cells = 100. Shaded cells = 5 + 5 + 5 + 5 = 20. Percent = 20/100 × 100% = 20%. Answer: 20%

Question: Exercise 7.1, Question 1 (ii): What percent of these shapes is shaded? (10 × 10 square with the top 2 full rows and 1 cell of the third row shaded)

Answer: Total cells = 100. Shaded cells = 10 + 10 + 1 = 21. Percent = 21/100 × 100% = 21%. Answer: 21%

Question: Exercise 7.1, Question 1 (iii): What percent of these shapes is shaded? (Square divided into 4 triangles, 2 of which are shaded)

Answer: Total triangles = 4. Shaded triangles = 2. Percent = 2/4 × 100% = 50%. Answer: 50%

Question: Exercise 7.1, Question 2 (i): Convert these into fractions in decimal form. 25%

Answer: 25% = 25/100 = 1/4. Answer: 1/4

Question: Exercise 7.1, Question 2 (ii): Convert these into fractions in decimal form. 1%

Answer: 1% = 1/100. Answer: 1/100

Question: Exercise 7.1, Question 2 (iii): Convert these into fractions in decimal form. 31%

Answer: 31% = 31/100. Answer: 31/100

Question: Exercise 7.1, Question 2 (iv): Convert these into fractions in decimal form. 15%

Answer: 15% = 15/100 = 3/20. Answer: 3/20

Question: Exercise 7.1, Question 2 (v): Convert these into fractions in decimal form. 8 1/2%

Answer: 8 1/2% = 17/2% = 17/2 × 1/100 = 17/200. Answer: 17/200

Question: Exercise 7.1, Question 2 (vi): Convert these into fractions in decimal form. 20.5%

Answer: 20.5% = 20.5/100 = 205/1000 = 41/200. Answer: 41/200

Question: Exercise 7.1, Question 2 (vii): Convert these into fractions in decimal form. 175%

Answer: 175% = 175/100 = 7/4. Answer: 7/4

Question: Exercise 7.1, Question 2 (viii): Convert these into fractions in decimal form. 115%

Answer: 115% = 115/100 = 23/20. Answer: 23/20

Question: Exercise 7.1, Question 2 (ix): Convert these into fractions in decimal form. 225%

Answer: 225% = 225/100 = 9/4. Answer: 9/4

Question: Exercise 7.1, Question 2 (x): Convert these into fractions in decimal form. 2 1/2%

Answer: 2 1/2% = 5/2% = 5/2 × 1/100 = 5/200 = 1/40. Answer: 1/40

Question: Exercise 7.1, Question 3 (i): Convert these fractions into percent. 1/4

Answer: 1/4 = 1/4 × 100% = 25%. Answer: 25%

Question: Exercise 7.1, Question 3 (ii): Convert these fractions into percent. 7/20

Answer: 7/20 = 7/20 × 100% = 35%. Answer: 35%

Question: Exercise 7.1, Question 3 (iii): Convert these fractions into percent. 8/25

Answer: 8/25 = 8/25 × 100% = 32%. Answer: 32%

Question: Exercise 7.1, Question 3 (iv): Convert these fractions into percent. 5/4

Answer: 5/4 = 5/4 × 100% = 125%. Answer: 125%

Question: Exercise 7.1, Question 4 (i): Write each given percent as a decimal fraction. 25%

Answer: 25% = 25/100 = 0.25. Answer: 0.25

Question: Exercise 7.1, Question 4 (ii): Write each given percent as a decimal fraction. 31%

Answer: 31% = 31/100 = 0.31. Answer: 0.31

Question: Exercise 7.1, Question 4 (iii): Write each given percent as a decimal fraction. 1%

Answer: 1% = 1/100 = 0.01. Answer: 0.01

Question: Exercise 7.1, Question 4 (iv): Write each given percent as a

decimal fraction. 2 1/2%

Answer: 2 1/2% = 2.5/100 = 0.025. Answer: 0.025

Question: Exercise 7.1, Question 4 (v): Write each given percent as a decimal fraction. 6.5%

Answer: 6.5% = 6.5/100 = 0.065. Answer: 0.065

Question: Exercise 7.1, Question 5 (i): Convert these decimal fractions into percent. 0.025

Answer: 0.025 = 0.025 × 100% = 2.5%. Answer: 2.5%

Question: Exercise 7.1, Question 5 (ii): Convert these decimal fractions into percent. 0.4

Answer: 0.4 = 0.4 × 100% = 40%. Answer: 40%

Question: Exercise 7.1, Question 5 (iii): Convert these decimal fractions into percent. 0.85

Answer: 0.85 = 0.85 × 100% = 85%. Answer: 85%

Question: Exercise 7.1, Question 5 (iv): Convert these decimal fractions into percent. 0.105

Answer: 0.105 = 0.105 × 100% = 10.5%. Answer: 10.5%

Question: Exercise 7.1, Question 5 (v): Convert these decimal fractions into percent. 12.5

Answer: 12.5 = 12.5 × 100% = 1250%. Answer: 1250%

Question: Exercise 7.1, Question 5 (vi): Convert these decimal fractions into percent. 125.5

Answer: 125.5 = 125.5 × 100% = 12550%. Answer: 12550%

Question: Exercise 7.1, Question 6 (i): What percent of 1 metre are these? 70 cm

Answer: 1 metre = 100 cm. Percent = 70/100 × 100% = 70%. Answer: 70%

Question: Exercise 7.1, Question 6 (ii): What percent of 1 metre are these? 50 cm

Answer: 1 metre = 100 cm. Percent = 50/100 × 100% = 50%. Answer: 50%

Question: Exercise 7.1, Question 6 (iii): What percent of 1 metre are these? 75 cm

Answer: 1 metre = 100 cm. Percent = 75/100 × 100% = 75%. Answer: 75%

Question: Exercise 7.1, Question 6 (iv): What percent of 1 metre are these? 125 cm

Answer: 1 metre = 100 cm. Percent = 125/100 × 100% = 125%. Answer: 125%

Question: Exercise 7.1, Question 6 (v): What percent of 1 metre are these? 100 cm

Answer: 1 metre = 100 cm. Percent = 100/100 × 100% = 100%. Answer: 100%

Question: Exercise 7.1, Question 7: In a shop, the sale of science books is 30%. Write it as a fraction.

Answer: 30% = 30/100 = 3/10. Answer: 3/10

Question: Exercise 7.1, Question 8: Khalida obtained 96 marks out of 150. What percent of the total marks did she obtain?

Answer: Percent = part/total × 100% = 96/150 × 100% = 64%. Answer: 64%

Question: Exercise 7.1, Question 9 (i): Find these percents. 50% of 10

Answer: 50% of 10 = 50/100 × 10 = 5. Answer: 5

Question: Exercise 7.1, Question 9 (ii): Find these percents. 30% of 70

Answer: 30% of 70 = 30/100 × 70 = 21. Answer: 21

Question: Exercise 7.1, Question 9 (iii): Find these percents. 25% of 250

Answer: 25% of 250 = 25/100 × 250 = 62 1/2. Answer: 62 1/2

Question: Exercise 7.1, Question 9 (iv): Find these percents. 50% of 50

Answer: 50% of 50 = 50/100 × 50 = 25. Answer: 25

Question: Exercise 7.1, Question 9 (v): Find these percents. 75% of 750

Answer: 75% of 750 = 75/100 × 750 = 562 1/2. Answer: 562 1/2

Question: Exercise 7.1, Question 9 (vi): Find these percents. 125% of 50 m.s

Answer: 125% of 50 = 125/100 × 50 = 62.5. Answer: 62.5 m.s

Question: Exercise 7.1, Question 9 (vii): Find these percents. 60% of 300 litres

Answer: 60% of 300 = 60/100 × 300 = 180. Answer: 180 litres

Question: Exercise 7.1, Question 9 (viii): Find these percents. 20% of 13.25 rupees

Answer: 20% of 13.25 = 20/100 × 13.25 = 2.65. Answer: 2.65 rupees

Question: Exercise 7.1, Question 9 (ix): Find these percents. 60.5% of 8 hours

Answer: 60.5% of 8 = 60.5/100 × 8 = 4.84. Answer: 4.84 hours

Question: Exercise 7.1, Question 9 (x): Find these percents. 25% of 172 metres

Answer: 25% of 172 = 25/100 × 172 = 43. Answer: 43 metres

Question: Exercise 7.1, Question 9 (xi): Find these percents. 15% of 210

Answer: 15% of 210 = 15/100 × 210 = 31.5. Answer: 31.5

Question: Exercise 7.1, Question 9 (xii): Find these percents. 75% of 1011

Answer: 75% of 1011 = 75/100 × 1011 = 758 1/4. Answer: 758 1/4

Question: Exercise 7.1, Question 9 (xiii): Find these percents. 35% of 301

Answer: 35% of 301 = 35/100 × 301 = 105.35. Answer: 105.35

Question: Exercise 7.1, Question 9 (xiv): Find these percents. 65% of 910

Answer: 65% of 910 = 65/100 × 910 = 591.5. Answer: 591.5

Question: Exercise 7.1, Question 10: Using percent, write these fractions in ascending and descending order. 2/5, 1/4, 3/4

Answer: 2/5 = 2/5 × 100% = 40%; 1/4 = 25%; 3/4 = 75%. Since 25% < 40% < 75%, ascending order: 1/4, 2/5, 3/4. Descending order: 3/4, 2/5, 1/4.

Question: Exercise 7.2, Question 1: I obtained 60 marks out of 75 marks. What will be the percent of my marks?

Answer: Percent = part/total × 100% = 60/75 × 100% = 80%. Answer: 80%

Question: Exercise 7.2, Question 2: Saim spent 300 rupees out of 500 rupees. Find the percent of his spending.

Answer: Percent = spending/total amount × 100% = 300/500 × 100% = 60%. Answer: 60%

Question: Exercise 7.2, Question 3: In a village there were 8500 voters. 34% of the people did not vote. Find the number of voters who cast their votes.

Answer: Non-voters = 34% of 8500 = 34/100 × 8500 = 2890. Voters = 8500 − 2890 = 5610. (Another way: voters = 66% of 8500 = 5610.) Answer: 5610 people

Question: Exercise 7.2, Question 4: The population of a village is 15000

people. If the population increases by 5% in a year, what will the population be after one year?

Answer: Increase = 5% of 15000 = 5/100 × 15000 = 750. Population after one year = 15000 + 750 = 15750. Answer: 15750 people

Question: Exercise 7.2, Question 5: Raheel gives 5% of her salary in charity in a month. Find her total monthly salary.

Answer: The book does not give the amount of charity, so the total salary cannot be found from the printed data. Method: total salary = charity amount ÷ 5%. For example, if the charity amount were known, dividing it by 5% would give the total salary.

Question: Exercise 7.2, Question 6: 13% of the distance between two cities is 36 kilometres. Find the total distance between the cities.

Answer: Total distance = given quantity ÷ rate = 36 ÷ 13% = 36 ÷ 13/100 = 36 × 100/13 = 3600/13 = 276 12/13 kilometres. Answer: 276 12/13 km

Question: Exercise 7.2, Question 7: Naeem's monthly salary is 8000 rupees and he spends 6000 rupees every month. Find the percent of his spending.

Answer: Spending percent = spending/salary × 100% = 6000/8000 × 100% = 75%. Answer: 75%

Question: Exercise 7.2, Question 8: On a rainy day, 600 out of 750 students were present in school. Tell what percent of the students were absent that day?

Answer: Absent students = 750 − 600 = 150. Absent percent = 150/750 × 100% = 20%. Answer: 20%

Question: Exercise 7.2, Question 9: If one angle of a triangle is 90°. Find the remaining two angles of the triangle, if these remaining two angles are respectively 30% and 70% of 90°.

Answer: First remaining angle = 30% of 90° = 30/100 × 90° = 27°. Second remaining angle = 70% of 90° = 70/100 × 90° = 63°. Check: 27° + 63° = 90°. Answer: 27° and 63°

Question: Exercise 7.2, Question 10: The area of a square is 100 m.s. Find the area of a rectangle whose area is 10% more than the area of the given square.

Answer: Increase = 10% of 100 = 10/100 × 100 = 10. Area of rectangle = 100 + 10 = 110. Answer: 110 m.s

Question: Exercise 7.2, Question 11: In a shop, every item has successive discounts of 15% and 20%. If the price of an item is 100 rupees, find the total discount given on that item.

Answer: First discount = 15% of 100 = 15 rupees, so the price becomes 100 − 15 = 85 rupees. Second discount = 20% of 85 = 17 rupees. Total discount = 15 + 17 = 32 rupees, i.e. 32%. Answer: 32 rupees (32%)

Question: Exercise 7.3, Question 1 (i): Find the percent profit or loss in each. Eggs bought at 120 rupees per dozen and sold at 130 rupees per dozen.

Answer: Cost price = 120 rupees, selling price = 130 rupees. Profit = 130 − 120 = 10 rupees. Profit percent = profit/cost price × 100% = 10/120 × 100% = 8 1/3%. Answer: 8 1/3% profit

Question: Exercise 7.3, Question 1 (ii): Find the percent profit or loss in each. A shopkeeper buys 25 double-rotis at 10 rupees each. He sells at 13 rupees each, while 7 double-rotis got spoiled.

Answer: Total cost price = 25 × 10 = 250 rupees. Rotis sold = 25 − 7 = 18. Total selling price = 18 × 13 = 234 rupees. Loss = 250 − 234 = 16 rupees. Loss percent = 16/250 × 100% = 6.4%. Answer: 6.4% loss

Question: Exercise 7.3, Question 1 (iii): Find the percent profit or loss in each. A person buys pencils at 60 rupees per dozen and sells them at 70 rupees per dozen.

Answer: Cost price = 60 rupees, selling price = 70 rupees. Profit = 70 − 60 = 10 rupees. Profit percent = 10/60 × 100% = 16 2/3%. Answer: 16 2/3% profit

Question: Exercise 7.3, Question 1 (iv): Find the percent profit or loss in each. A person buys 10 dozen eggs at 120 rupees per dozen. 3 dozen eggs break. He sells the remaining eggs at 130 rupees per dozen.

Answer: Total cost price = 10 × 120 = 1200 rupees. Eggs sold = 10 − 3 = 7 dozen. Total selling price = 7 × 130 = 910 rupees. Loss = 1200 − 910 = 290 rupees. Loss percent = 290/1200 × 100% = 24 1/6%. Answer: 24 1/6% loss

Question: Exercise 7.3, Question 1 (v): Find the percent profit or loss in each. A milkman buys milk at 75.50 rupees per litre and sells it at 80 rupees per litre.

Answer: Cost price = 75.50 rupees, selling price = 80 rupees. Profit = 80 − 75.50 = 4.50 rupees. Profit percent = 4.50/75.50 × 100% = 900/151% = 5 145/151%. Answer: 5 145/151% profit

Question: Exercise 7.3, Question 2: A bookseller buys 20 mathematics books at 120 rupees per book and sells them at 125 rupees per book. Find his percent profit.

Answer: Cost price = 120 rupees, selling price = 125 rupees. Profit = 125 − 120 = 5 rupees. Profit percent = 5/120 × 100% = 4 1/6%. Answer: 4 1/6% profit

Question: Exercise 7.3, Question 3: A trader buys 10 kilograms of rice at 120 rupees per kilogram and sells it at 125 rupees per kilogram. Find his percent profit.

Answer: Cost price = 120 rupees, selling price = 125 rupees. Profit = 125 − 120 = 5 rupees. Profit percent = 5/120 × 100% = 4 1/6%. Answer: 4 1/6% profit

Question: Exercise 7.3, Question 4: A shopkeeper bears a 15% loss on the sale of a jacket. If the cost price of the jacket is 1000 rupees, find his percent loss.

Answer: Loss = 15% of 1000 = 15/100 × 1000 = 150 rupees. Loss percent = loss/cost price × 100% = 150/1000 × 100% = 15%. Answer: 15% loss

Question: Exercise 7.3, Question 5: At a 20% discount, Umeed bought a dinner set for 52000 rupees. Find the original price of the dinner set.

Answer: Rate paid after discount = 100% − 20% = 80%. Original price × 80% = 52000, so original price = 52000 ÷ 80% = 52000 × 100/80 = 65000 rupees. Answer: 65000 rupees

Question: Exercise 7.3, Question 6: The original selling price of a ladies' purse is 1500 rupees; it was sold at a 15% discount. Find the total selling price of the purse.

Answer: Discount = 15% of 1500 = 15/100 × 1500 = 225 rupees. Total selling price = original selling price − discount = 1500 − 225 = 1275 rupees. Answer: 1275 rupees

Question: Review Exercise 7, Question 1: Karim bought 50 dozen pencils at 60 rupees per dozen. He sold all the pencils at 13 rupees per pencil. Find his percent profit or loss.

Answer: Total cost price = 50 × 60 = 3000 rupees. Total selling price = 50 × 12 × 13 = 7800 rupees. Profit = 7800 − 3000 = 4800 rupees. Profit percent = 4800/3000 × 100% = 160%. Answer: 160% profit

Question: Review Exercise 7, Question 2: A television manufacturing company announces that a Hathi colour TV is available for 13050 rupees. Its previous price was 17400 rupees. Find the percent discount offered by the company on the colour TV.

Answer: Discount = 17400 − 13050 = 4350 rupees. Discount percent = discount/previous price × 100% = 4350/17400 × 100% = 25%. Answer: 25% discount

Question: Review Exercise 7, Question 3: Imran obtained 548 marks out of 800 marks, and the other student obtained 459 marks out of 600. Whose performance is better?

Answer: Imran's percent = 548/800 × 100% = 68.5%. The other student's percent = 459/600 × 100% = 76.5%. Since 76.5% > 68.5%, the other student's performance is better. Answer: The other student's performance is better

Question: Review Exercise 7, Question 4: A fruit seller buys 50 dozen bananas at 60 rupees per dozen. 10 dozen bananas go bad. He sells the remaining bananas at 70 rupees per dozen. Find his percent profit or loss.

Answer: Total cost price = 50 × 60 = 3000 rupees. Bananas sold = 50 − 10 = 40 dozen. Total selling price = 40 × 70 = 2800 rupees. Loss = 3000 − 2800 = 200 rupees. Loss percent = 200/3000 × 100% = 6 2/3%. Answer: 6 2/3% loss

Question: Review Exercise 7, Question 5: A car was sold for 87500 rupees with a 2 1/2% discount. What is the original selling price of the car?

Answer: Rate paid after discount = 100% − 2 1/2% = 97 1/2%. Original price × 97 1/2% = 87500, so original price = 87500 ÷ 97.5% = 87500 × 1000/975 = 3500000/39 = 89743 23/39 rupees, i.e. about 89743.59 rupees. Answer: 89743 23/39 rupees (about 89743.59 rupees)

Question: Review Exercise 7, Question 6 (i): Which is greater. 0.045 or 5%

Answer: 0.045 = 0.045 × 100% = 4.5%. Since 4.5% < 5%, 5% is greater. Answer: 5% is greater ✓

Question: Review Exercise 7, Question 6 (ii): Which is greater. 0.64 or 62%

Answer: 0.64 = 0.64 × 100% = 64%. Since 64% > 62%, 0.64 is greater. Answer: 0.64 is greater ✓

Question: Review Exercise 7, Question 6 (iii): Which is greater. 0.8 or 8%

Answer: 0.8 = 0.8 × 100% = 80%. Since 80% > 8%, 0.8 is greater. Answer: 0.8 is greater ✓

Question: Review Exercise 7, Question 6 (iv): Which is greater. 0.15 or 19%

Answer: 0.15 = 0.15 × 100% = 15%. Since 15% < 19%, 19% is greater. Answer: 19% is greater ✓

Question: Review Exercise 7, Question 6 (v): Which is greater. 1/2 or 40%

Answer: 1/2 = 1/2 × 100% = 50%. Since 50% > 40%, 1/2 is greater. Answer: 1/2 is greater ✓

Question: Review Exercise 7, Question 6 (vi): Which is greater. 1/5 or 25%

Answer: 1/5 = 1/5 × 100% = 20%. Since 20% < 25%, 25% is greater. Answer: 25% is greater ✓

Question: Review Exercise 7, Question 6 (vii): Which is greater. 3/4 or 70%

Answer: 3/4 = 3/4 × 100% = 75%. Since 75% > 70%, 3/4 is greater. Answer: 3/4 is greater ✓

Question: Review Exercise 7, Question 6 (viii): Which is greater. 1 1/4 or 130%

Answer: 1 1/4 = 5/4 = 5/4 × 100% = 125%. Since 125% < 130%, 130% is greater. Answer: 130% is greater ✓

Question: Review Exercise 7, Question 6 (ix): Which is greater. 1/10 or 9%

Answer: 1/10 = 1/10 × 100% = 10%. Since 10% > 9%, 1/10 is greater. Answer: 1/10 is greater ✓

Review Exercise 7, Question 7 (a) (i): Fill in the blank.

3/4

% = 3/4

Review Exercise 7, Question 7 (a) (ii): Fill in the blank. 5/5 % = 5/5

Review Exercise 7, Question 7 (a) (iii): Fill in the blank. _____ (common fraction) = 30%

Answer: 30%

Review Exercise 7, Question 7 (a) (iv): Fill in the blank. _____ = 0.125

Answer: 0.125

Review Exercise 7, Question 7 (a) (v): Fill in the blank. _____ (decimal fraction) = 38%

Answer: 38%

Review Exercise 7, Question 7 (a) (vi): Fill in the blank. _____ = 36% of 120

Answer: 36% of 120 = 36/100 × 120 = 43.2. Completed sentence: 43.2

Review Exercise 7, Question 7 (a) (vii): Fill in the blank. _____ = loss + selling price

Answer: Cost price

Review Exercise 7, Question 7 (a) (viii): Fill in the blank. _____ = selling price − cost price

Answer: Profit

✗ Review Exercise 7, Question 7 (b) (i): Write true or false for these. If the cost price is more than the selling price, then there will be profit.

✓ Review Exercise 7, Question 7 (b) (ii): Write true or false for these. If the cost price is more than the selling price, then there will be a loss.

✓ Review Exercise 7, Question 7 (b) (iii): Write true or false for these. Loss = cost price − selling price

✓ Review Exercise 7, Question 7 (b) (iv): Write true or false for these. Cost price = loss + selling price

✓ Review Exercise 7, Question 7 (b) (v): Write true or false for these. Loss percent = loss/cost price × 100%

Review Exercise 7, Question 7 (b) (vi): Write true or false for these.

Profit = selling price − cost price

Review Exercise 7, Question 7 (b) (vii): Write true or false for these.

Profit percent = profit/cost price × 100%

8. Introduction to Algebra

Question: Exercise 8.1, Question 1 (i): Write these sentences in symbols. A number x added to 4.

Answer: x + 4. Answer: x + 4

Question: Exercise 8.1, Question 1 (ii): Write these sentences in symbols. A number y multiplied by seven than.

Answer: The book's printed sentence ends with "than" without giving the number compared; on the plain reading it is seven times y. Answer: 7y

Question: Exercise 8.1, Question 1 (iii): Write these sentences in symbols. Six added to a number z is less than four.

Answer: Six plus z is less than four. Answer: 6 + z < 4

Question: Exercise 8.1, Question 2 (i): Which of these statements are true or false? 5 + 9 = 18

Answer: 5 + 9 = 14, not 18. Answer: False ✗

Question: Exercise 8.1, Question 2 (ii): Which of these statements are true or false? 8 + 3 = 11

Answer: 8 + 3 = 11. Answer: True ✓

Question: Exercise 8.1, Question 2 (iii): Which of these statements are true or false? 20 + 3 = 26

Answer: 20 + 3 = 23, not 26. Answer: False ✗

Question: Exercise 8.1, Question 2 (iv): Which of these statements are true or false? 21 − 8 = 13

Answer: 21 − 8 = 13. Answer: True ✓

Question: Exercise 8.1, Question 2 (v): Which of these statements are true or false? 18 − 13 = 15

Answer: 18 − 13 = 5, not 15. Answer: False ✗

Question: Exercise 8.1, Question 2 (vi): Which of these statements are true or false? 16 − 10 = 6

Answer: 16 − 10 = 6. Answer: True ✓

Question: Exercise 8.1, Question 3 (i): Which of these are open statements? D + 3 = 9

Answer: D is a variable whose value is not fixed, so this is an open statement. Answer: Open statement

Question: Exercise 8.1, Question 3 (ii): Which of these are open statements? 3a + 4 = 8

Answer: a is a variable whose value is not fixed, so this is an open statement. Answer: Open statement

Question: Exercise 8.1, Question 3 (iii): Which of these are open statements? 9 + 4 = 13

Answer: There is no variable in it and the statement is true, so this is a closed statement. Answer: Closed statement (true statement)

Question: Exercise 8.1, Question 4 (i): Find the value of the number Δ or □ that makes these statements true. 6 + Δ = 15

Answer: Δ = 15 − 6 = 9. Check: 6 + 9 = 15. Answer: Δ = 9

Question: Exercise 8.1, Question 4 (ii): Find the value of the number Δ or □ that makes these statements true. □ + 5 = 8

Answer: □ = 8 − 5 = 3. Check: 3 + 5 = 8. Answer: □ = 3

Question: Exercise 8.1, Question 4 (iii): Find the value of the number Δ or □ that makes these statements true. 14 − Δ = 8

Answer: Δ = 14 − 8 = 6. Check: 14 − 6 = 8. Answer: Δ = 6

Question: Exercise 8.1, Question 4 (iv): Find the value of the number Δ or □ that makes these statements true. 24 − Δ = 20

Answer: Δ = 24 − 20 = 4. Check: 24 − 4 = 20. Answer: Δ = 4

Question: Exercise 8.1, Question 4 (v): Find the value of the number Δ or □ that makes these statements true. 3□ + 1 = 10

Answer: 3□ = 10 − 1 = 9, so □ = 9 ÷ 3 = 3. Check: 3 × 3 + 1 = 10. Answer: □ = 3

Question: Exercise 8.1, Question 4 (vi): Find the value of the number Δ or □ that makes these statements true. 35 − □ = 29

Answer: □ = 35 − 29 = 6. Check: 35 − 6 = 29. Answer: □ = 6

Question: Exercise 8.2, Question 1 (i): Write the number of terms of these expressions. 3 × m × n

Answer: This is a single term, because terms are separated by "+" or "−". Answer: 1 term

Question: Exercise 8.2, Question 1 (ii): Write the number of terms of these expressions. p

Answer: This is a single term. Answer: 1 term

Question: Exercise 8.2, Question 1 (iii): Write the number of terms of these expressions. 2x + 3y

Answer: The terms are 2x and 3y. Answer: 2 terms

Question: Exercise 8.2, Question 1 (iv): Write the number of terms of these expressions. 4y + 7

Answer: The terms are 4y and 7. Answer: 2 terms

Question: Exercise 8.2, Question 1 (v): Write the number of terms of these expressions. 4ℓ + 3m + 9

Answer: The terms are 4ℓ, 3m and 9. Answer: 3 terms

Question: Exercise 8.2, Question 1 (vi): Write the number of terms of these expressions. 7x + 5 + 8

Answer: The terms are 7x, 5 and 8. Answer: 3 terms

Question: Exercise 8.2, Question 2 (i): Write the variables of these expressions. 8x + 3

Answer: The variable is x. Answer: x

Question: Exercise 8.2, Question 2 (ii): Write the variables of these expressions. −7x + 1

Answer: The variable is x. Answer: x

Question: Exercise 8.2, Question 2 (iii): Write the variables of these expressions. 6x + y

Answer: The variables are x and y. Answer: x, y

Question: Exercise 8.2, Question 2 (iv): Write the variables of these expressions. 2x/y

Answer: The variables are x and y. Answer: x, y

Question: Exercise 8.2, Question 2 (v): Write the variables of these expressions. 2xy

Answer: The variables are x and y. Answer: x, y

Question: Exercise 8.2, Question 2 (vi): Write the variables of these expressions. 5, 3yz

Answer: 5 has no variable; 3yz has variables y and z. Answer: y, z (none in 5)

Question: Exercise 8.2, Question 2 (vii): Write the variables of these expressions. 3xyz

Answer: The variables are x, y and z. Answer: x, y, z

Question: Exercise 8.2, Question 2 (viii): Write the variables of these expressions. 3xy + 9yz

Answer: The variables are x, y and z. Answer: x, y, z

Question: Exercise 8.2, Question 3 (i): Write the constant terms of these expressions. 2x + 2

Answer: The constant term is 2. Answer: 2

Question: Exercise 8.2, Question 3 (ii): Write the constant terms of these expressions. −3 + x

Answer: The constant term is −3. Answer: −3

Question: Exercise 8.2, Question 3 (iii): Write the constant terms of these expressions. xy + 1

Answer: The constant term is 1. Answer: 1

Question: Exercise 8.2, Question 3 (iv): Write the constant terms of these expressions. xy + yz + 1/2

Answer: The constant term is 1/2. Answer: 1/2

Question: Exercise 8.2, Question 3 (v): Write the constant terms of these expressions. 5x + 9

Answer: The constant term is 9. Answer: 9

Question: Exercise 8.2, Question 3 (vi): Write the constant terms of these expressions. 2x + 2y + 10

Answer: The constant term is 10. Answer: 10

Question: Exercise 8.2, Question 4 (i): Write the numerical coefficients of the variables of these expressions. −7x

Answer: The numerical coefficient of x is −7. Answer: −7

Question: Exercise 8.2, Question 4 (ii): Write the numerical coefficients of the variables of these expressions. 5y

Answer: The numerical coefficient of y is 5. Answer: 5

Question: Exercise 8.2, Question 4 (iii): Write the numerical coefficients of the variables of these expressions. 2x + 3y

Answer: The numerical coefficient of x is 2 and of y is 3. Answer: 2, 3

Question: Exercise 8.2, Question 4 (iv): Write the numerical coefficients of the variables of these expressions. x + 1/2 y − 1/4 z

Answer: The numerical coefficient of x is 1, of y is 1/2 and of z is −1/4. Answer: 1, 1/2, −1/4

Question: Exercise 8.2, Question 4 (v): Write the numerical coefficients of the variables of these expressions. 5x + 9

Answer: The numerical coefficient of x is 5 (9 is constant). Answer: 5

Question: Exercise 8.2, Question 4 (vi): Write the numerical coefficients of the variables of these expressions. 1/4 x + 6y + 1

Answer: The numerical coefficient of x is 1/4 and of y is 6 (1 is constant). Answer: 1/4, 6

Question: Exercise 8.2, Question 5: Write the like terms from these. 2xy, 4ℓm, −7xz, 140xy, 13mn, xz, 1/5 xy, −9xyz, 1/4 xy, −2p, 1/3 xy, −2xy, 5xy, xyz, 46p, 5ℓm

Answer: xy-terms: 2xy, 140xy, 1/5 xy, 1/4 xy, 1/3 xy, −2xy, 5xy. ℓm-terms: 4ℓm, 5ℓm. xz-terms: −7xz, xz. xyz-terms: −9xyz, xyz. p-terms: −2p, 46p. 13mn stands alone.

Question: Exercise 8.3, Question 1 (i): Add these expressions. 2x, x, 4x

Answer: 2x + x + 4x = (2 + 1 + 4)x = 7x. Answer: 7x

Question: Exercise 8.3, Question 1 (ii): Add these expressions. 2a, 3a, 6a, a

Answer: 2a + 3a + 6a + a = (2 + 3 + 6 + 1)a = 12a. Answer: 12a

Question: Exercise 8.3, Question 1 (iii): Add these expressions. 8ℓm, 4ℓm, ℓm, 6ℓm

Answer: 8ℓm + 4ℓm + ℓm + 6ℓm = (8 + 4 + 1 + 6)ℓm = 19ℓm. Answer: 19ℓm

Question: Exercise 8.3, Question 1 (iv): Add these expressions. 2xy, 4xy, xy, 6xy, 3xy

Answer: 2xy + 4xy + xy + 6xy + 3xy = (2 + 4 + 1 + 6 + 3)xy = 16xy. Answer: 16xy

Question: Exercise 8.3, Question 2 (i): Bring into simplest form. 2x + 9x

Answer: 2x + 9x = (2 + 9)x = 11x. Answer: 11x

Question: Exercise 8.3, Question 2 (ii): Bring into simplest form. a + 2a + 3a

Answer: a + 2a + 3a = (1 + 2 + 3)a = 6a. Answer: 6a

Question: Exercise 8.3, Question 2 (iii): Bring into simplest form. x + 3x + 6x + 10x

Answer: x + 3x + 6x + 10x = (1 + 3 + 6 + 10)x = 20x. Answer: 20x

Question: Exercise 8.3, Question 2 (iv): Bring into simplest form. 2st + 3st + 5st + 7st

Answer: 2st + 3st + 5st + 7st = (2 + 3 + 5 + 7)st = 17st. Answer: 17st

Question: Exercise 8.3, Question 3 (i): Add. 3a + 2b, a + b, 4a

Answer: (3a + 2b) + (a + b) + 4a = (3a + a + 4a) + (2b + b) = 8a + 3b. Answer: 8a + 3b

Question: Exercise 8.3, Question 3 (ii): Add. 6x + 5y + 7z, 2x + 3y + z, y + 2x

Answer: (6x + 5y + 7z) + (2x + 3y + z) + (y + 2x) = (6x + 2x + 2x) + (5y + 3y + y) + (7z + z) = 10x + 9y + 8z. Answer: 10x + 9y + 8z

Question: Exercise 8.3, Question 3 (iii): Add. pq + qr + pr, qr + 4pr + 2pq, 3pq + 2pr

Answer: (pq + qr + pr) + (qr + 4pr + 2pq) + (3pq + 2pr) = (pq + 2pq + 3pq) + (qr + qr) + (pr + 4pr + 2pr) = 6pq + 2qr + 7pr. Answer: 6pq + 2qr + 7pr

Question: Exercise 8.3, Question 3 (iv): Add. 3c + 4d + 5f, 5c + 7d − 6f

Answer: (3c + 4d + 5f) + (5c + 7d − 6f) = (3c + 5c) + (4d + 7d) + (5f − 6f) = 8c + 11d − f. Answer: 8c + 11d − f

Question: Exercise 8.3, Question 4 (i): Subtract. 5x from 8x

Answer: 8x − 5x = (8 − 5)x = 3x. Answer: 3x

Question: Exercise 8.3, Question 4 (ii): Subtract. −3y from 9y

Answer: 9y − (−3y) = 9y + 3y = 12y. Answer: 12y

Question: Exercise 8.3, Question 4 (iii): Subtract. 2x + 3y from 6x + 8y

Answer: (6x + 8y) − (2x + 3y) = 6x + 8y − 2x − 3y = 4x + 5y. Answer: 4x + 5y

Question: Exercise 8.3, Question 4 (iv): Subtract. Subtract (−20f + 30g + 40) from (20 + 10f + 20g).

Answer: (20 + 10f + 20g) − (−20f + 30g + 40) = 20 + 10f + 20g + 20f − 30g − 40 = −20 + 30f − 10g. Answer: −20 + 30f − 10g

Question: Exercise 8.3, Question 5 (i): Perform the subtraction operation. 8x + 5x

Answer: 8x + 5x = (8 + 5)x = 13x. Answer: 13x

Question: Exercise 8.3, Question 5 (ii): Perform the subtraction operation. 12ab − 9ab

Answer: 12ab − 9ab = (12 − 9)ab = 3ab. Answer: 3ab

Question: Exercise 8.3, Question 5 (iii): Perform the subtraction operation. 5z + 3z

Answer: $5z + 3z = (5 + 3)z = 8z$. Answer: $8z$

Question: Exercise 8.3, Question 5 (iv): Perform the subtraction operation. $10xy - 17xy$

Answer: $10xy - 17xy = (10 - 17)xy = -7xy$. Answer: $-7xy$

Question: Exercise 8.3, Question 5 (v): Perform the subtraction operation. $(2x + 7y) - (3x + 5y)$

Answer: $(2x + 7y) - (3x + 5y) = 2x + 7y - 3x - 5y = -x + 2y$. Answer: $-x + 2y$

Question: Exercise 8.3, Question 5 (vi): Perform the subtraction operation. $(2x + 15xb + 9y) + (x + 20xb + 3y)$

Answer: $(2x + 15xb + 9y) + (x + 20xb + 3y) = (2x + x) + (15xb + 20xb) + (9y + 3y) = 3x + 35xb + 12y$. Answer: $3x + 35xb + 12y$

Question: Exercise 8.4, Question 1 (i): Bring these expressions into simplest form. $[x + x + (y + y + 2x)]$

Answer: $[x + x + (y + y + 2x)] = [2x + 2y + 2x] = 4x + 2y$. Answer: $4x + 2y$

Question: Exercise 8.4, Question 1 (ii): Bring these expressions into simplest form. $[5a - \{3a + (3b - 6b + 4a)\}]$

Answer: From the innermost bracket: $3b - 6b + 4a = 4a - 3b$. Then $\{3a + 4a - 3b\} = 7a - 3b$. Then $5a - (7a - 3b) = 5a - 7a + 3b = -2a + 3b$. Answer: $-2a + 3b$

Question: Exercise 8.4, Question 1 (iii): Bring these expressions into simplest form. $[2a - \{5b - 3(2a + b - 3a)\}]$

Answer: From the innermost bracket: $2a + b - 3a = -a + b$. $3(-a + b) = -3a + 3b$. Then $5b - (-3a + 3b) = 3a + 2b$. Then $2a - (3a + 2b) = -a - 2b$. Answer: $-a - 2b$

Question: Exercise 8.4, Question 1 (iv): Bring these expressions into simplest form. $5\ell - [2 - (3 + 11\ell + 3m + 8\ell)]$

Answer: From the innermost bracket: $3 + 11\ell + 3m + 8\ell = 3 + 19\ell + 3m$. Then $2 - (3 + 19\ell + 3m) = -1 - 19\ell - 3m$. Then $5\ell - (-1 - 19\ell - 3m) = 5\ell + 1 + 19\ell + 3m = 24\ell + 3m + 1$. Answer: $24\ell + 3m + 1$

Question: Exercise 8.4, Question 1 (v): Bring these expressions into simplest form. $2(x + y) - 3[4x + \{7x + 3y + (5x - 4y)\}]$

Answer: From the innermost bracket: $\{7x + 3y + 5x - 4y\} = 12x - y$. Then $[4x + 12x - y] = 16x - y$. $3(16x - y) = 48x - 3y$. Then $2x + 2y - (48x - 3y) = -46x + 5y$. Answer: $-46x + 5y$

Question: Exercise 8.4, Question 2 (i): When $x = 2$, find the values of $3x$ and $-5x$.

Answer: 3x = 3 × 2 = 6. −5x = −5 × 2 = −10. Answer: 6 and −10

Question: Exercise 8.4, Question 2 (ii) (a): When y = −3, find the values of these. −3y

Answer: −3y = −3 × (−3) = 9. Answer: 9

Question: Exercise 8.4, Question 2 (ii) (b): When y = −3, find the values of these. 5y + 7

Answer: 5y + 7 = 5 × (−3) + 7 = −15 + 7 = −8. Answer: −8

Question: Exercise 8.4, Question 2 (ii) (c): When y = −3, find the values of these. 5/3 y

Answer: 5/3 y = 5/3 × (−3) = −5. Answer: −5

Question: Exercise 8.4, Question 2 (ii) (d): When y = −3, find the values of these. 6/24 y + 9

Answer: 6/24 y + 9 = 1/4 × (−3) + 9 = −3/4 + 9 = 8 1/4. Answer: 8 1/4

Question: Exercise 8.4, Question 3 (i): When a = 3, b = 5 and c = 2, find the value of these. a + b + c

Answer: a + b + c = 3 + 5 + 2 = 10. Answer: 10

Question: Exercise 8.4, Question 3 (ii): When a = 3, b = 5 and c = 2, find the value of these. 2a − 3b + c

Answer: 2a − 3b + c = 2 × 3 − 3 × 5 + 2 = 6 − 15 + 2 = −7. Answer: −7

Question: Exercise 8.4, Question 3 (iii): When a = 3, b = 5 and c = 2, find the value of these. (a − bc)/c

Answer: (a − bc)/c = (3 − 5 × 2)/2 = (3 − 10)/2 = −7/2 = −3 1/2. Answer: −3 1/2

Question: Exercise 8.4, Question 3 (iv): When a = 3, b = 5 and c = 2, find the value of these. (2a + b + c)/abc

Answer: (2a + b + c)/abc = (6 + 5 + 2)/(3 × 5 × 2) = 13/30. Answer: 13/30

Question: Exercise 8.4, Question 3 (v): When a = 3, b = 5 and c = 2, find the value of these. (a + b)/c + (b + c)/a

Answer: (a + b)/c + (b + c)/a = (3 + 5)/2 + (5 + 2)/3 = 8/2 + 7/3 = 4 + 7/3 = 19/3 = 6 1/3. Answer: 6 1/3

Question: Exercise 8.4, Question 3 (vi): When a = 3, b = 5 and c = 2, find the value of these. (2a + b − c)/(a − 3b + c)

Answer: (2a + b − c)/(a − 3b + c) = (6 + 5 − 2)/(3 − 15 + 2) = 9/(−10) = −9/10. Answer: −9/10

Question: Exercise 8.4, Question 3 (vii): When a = 3, b = 5 and c = 2, find the value of these. a(2b + 3c)/3ca

Answer: a(2b + 3c)/3ca = 3 × (10 + 6)/(3 × 2 × 3) = 48/18 = 8/3 = 2 2/3. Answer: 2 2/3

Question: Exercise 8.4, Question 3 (viii): When a = 3, b = 5 and c = 2, find the value of these. a + 3abc − b + c

Answer: a + 3abc − b + c = 3 + 3 × 3 × 5 × 2 − 5 + 2 = 3 + 90 − 5 + 2 = 90. Answer: 90

Question: Exercise 8.4, Question 3 (ix): When a = 3, b = 5 and c = 2, find the value of these. (3a + 4b + 5c)/(a + b − c)

Answer: (3a + 4b + 5c)/(a + b − c) = (9 + 20 + 10)/(3 + 5 − 2) = 39/6 = 13/2 = 6 1/2. Answer: 6 1/2

Question: Exercise 8.4, Question 3 (x): When a = 3, b = 5 and c = 2, find the value of these. a/b − b/c + c/a − a/c

Answer: a/b − b/c + c/a − a/c = 3/5 − 5/2 + 2/3 − 3/2 = 18/30 − 75/30 + 20/30 − 45/30 = −82/30 = −41/15 = −2 11/15. Answer: −2 11/15

Question: Exercise 8.4, Question 4: When x = 2, y = 3, find the value of these expressions. 8x + [3y − {6x + (5 − 4x)}]

Answer: 8x = 16, 3y = 9, 6x = 12, (5 − 4x) = 5 − 8 = −3. {12 + (−3)} = 9. [9 − 9] = 0. Total = 16 + 0 = 16. Answer: 16

Question: Exercise 8.4, Question 5 (i): Abid bought 5 bags at x rupees per bag. After paying the price, 3 rupees remained with him. Find the price of 5 bags.

Answer: Price of one bag = x rupees. Price of 5 bags = 5 × x = 5x rupees. Answer: 5x rupees

Question: Exercise 8.4, Question 5 (ii): Abid bought 5 bags at x rupees per bag. After paying the price, 3 rupees remained with him. Write the algebraic expression for paying the price of 5 bags.

Answer: Total money with Abid = price of bags + remaining money = 5x + 3. Answer: 5x + 3 rupees

Question: Exercise 8.4, Question 5 (iii): Abid bought 5 bags at x rupees per bag. After paying the price, 3 rupees remained with him. When x = 30, find the total money that Abid had.

Answer: Total money = 5x + 3 = 5 × 30 + 3 = 150 + 3 = 153 rupees. Answer: 153 rupees

Question: Review Exercise 8, Question 1 (i): Write in symbols. The price of x books, when the price of one book is 18 rupees.

Answer: Price of x books = 18 × x = 18x rupees. Answer: 18x

Question: Review Exercise 8, Question 1 (ii): Write in symbols. The sum of 5a and 9b.

Answer: 5a + 9b. Answer: 5a + 9b

Question: Review Exercise 8, Question 1 (iii): Write in symbols. 3 times x plus 5.

Answer: 3x + 5. Answer: 3x + 5

Question: Review Exercise 8, Question 1 (iv): Write in symbols. From the five of y, cut on leaves.

Answer: The book's printed words do not join up; the intended reading is "5 less than y". Answer: y − 5

Question: Review Exercise 8, Question 1 (v): Write in symbols. 6 more than p.

Answer: p + 6. Answer: p + 6

Question: Review Exercise 8, Question 2 (i): Write these in words. x + 2

Answer: 2 added to a number x. Answer: 2 added to x

Question: Review Exercise 8, Question 2 (ii): Write these in words. 3y − 4

Answer: 4 subtracted from 3y. Answer: 4 subtracted from 3y

Question: Review Exercise 8, Question 2 (iii): Write these in words. x + 5

Answer: 5 added to a number x. Answer: 5 added to x

Question: Review Exercise 8, Question 2 (iv): Write these in words. s + 2t

Answer: 2t added to s. Answer: 2t added to s

Question: Review Exercise 8, Question 2 (v): Write these in words. (x − y)/2

Answer: Half of the difference of x and y. Answer: Half of the difference of x and y

Question: Review Exercise 8, Question 3 (i): Find the number of terms of each expression below. 2a + 5b

Answer: The terms are 2a and 5b. Answer: 2 terms

Question: Review Exercise 8, Question 3 (ii): Find the number of terms of each expression below. 16x

Answer: This is a single term. Answer: 1 term

Question: Review Exercise 8, Question 3 (iii): Find the number of terms of each expression below. 2bc + 2a − c

Answer: The terms are 2bc, 2a and c. Answer: 3 terms

Question: Review Exercise 8, Question 3 (iv): Find the number of terms of each expression below. 3xy − 8x + 6y − 2

Answer: The terms are 3xy, 8x, 6y and 2. Answer: 4 terms

Question: Review Exercise 8, Question 3 (v): Find the number of terms of each expression below. 6x, 5y + 2z − 6

Answer: 6x has 1 term. 5y + 2z − 6 has terms 5y, 2z and 6, i.e. 3 terms. Answer: 1 term and 3 terms

Question: Review Exercise 8, Question 4: Write expressions with one term, two terms and three terms separately.

Answer: One-term expression: 5x. Two-term expression: 3x + 2y. Three-term expression: 2x + 3y + 4z. (The book gives no fixed expressions, so these are examples.)

Question: Review Exercise 8, Question 5 (i): Bring into simplest form. 4xy + 2xy + 2/3 xy

Answer: 4xy + 2xy + 2/3 xy = (4 + 2 + 2/3)xy = 20/3 xy = 6 2/3 xy. Answer: 6 2/3 xy

Question: Review Exercise 8, Question 5 (ii): Bring into simplest form. 9ab + 3ab + 3/5 ab

Answer: 9ab + 3ab + 3/5 ab = (9 + 3 + 3/5)ab = 63/5 ab = 12 3/5 ab. Answer: 12 3/5 ab

Question: Review Exercise 8, Question 6 (i): Add. 3x + 4y, 5x + 9y

Answer: (3x + 4y) + (5x + 9y) = (3x + 5x) + (4y + 9y) = 8x + 13y. Answer: 8x + 13y

Question: Review Exercise 8, Question 6 (ii): Add. 2d + 5c + 3b, 7c + 5d + 9b

Answer: (2d + 5c + 3b) + (7c + 5d + 9b) = (2d + 5d) + (5c + 7c) + (3b + 9b) = 7d + 12c + 12b. Answer: 7d + 12c + 12b

Question: Review Exercise 8, Question 6 (iii): Add. 12xy + 3x + 4y, 5x + 6y + 8xy

Answer: (12xy + 3x + 4y) + (5x + 6y + 8xy) = (12xy + 8xy) + (3x + 5x) + (4y + 6y) = 20xy + 8x + 10y. Answer: 20xy + 8x + 10y

Question: Review Exercise 8, Question 7: Subtract x + 3y from 2x + 7y.

Answer: (2x + 7y) − (x + 3y) = 2x + 7y − x − 3y = x + 4y. Answer: x + 4y

Question: Review Exercise 8, Question 8: Subtract x + 3y + 5z from 2x − 15y − 9z.

Answer: (2x − 15y − 9z) − (x + 3y + 5z) = 2x − 15y − 9z − x − 3y − 5z = x − 18y − 14z. Answer: x − 18y − 14z

Question: Review Exercise 8, Question 9 (i): Which of these are true or false statements. 13 − 5 = 3

Answer: 13 − 5 = 8, not 3. Answer: False ✗

Question: Review Exercise 8, Question 9 (ii): Which of these are true or false statements. 13 − 5 = 5

Answer: 13 − 5 = 8, not 5. Answer: False ✗

Question: Review Exercise 8, Question 9 (iii): Which of these are true or false statements. 13 − 5 = 8

Answer: 13 − 5 = 8. Answer: True ✓

Question: Review Exercise 8, Question 10 (i): Find the value of the variable that makes the statement true. x = 15 − 6

Answer: x = 15 − 6 = 9. Answer: x = 9

Question: Review Exercise 8, Question 10 (ii): Find the value of the variable that makes the statement true. y − 8 = 5

Answer: y = 5 + 8 = 13. Check: 13 − 8 = 5. Answer: y = 13

Question: Review Exercise 8, Question 10 (iii): Find the value of the variable that makes the statement true. 14 = b + 8

Answer: b = 14 − 8 = 6. Check: 6 + 8 = 14. Answer: b = 6

Question: Review Exercise 8, Question 10 (iv): Find the value of the variable that makes the statement true. 4x − 10 = 2

Answer: 4x = 2 + 10 = 12, so x = 12 ÷ 4 = 3. Check: 4 × 3 − 10 = 2. Answer: x = 3

Question: Review Exercise 8, Question 10 (v): Find the value of the variable that makes the statement true. 12 − 2m = 8

Answer: 2m = 12 − 8 = 4, so m = 4 ÷ 2 = 2. Check: 12 − 2 × 2 = 8. Answer: m = 2

9. Linear Equations

Question: Exercise 9.1, Question 1: Form one-degree equations from these. The difference of a number and 20 is two.

Answer: Suppose the required number is x. x − 20 = 2. Answer: x − 20 = 2

Question: Exercise 9.1, Question 2: Form one-degree equations from these. The sum of a number and 8 is 15.

Answer: Suppose the required number is x. x + 8 = 15. Answer: x + 8 = 15

Question: Exercise 9.1, Question 3: Form one-degree equations from these. A number multiplied by 5 is 35.

Answer: Suppose the required number is x. 5x = 35. Answer: 5x = 35

Question: Exercise 9.1, Question 4: Form one-degree equations from these. Sixteen divided by four equals 4.

Answer: 16 ÷ 4 = 4. (This printed item has no variable.) Answer: 16 ÷ 4 = 4

Question: Exercise 9.1, Question 5: Form one-degree equations from these. Adding two to a number gives the answer 4.

Answer: Suppose the required number is x. x + 2 = 4. Answer: x + 2 = 4

Question: Exercise 9.1, Question 6: Form one-degree equations from these. Subtracting fifteen from a number gives the answer 5.

Answer: Suppose the required number is x. x − 15 = 5. Answer: x − 15 = 5

Question: Exercise 9.1, Question 7: Form one-degree equations from these. Adding 4 to twice a number gives 20.

Answer: Suppose the required number is x. 2x + 4 = 20. Answer: 2x + 4 = 20

Question: Exercise 9.1, Question 8: Form one-degree equations from these. My age is fifteen years more than my father's age.

Answer: Suppose my age is m and my father's age is f. m = f + 15. Answer: m = f + 15

Question: Exercise 9.2, Question 1: Solve these equations. x − 7 = 8

Answer: Adding 7 to both sides: x − 7 + 7 = 8 + 7, so x = 15. Check: 15 −

7 = 8. Answer: x = 15

Question: Exercise 9.2, Question 2: Solve these equations. y + 5 = 12

Answer: Subtracting 5 from both sides: y + 5 − 5 = 12 − 5, so y = 7. Check: 7 + 5 = 12. Answer: y = 7

Question: Exercise 9.2, Question 3: Solve these equations. x + 2 = 8

Answer: Subtracting 2 from both sides: x + 2 − 2 = 8 − 2, so x = 6. Check: 6 + 2 = 8. Answer: x = 6

Question: Exercise 9.2, Question 4: Solve these equations. 7y = 14

Answer: Dividing both sides by 7: 7y/7 = 14/7, so y = 2. Check: 7 × 2 = 14. Answer: y = 2

Question: Exercise 9.2, Question 5: Solve these equations. 3x = 30

Answer: Dividing both sides by 3: 3x/3 = 30/3, so x = 10. Check: 3 × 10 = 30. Answer: x = 10

Question: Exercise 9.2, Question 6: Solve these equations. y − 16 = 2

Answer: Adding 16 to both sides: y − 16 + 16 = 2 + 16, so y = 18. Check: 18 − 16 = 2. Answer: y = 18

Question: Exercise 9.2, Question 7: Solve these equations. x/4 = 3

Answer: Multiplying both sides by 4: x/4 × 4 = 3 × 4, so x = 12. Check: 12/4 = 3. Answer: x = 12

Question: Exercise 9.2, Question 8: Solve these equations. 6x + 1 = 49

Answer: Subtracting 1 from both sides: 6x = 48. Dividing both sides by 6: x = 48 ÷ 6 = 8. Check: 6 × 8 + 1 = 49. Answer: x = 8

Question: Exercise 9.2, Question 9: Solve these equations. 8x = 32

Answer: Dividing both sides by 8: 8x/8 = 32/8, so x = 4. Check: 8 × 4 = 32. Answer: x = 4

Question: Exercise 9.2, Question 10: Solve these equations. 7y + 2 = 44

Answer: Subtracting 2 from both sides: 7y = 42. Dividing both sides by 7: y = 42 ÷ 7 = 6. Check: 7 × 6 + 2 = 44. Answer: y = 6

Question: Exercise 9.2, Question 11: Solve these equations. 6 = x + 9

Answer: Subtracting 9 from both sides: 6 − 9 = x, so x = −3. Check: −3 + 9 = 6. Answer: x = −3

Question: Exercise 9.2, Question 12: Solve these equations. 9x − 21 = 78

Answer: Adding 21 to both sides: 9x = 99. Dividing both sides by 9: x = 99 ÷ 9 = 11. Check: 9 × 11 − 21 = 78. Answer: x = 11

Question: Exercise 9.2, Question 13: Solve these equations. (6x − 4)/(2x + 2) = 2

Answer: Multiplying both sides by (2x + 2): 6x − 4 = 2(2x + 2) = 4x + 4. Subtracting 4x from both sides: 2x − 4 = 4. Adding 4 to both sides: 2x = 8, so x = 4. Check: (24 − 4)/(8 + 2) = 20/10 = 2. Answer: x = 4

Question: Exercise 9.2, Question 14: Solve these equations. x/2 + 4 = x − 1/3

Answer: Multiplying both sides by 6: 3x + 24 = 6x − 2. Subtracting 3x from both sides: 24 = 3x − 2. Adding 2 to both sides: 26 = 3x, so x = 26/3 = 8 2/3. Check: 13/3 + 4 = 25/3 and 26/3 − 1/3 = 25/3. Answer: x = 8 2/3

Question: Exercise 9.2, Question 15: Solve these equations. (5x − 4)/8 = (x + 6)/4

Answer: Multiplying both sides by 8: 5x − 4 = 2(x + 6) = 2x + 12. Subtracting 2x from both sides: 3x − 4 = 12. Adding 4 to both sides: 3x = 16, so x = 16/3 = 5 1/3. Check: (80/3 − 4)/8 = 17/6 and (16/3 + 6)/4 = 17/6. Answer: x = 5 1/3

Question: Exercise 9.3, Question 1: 44 pieces of glass are sewn in a red ghagra and a green choli. 6 more pieces of glass are sewn in the ghagra than in the green one. Find the number of pieces sewn in the green choli.

Answer: Suppose the pieces in the green choli are x. The pieces in the ghagra are x + 6. Equation: x + (x + 6) = 44. 2x + 6 = 44, so 2x = 38 and x = 19. Answer: 19 pieces in the green choli (and 25 pieces in the ghagra)

Question: Exercise 9.3, Question 2: In a cricket match, Jamil and Salim raised the team's score by 84 runs. If Jamil scores 12 runs more than Salim, find Salim's score.

Answer: Suppose Salim's score is x. Jamil's score is x + 12. Equation: x + (x + 12) = 84. 2x + 12 = 84, so 2x = 72 and x = 36. Answer: Salim's score is 36 runs (Jamil's is 48 runs)

Question: Exercise 9.3, Question 3: The sum of a number and its double is 9. Find the number.

Answer: Suppose the required number is x. Equation: x + 2x = 9. 3x = 9, so x = 3. Answer: 3

Question: Exercise 9.3, Question 4: Multiplying the sum of a number and 6 by 7 gives 77. Find the number.

Answer: Suppose the required number is x. Equation: 7(x + 6) = 77. Dividing both sides by 7: x + 6 = 11, so x = 5. Answer: 5

Question: Exercise 9.3, Question 5: The sum of a quarter of a number and 6 is 42. Find the number.

Answer: Suppose the required number is x. Equation: x/4 + 6 = 42. Subtracting 6 from both sides: x/4 = 36, so x = 144. Answer: 144

Question: Exercise 9.3, Question 6: The sum of the ages of a mother and her daughter is 22 years. The daughter is 20 years younger than the mother. Find the daughter's age.

Answer: Suppose the daughter's age is x years. The mother's age is x + 20 years. Equation: x + (x + 20) = 22. 2x + 20 = 22, so 2x = 2 and x = 1. Answer: The daughter is 1 year old (mother is 21 years)

Question: Exercise 9.3, Question 7: When Dua opens a book, two pages face her. The sum of the page numbers is 155. If the number of one page is 50, find the number of the other page.

Answer: Facing pages have consecutive numbers. Suppose one page is x, the other is x + 1. Equation: x + (x + 1) = 155. 2x + 1 = 155, so 2x = 154 and x = 77. The pages are 77 and 78. Note: the printed condition "one page's number is 50" does not go with the sum 155, because 50 + 51 = 101, not 155; so the "50" part is wrong in the printed question.

Question: Exercise 9.3, Question 8: The price of a toy has decreased by 7 rupees. If the new price is 25 rupees, find the original price.

Answer: Suppose the original price is x rupees. Equation: x − 7 = 25. Adding 7 to both sides: x = 32. Answer: 32 rupees

Question: Review Exercise 9, Question 1 (i): Write the algebraic expressions of these. The sum of x and 4

Answer: x + 4. Answer: x + 4

Question: Review Exercise 9, Question 1 (ii): Write the algebraic expressions of these. 7 less than a number x

Answer: x − 7. Answer: x − 7

Question: Review Exercise 9, Question 1 (iii): Write the algebraic expressions of these. A number x multiplied by 9

Answer: 9x. Answer: 9x

Question: Review Exercise 9, Question 1 (iv): Write the algebraic expressions of these. A number y divided by 6

Answer: y/6. Answer: y/6

Question: Review Exercise 9, Question 1 (v): Write the algebraic expressions of these. The sum of two numbers x and y

Answer: x + y. Answer: x + y

Question: Review Exercise 9, Question 1 (vi): Write the algebraic expressions of these. The product of a variable x and 8

Answer: 8x. Answer: 8x

Question: Review Exercise 9, Question 2 (i): Write an equation for each of these. Subtracting 3 from a number x gives 10.

Answer: x − 3 = 10. Answer: x − 3 = 10

Question: Review Exercise 9, Question 2 (ii): Write an equation for each of these. Adding 5 to three times a number y gives 17.

Answer: 3y + 5 = 17. Answer: 3y + 5 = 17

Question: Review Exercise 9, Question 2 (iii): Write an equation for each of these. Twice a number z plus 4 is 20.

Answer: 2z + 4 = 20. Answer: 2z + 4 = 20

Question: Review Exercise 9, Question 2 (iv): Write an equation for each of these. Subtracting half of a number m from 7 gives 4.

Answer: 7 − m/2 = 4. Answer: 7 − m/2 = 4

Question: Review Exercise 9, Question 2 (v): Write an equation for each of these. The sum of two consecutive even numbers is 6.

Answer: Suppose the smaller even number is x, the other is x + 2. x + (x + 2) = 6. Answer: x + (x + 2) = 6

Question: Review Exercise 9, Question 2 (vi): Write an equation for each of these. After 8 years my age will be three times my present age.

Answer: Suppose the present age is x. After 8 years the age will be x + 8 and it will be three times the present age: x + 8 = 3x. Answer: x + 8 = 3x

Question: Review Exercise 9, Question 3 (i): Solve these equations. 5x − 3 = 18

Answer: Adding 3 to both sides: 5x = 21. Dividing both sides by 5: x = 21/5 = 4 1/5. Check: 5 × 21/5 − 3 = 18. Answer: x = 4 1/5

Question: Review Exercise 9, Question 3 (ii): Solve these equations. (y + 1)/2 = 5

Answer: Multiplying both sides by 2: y + 1 = 10. Subtracting 1 from both sides: y = 9. Check: (9 + 1)/2 = 5. Answer: y = 9

Question: Review Exercise 9, Question 3 (iii): Solve these equations. (m − 1)/3 = 1

Answer: Multiplying both sides by 3: m − 1 = 3. Adding 1 to both sides: m = 4. Check: (4 − 1)/3 = 1. Answer: m = 4

Question: Review Exercise 9, Question 3 (iv): Solve these equations. 4y − 10 = 1

Answer: Adding 10 to both sides: 4y = 11. Dividing both sides by 4: y = 11/4 = 2 3/4. Check: 4 × 11/4 − 10 = 1. Answer: y = 2 3/4

Question: Review Exercise 9, Question 3 (v): Solve these equations. p/7 − 3 = 2

Answer: Adding 3 to both sides: p/7 = 5. Multiplying both sides by 7: p = 35. Check: 35/7 − 3 = 2. Answer: p = 35

Question: Review Exercise 9, Question 3 (vi): Solve these equations. 1/2 q = 3

Answer: Multiplying both sides by 2: q = 6. Check: 1/2 × 6 = 3. Answer: q = 6

Question: Review Exercise 9, Question 4: A rope n metres long is cut into two parts. One part is 3 metres long and the other part is 2x − 17 metres long. Find the original length of the rope.

Answer: Original length = sum of both parts = 3 + (2x − 17) = 2x − 14 metres. The book does not give the value of x, so no numerical answer for the original length can be found from the printed data.

Question: Review Exercise 9, Question 5: If a number is tripled and 5 is added to the answer, we get 44. Find the number.

Answer: Suppose the required number is x. Equation: 3x + 5 = 44. Subtracting 5 from both sides: 3x = 39, so x = 13. Answer: 13

Question: Review Exercise 9, Question 6: Adding half to twice a number gives 20. Find the number.

Answer: Suppose the required number is x. Equation: 2x + x/2 = 20. Multiplying both sides by 2: 4x + x = 40, i.e. 5x = 40, so x = 8. Answer: 8

Question: Review Exercise 9, Question 7: Subtracting 5 from three times a number gives one more than the original number. Find the number.

Answer: Suppose the required number is x. Equation: 3x − 5 = x + 1. Subtracting x from both sides: 2x − 5 = 1. Adding 5 to both sides: 2x = 6, so x = 3. Answer: 3

Question: Review Exercise 9, Question 8: The sum of two consecutive even numbers is 34. Find the smaller of them.

Answer: Suppose the smaller even number is x, the other is x + 2. Equation: x + (x + 2) = 34. 2x + 2 = 34, so 2x = 32 and x = 16. Answer: 16

10. Geometry

Question: Exercise 10.1, Question 1 (i): Join the given line segments together by geometry, and find their sum, i.e. the total measure. PQ = 2 cm and RS = 3 cm.

Answer: Draw a straight line and mark a point A on it. Open the compass to PQ (2 cm) and, with A as centre, draw an arc cutting the line at point B. Now open the compass to RS (3 cm) and, with B as centre, draw an arc cutting the line at point C. AC is the required joined line segment. Measure: mAC = mPQ + mRS = 2 cm + 3 cm = 5 cm.

Question: Exercise 10.1, Question 1 (ii): Join the given line segments together by geometry, and find their sum, i.e. the total measure. AB = 3.6 cm and CD = 4.7 cm.

Answer: Mark a point A on a straight line. Open the compass to AB (3.6 cm) and, with A as centre, draw an arc to get point B. Then open the compass to CD (4.7 cm) and, with B as centre, draw an arc to get point C. AC is the required joined line segment. Measure: mAC = mAB + mCD = 3.6 cm + 4.7 cm = 8.3 cm.

Question: Exercise 10.1, Question 1 (iii): Join the given line segments together by geometry, and find their sum, i.e. the total measure. EF = 2.3 cm, GH = 4 cm and IJ = 1.5 cm.

Answer: Mark a point A on a straight line. With the compass take EF (2.3 cm) and draw an arc from A to get point B. Then take GH (4 cm) and draw an arc from B to get point C. Then take IJ (1.5 cm) and draw an arc from C to get point D. AD is the required joined line segment. Measure: mAD = mEF + mGH + mIJ = 2.3 cm + 4 cm + 1.5 cm = 7.8 cm.

Question: Exercise 10.1, Question 1 (iv): Join the given line segments together by geometry, and find their sum, i.e. the total measure. JR = 3.2 cm, LM = 2.5 cm and NO = 4.3 cm.

Answer: Mark a point A on a straight line. With the compass take JR (3.2 cm) and draw an arc from A to get point B. Then take LM (2.5 cm) and draw an arc from B to get point C. Then take NO (4.3 cm) and draw an arc from C to get point D. AD is the required joined line segment. Measure: mAD = mJR + mLM + mNO = 3.2 cm + 2.5 cm + 4.3 cm = 10.0 cm.

Question: Exercise 10.1, Question 2 (i): Find the difference in the measures of the given line segments by geometry. LM = 2 cm and OP = 4 cm.

Answer: Mark a point A on a straight line. With the compass take the larger segment OP (4 cm) and draw an arc from A to get point B. Then open the compass to the smaller segment LM (2 cm) and, with A as centre, draw an arc to get point C. CB is the remaining difference segment. Measure: mCB = mOP − mLM = 4 cm − 2 cm = 2 cm.

Question: Exercise 10.1, Question 2 (ii): Find the difference in the measures of the given line segments by geometry. PQ = 3.5 cm and RS = 5 cm.

Answer: Mark a point A on a straight line. With the compass take the larger segment RS (5 cm) and draw an arc from A to get point B. Then open the compass to PQ (3.5 cm) and, with A as centre, draw an arc to get point C. CB is the difference segment. Measure: mCB = mRS − mPQ = 5 cm − 3.5 cm = 1.5 cm.

Question: Exercise 10.1, Question 2 (iii): Find the difference in the measures of the given line segments by geometry. AB = 4.3 cm and CD = 6.4 cm.

Answer: Mark a point A on a straight line. With the compass take the larger segment CD (6.4 cm) and draw an arc from A to get point B. Then open the compass to AB (4.3 cm) and, with A as centre, draw an arc to get point C. CB is the difference segment. Measure: mCB = mCD − mAB = 6.4 cm − 4.3 cm = 2.1 cm.

Question: Exercise 10.1, Question 2 (iv): Find the difference in the measures of the given line segments by geometry. JK = 4 cm and LM = 5.8 cm.

Answer: Mark a point A on a straight line. With the compass take the larger segment LM (5.8 cm) and draw an arc from A to get point B. Then open the compass to JK (4 cm) and, with A as centre, draw an arc to get point C. CB is the difference segment. Measure: mCB = mLM − mJK = 5.8 cm − 4 cm = 1.8 cm.

Question: Exercise 10.1, Question 2 (v): Find the difference in the measures of the given line segments by geometry. LM = 6.7 cm and LN = 5.2 cm.

Answer: Mark a point A on a straight line. With the compass take the larger segment LM (6.7 cm) and draw an arc from A to get point B. Then open the compass to LN (5.2 cm) and, with A as centre, draw an arc to get point C. CB is the difference segment. Measure: mCB = mLM − mLN = 6.7 cm − 5.2 cm = 1.5 cm.

Question: Exercise 10.1, Question 3 (i): Draw line segments of the given measures and construct their perpendicular bisectors with the help of a compass. 5 cm.

Answer: First draw a line segment AB of 5 cm with a scale. Then open the compass to more than half of 5 cm. With A as centre, draw two arcs above and below the segment. With the same opening and B as centre, draw two more arcs above and below cutting the first arcs at two points. Join those two points with a line. This line is the perpendicular bisector of AB and divides AB at its midpoint into two equal parts (each 2.5 cm).

Question: Exercise 10.1, Question 3 (ii): Draw line segments of the given measures and construct their perpendicular bisectors with the help of a compass. 6.3 cm.

Answer: First draw a line segment AB of 6.3 cm with a scale. Then open

the compass to more than half of 6.3 cm (3.15 cm). With A as centre, draw two arcs above and below the segment. With the same opening and B as centre, draw two more arcs above and below cutting the first arcs at two points. Join those two points with a line. This line is the perpendicular bisector of AB and divides AB at its midpoint into two equal parts (each 3.15 cm).

Question: Exercise 10.1, Question 3 (iii): Draw line segments of the given measures and construct their perpendicular bisectors with the help of a compass. 8 cm.

Answer: First draw a line segment AB of 8 cm with a scale. Then open the compass to more than half of 8 cm (4 cm). With A as centre, draw two arcs above and below the segment. With the same opening and B as centre, draw two more arcs above and below cutting the first arcs at two points. Join those two points with a line. This line is the perpendicular bisector of AB and divides AB at its midpoint into two equal parts (each 4 cm).

Question: Exercise 10.1, Question 4 (i): Construct a perpendicular from the given point on the given line segment with the help of a compass. Point C on line segment AB.

Answer: Open the compass to a suitable radius and, with C as centre, draw an arc cutting AB at two points P and Q. Then open the compass to more than half of PQ. With P as centre draw an arc above AB, and with the same opening and Q as centre draw another arc cutting the first arc at point R. Join C and R with a line. CR is the required perpendicular to AB from C.

Question: Exercise 10.1, Question 4 (ii): Construct a perpendicular from the given point on the given line segment with the help of a compass. Point F on line segment DE.

Answer: Open the compass to a suitable radius and, with F as centre, draw an arc cutting DE at two points P and Q. Then open the compass to more than half of PQ. With P as centre draw an arc on one side of DE, and with the same opening and Q as centre draw another arc cutting the first arc at point R. Join F and R with a line. FR is the required perpendicular to DE from F.

Question: Exercise 10.1, Question 4 (iii): Construct a perpendicular from the given point on the given line segment with the help of a compass. Point K on line segment IJ.

Answer: Open the compass to a suitable radius and, with K as centre, draw an arc cutting IJ at two points P and Q. Then open the compass to more than half of PQ. With P as centre draw an arc on one side of IJ, and with the same opening and Q as centre draw another arc cutting the first arc at point R. Join K and R with a line. KR is the required perpendicular to IJ from K.

Question: Exercise 10.1, Question 5 (i): Construct a perpendicular from a point outside the given line segment with the help of a compass. Line segment AB, outside point C.

Answer: Open the compass wide enough that an arc drawn with C as centre cuts AB at two points P and Q. Then open the compass to more

than half of PQ. With P as centre draw an arc on the side of AB opposite to C, and with the same opening and Q as centre draw another arc cutting the first arc at point R. Join C and R with a line. CR is the required perpendicular to AB from C.

Question: Exercise 10.1, Question 5 (ii): Construct a perpendicular from a point outside the given line segment with the help of a compass. Line segment DE, outside point F.

Answer: Open the compass wide enough that an arc drawn with F as centre cuts DE at two points P and Q. Then open the compass to more than half of PQ. With P as centre draw an arc on the side of DE opposite to F, and with the same opening and Q as centre draw another arc cutting the first arc at point R. Join F and R with a line. FR is the required perpendicular to DE from F.

Question: Exercise 10.1, Question 5 (iii): Construct a perpendicular from a point outside the given line segment with the help of a compass. Line segment IJ, outside point K.

Answer: Open the compass wide enough that an arc drawn with K as centre cuts IJ at two points P and Q. Then open the compass to more than half of PQ. With P as centre draw an arc on the side of IJ opposite to K, and with the same opening and Q as centre draw another arc cutting the first arc at point R. Join K and R with a line. KR is the required perpendicular to IJ from K.

Question: Exercise 10.2, Question 1 (i): Bisect the following angles with the help of a compass. Angle ABC = 80°.

Answer: With the vertex B of the angle as centre, draw an arc of a suitable radius cutting both arms of the angle at points P and Q. Then open the compass to more than half of PQ. With P as centre draw an arc inside the angle, and with the same opening and Q as centre draw another arc cutting the first arc at point R. Join B and R with a ray. This ray is the bisector of angle ABC and divides the angle into two equal parts: each part 80° ÷ 2 = 40°.

Question: Exercise 10.2, Question 1 (ii): Bisect the following angles with the help of a compass. Angle ABC = 110°.

Answer: With the vertex B as centre, draw an arc of a suitable radius cutting both arms of the angle at points P and Q. Then open the compass to more than half of PQ. With P and Q as centres draw two arcs inside the angle cutting each other at point R. Join B and R with a ray. This ray is the bisector of angle ABC: each part 110° ÷ 2 = 55°.

Question: Exercise 10.2, Question 1 (iii): Bisect the following angles with the help of a compass. Angle ABC = 125°.

Answer: With the vertex B as centre, draw an arc of a suitable radius cutting both arms of the angle at points P and Q. Then open the compass to more than half of PQ. With P and Q as centres draw two arcs inside the angle cutting each other at point R. Join B and R with a ray. This ray is the bisector of angle ABC: each part 125° ÷ 2 = 62.5°.

Question: Exercise 10.2, Question 2 (i): Bisect the given angles with a compass, and then bisect one part again to divide the angles into four equal parts. Angle ABC = 95°.

Answer: First bisect angle ABC by the book's method: equal arcs on both arms and a ray through the crossing of two arcs inside. This divides the angle into two equal parts (each 95° ÷ 2 = 47.5°). Then bisect each of these two parts again. This divides the angle into four equal parts: each part 95° ÷ 4 = 23.75°.

Question: Exercise 10.2, Question 2 (ii): Bisect the given angles with a compass, and then bisect one part again to divide the angles into four equal parts. Angle ABC = 115°.

Answer: First bisect angle ABC: the angle is divided into two equal parts (each 115° ÷ 2 = 57.5°). Then bisect each part again. This divides the angle into four equal parts: each part 115° ÷ 4 = 28.75°.

Question: Exercise 10.2, Question 2 (iii): Bisect the given angles with a compass, and then bisect one part again to divide the angles into four equal parts. Angle ABC = 145°.

Answer: First bisect angle ABC: the angle is divided into two equal parts (each 145° ÷ 2 = 72.5°). Then bisect each part again. This divides the angle into four equal parts: each part 145° ÷ 4 = 36.25°.

Question: Exercise 10.2, Question 3 (i): First construct angles equal in measure to the given angles with the help of a compass. Then state the measures of the bisected angles. Angle ABC (given in the figure).

Answer: The book prints no numerical measure for this angle; the student first measures angle ABC in the printed figure with a protractor (in the printed figure it is approximately 45°). Then construct an equal angle with a compass: draw a ray, draw an arc with the vertex as centre, transfer the same arc to the other ray, take the distance between the arm-cut points and draw the second ray. Then bisect that angle. Measure of the bisected angle = approximately 45° ÷ 2 = 22.5° (22½°).

Question: Exercise 10.2, Question 3 (ii): First construct angles equal in measure to the given angles with the help of a compass. Then state the measures of the bisected angles. Angle EFG (given in the figure).

Answer: The book prints no numerical measure for this angle; the student first measures angle EFG in the printed figure with a protractor (in the printed figure it is approximately 30°). Then construct an equal angle with a compass (by the method above) and bisect it. Measure of the bisected angle = approximately 30° ÷ 2 = 15°.

Question: Exercise 10.2, Question 3 (iii): First construct angles equal in measure to the given angles with the help of a compass. Then state the measures of the bisected angles. Angle RST (given in the figure).

Answer: The angle RST given in the figure is a right (90°) angle. Construct an equal 90° angle with a compass (by the method of making 90° with two 60° arcs on a ray) and bisect it. Measure of the bisected angle = 90° ÷ 2 = 45°.

Question: Exercise 10.2, Question 4 (i): Construct angles of the given measures with the help of a compass. 30°.

Answer: Draw a ray OP. With O as centre, draw an arc of a suitable radius cutting OP at A. With the same radius and A as centre, draw an arc cutting the first arc at B (this gives a 60° angle). Then bisect angle AOB: with A and B as centres draw two arcs inside, and join O to their crossing point with ray OR. m∠POR = 30°.

Question: Exercise 10.2, Question 4 (ii): Construct angles of the given measures with the help of a compass. (22½)°.

Answer: First construct a 90° angle by the method above. Then bisect it to get a 45° angle. Then bisect the 45° angle again. The resulting angle = 45° ÷ 2 = (22½)°.

Question: Exercise 10.2, Question 4 (iii): Construct angles of the given measures with the help of a compass. (67½)°.

Answer: First construct a 90° angle and bisect it to get a 45° angle. Then bisect the 45° angle to get a (22½)° angle. Then add 45° and (22½)°: 45° + (22½)° = (67½)°. Alternatively: take away a (22½)° angle from the 90° angle: 90° − (22½)° = (67½)°.

Question: Exercise 10.2, Question 4 (iv): Construct angles of the given measures with the help of a compass. 75°.

Answer: First construct a 90° angle. Then bisect the remaining 30° part after the 60° part of the 90° to get 15° (the book's method: take 15° away from 90°). 90° − 15° = 75°. Alternatively: add 15° (got by bisecting 30°) to a 60° angle: 60° + 15° = 75°.

Question: Exercise 10.2, Question 4 (v): Construct angles of the given measures with the help of a compass. 105°.

Answer: First construct a 90° angle. Then add a 15° angle outside it (got by bisecting 30°): 90° + 15° = 105°. Alternatively: take 15° away from a 120° angle: 120° − 15° = 105°.

Question: Exercise 10.2, Question 4 (vi): Construct angles of the given measures with the help of a compass. 120°.

Answer: Draw a ray OP. With O as centre, draw an arc of a suitable radius cutting OP at A. With the same radius and A as centre, draw an arc cutting the first arc at B (60°). Then with the same radius and B as centre, draw an arc cutting the first arc at C (120°). Join O and C with ray OR. m∠POR = 60° + 60° = 120°.

Question: Exercise 10.2, Question 4 (vii): Construct angles of the given measures with the help of a compass. 135°.

Answer: First construct a 90° angle. Then add a 45° angle outside it (got by bisecting 90°): 90° + 45° = 135°.

Question: Exercise 10.2, Question 4 (viii): Construct angles of the given measures with the help of a compass. 165°.

Answer: First construct a 150° angle (by adding 30° to 120°, or taking 30° away from 180°). Then add a 15° angle outside it (got by bisecting 30°): 150° + 15° = 165°.

Question: Exercise 10.3, Question 1: Construct triangle ABC, where: mAB = 3 cm, mBC = 5 cm, mCA = 7 cm.

Answer: This is a three-sides (SSS) construction. First draw the longest side AC = 7 cm with a scale. Then open the compass to AB = 3 cm and, with A as centre, draw an arc. Then open the compass to BC = 5 cm and, with C as centre, draw another arc cutting the first arc at point B. Join B to A and C. The resulting △ABC is the required triangle.

Question: Exercise 10.3, Question 2: Construct triangle ABC, where: mAB = 2 cm, mBC = 5 cm, mCA = 5 cm.

Answer: This is a three-sides (SSS) construction. First draw AC = 5 cm. Then open the compass to AB = 2 cm and, with A as centre, draw an arc. Then open the compass to BC = 5 cm and, with C as centre, draw another arc cutting the first arc at point B. Join B to A and C. The resulting △ABC is the required triangle.

Question: Exercise 10.3, Question 3: Construct triangle ABC, where: mAB = 5 cm, mBC = 7 cm, mCA = 9 cm.

Answer: This is a three-sides (SSS) construction. First draw AC = 9 cm. Then open the compass to AB = 5 cm and, with A as centre, draw an arc. Then open the compass to BC = 7 cm and, with C as centre, draw another arc cutting the first arc at point B. Join B to A and C. The resulting △ABC is the required triangle.

Question: Exercise 10.3, Question 4: Construct triangle ABC, where: mAB = 8 cm, mBC = 6 cm, mCA = 10 cm.

Answer: This is a three-sides (SSS) construction. First draw AC = 10 cm. Then open the compass to AB = 8 cm and, with A as centre, draw an arc. Then open the compass to BC = 6 cm and, with C as centre, draw another arc cutting the first arc at point B. Join B to A and C. The resulting △ABC is the required triangle.

Question: Exercise 10.3, Question 5: Construct triangle MNO, where: m∠N = 63°, mNO = 3 cm, mMN = 5 cm.

Answer: This is a two-sides-and-included-angle (SAS) construction. First draw NO = 3 cm. Then construct a 63° angle at N (in practice with a protractor, since 63° is not a standard compass angle). Then on the other arm of the angle take MN = 5 cm from N to get point M. Join M to O. The resulting △MNO is the required triangle.

Question: Exercise 10.3, Question 6: Construct triangle MNO, where: m∠N = 75°, mNO = 3.4 cm, mMN = 4 cm.

Answer: This is a two-sides-and-included-angle (SAS) construction. First draw NO = 3.4 cm. Then construct a 75° angle at N with a compass (take 15° away from 90°: 90° − 15° = 75°). Then on the other arm of the angle take MN = 4 cm from N to get point M. Join M to O. The resulting △MNO is

the required triangle.

Question: Exercise 10.3, Question 7: Construct triangle MNO, where: m∠N = 80°, mNO = 3.5 cm, mMN = 4.3 cm.

Answer: This is a two-sides-and-included-angle (SAS) construction. First draw NO = 3.5 cm. Then construct an 80° angle at N (in practice with a protractor, since 80° is not one of the standard compass angles). Then on the other arm of the angle take MN = 4.3 cm from N to get point M. Join M to O. The resulting △MNO is the required triangle.

Question: Exercise 10.3, Question 8: Construct triangle MNO, where: m∠N = 90°, mNO = 4 cm, mMN = 3 cm.

Answer: This is a two-sides-and-included-angle (SAS) construction. First draw NO = 4 cm. Then construct a 90° angle at N with a compass (with two 60° arcs). Then on the other arm of the angle take MN = 3 cm from N to get point M. Join M to O. The resulting △MNO is the required triangle.

Question: Exercise 10.3, Question 9: Construct triangle PQR, where: mPQ = 5 cm, m∠Q = 40°, m∠P = 60°.

Answer: This is a one-side-and-two-angles (ASA) construction. First draw PQ = 5 cm. Then construct a 60° angle at P (with a compass) and a 40° angle at Q (in practice with a protractor, since 40° is not a standard compass angle). Extend the arms of both angles; where they meet is point R. The resulting △PQR is the required triangle.

Question: Exercise 10.3, Question 10: Construct triangle PQR, where: mPQ = 7 cm, m∠P = 65°, m∠Q = 60°.

Answer: This is a one-side-and-two-angles (ASA) construction. First draw PQ = 7 cm. Then construct a 65° angle at P (in practice with a protractor) and a 60° angle at Q (with a compass). Extend the arms of both angles; where they meet is point R. The resulting △PQR is the required triangle.

Question: Exercise 10.3, Question 11: Construct triangle PQR, where: mQR = 6.5 cm, m∠Q = 45°, m∠R = 30°.

Answer: This is a one-side-and-two-angles (ASA) construction. First draw QR = 6.5 cm. Then construct a 45° angle at Q (by bisecting 90°) and a 30° angle at R (by bisecting 60°). Extend the arms of both angles; where they meet is point P. The resulting △PQR is the required triangle.

Question: Exercise 10.3, Question 12: Construct triangle PQR, where: mQR = 7.5 cm, m∠Q = 35°, m∠R = 40°.

Answer: This is a one-side-and-two-angles (ASA) construction. First draw QR = 7.5 cm. Then construct a 35° angle at Q and a 40° angle at R (in practice with a protractor, since 35° and 40° are not standard compass angles). Extend the arms of both angles; where they meet is point P. The resulting △PQR is the required triangle.

Question: Exercise 10.3, Question 13: Construct right triangle XYZ, where m∠Z = 90°, mXY = 7 cm, mXZ = 5 cm.

Answer: This is a hypotenuse-and-one-side (RHS) construction. First draw XZ = 5 cm. Then construct a 90° angle at Z with a compass (with two 60° arcs). Then open the compass to the hypotenuse XY = 7 cm and, with X as centre, draw an arc cutting the 90° ray at point Y. Join Y to X. The resulting △XYZ is the required right triangle.

Question: Exercise 10.3, Question 14: Construct right triangle XYZ, where m∠Z = 90°, mXY = 10 cm, mXZ = 6 cm.

Answer: This is a hypotenuse-and-one-side (RHS) construction. First draw XZ = 6 cm. Then construct a 90° angle at Z. Then open the compass to the hypotenuse XY = 10 cm and, with X as centre, draw an arc cutting the 90° ray at point Y. Join Y to X. The resulting △XYZ is the required right triangle.

Question: Exercise 10.3, Question 15: Construct right triangle XYZ, where m∠Z = 90°, mXY = 14 cm, mXZ = 10 cm.

Answer: This is a hypotenuse-and-one-side (RHS) construction. First draw XZ = 10 cm. Then construct a 90° angle at Z. Then open the compass to the hypotenuse XY = 14 cm and, with X as centre, draw an arc cutting the 90° ray at point Y. Join Y to X. The resulting △XYZ is the required right triangle.

Question: Exercise 10.3, Question 16: Construct right triangle XYZ, where m∠Z = 90°, mXY = 8 cm, mXZ = 6.5 cm.

Answer: This is a hypotenuse-and-one-side (RHS) construction. First draw XZ = 6.5 cm. Then construct a 90° angle at Z. Then open the compass to the hypotenuse XY = 8 cm and, with X as centre, draw an arc cutting the 90° ray at point Y. Join Y to X. The resulting △XYZ is the required right triangle.

Question: Review Exercise 10, Question 1: Give any four examples from the environment of your daily life and explain that these are a few example things which represent a line segment.

Answer: Four examples of a line segment from daily life: (1) the edge (side) of a book, (2) a pencil or pen, (3) the side of a room wall or of a door, (4) the side of a scale (ruler). Each of these has two fixed ends and a fixed length, so they are examples of a line segment.

Question: Review Exercise 10, Question 2 (i): If mAB = 4.3 cm, mCD = 3.5 cm and mEF = 2 cm, then by the principle of geometry find the measure of a line segment equal to: (i) mAB + mCD + mEF

Answer: Take the given measures one after another on a straight line with a compass and join them. mAB + mCD + mEF = 4.3 cm + 3.5 cm + 2 cm = 9.8 cm. So the required line segment measures 9.8 cm.

Question: Review Exercise 10, Question 2 (ii): If mAB = 4.3 cm, mCD = 3.5 cm and mEF = 2 cm, then by the principle of geometry find the measure of a line segment equal to: (ii) mAB − mCD

Answer: Take the larger segment AB (4.3 cm) on a straight line and take CD (3.5 cm) away from it. The remaining part = mAB − mCD = 4.3 cm −

3.5 cm = 0.8 cm. So the required line segment measures 0.8 cm.

Question: Review Exercise 10, Question 2 (iii): If mAB = 4.3 cm, mCD = 3.5 cm and mEF = 2 cm, then by the principle of geometry find the measure of a line segment equal to: (iii) mAB − mEF

Answer: Take the larger segment AB (4.3 cm) on a straight line and take EF (2 cm) away from it. The remaining part = mAB − mEF = 4.3 cm − 2 cm = 2.3 cm. So the required line segment measures 2.3 cm.

Question: Review Exercise 10, Question 3: A line segment measuring 7.6 cm. Construct its perpendicular bisector with the help of a compass. After that, again construct the bisector of each part obtained with the help of a compass.

Answer: First draw a line segment AB of 7.6 cm. Then open the compass to more than half of 7.6 cm (3.8 cm) and, with A and B as centres, draw arcs above and below, and join their crossings to make the perpendicular bisector of AB. This divides AB into two equal parts (each 7.6 ÷ 2 = 3.8 cm). Then bisect each 3.8 cm part again in the same way. This divides AB into four equal parts: each part 7.6 ÷ 4 = 1.9 cm.

Question: Review Exercise 10, Question 4 (i): Construct angles of the following measures with a compass, and then construct the bisector of each angle. (i) 30°

Answer: First construct a 30° angle with a compass (by bisecting a 60° angle). Then bisect that 30° angle. Each bisected part measures 30° ÷ 2 = 15°.

Question: Review Exercise 10, Question 4 (ii): Construct angles of the following measures with a compass, and then construct the bisector of each angle. (ii) 75°

Answer: First construct a 75° angle with a compass (take 15° away from 90°). Then bisect that 75° angle. Each bisected part measures 75° ÷ 2 = 37.5°.

Question: Review Exercise 10, Question 4 (iii): Construct angles of the following measures with a compass, and then construct the bisector of each angle. (iii) 105°

Answer: First construct a 105° angle with a compass (add 15° to 90°). Then bisect that 105° angle. Each bisected part measures 105° ÷ 2 = 52.5°.

Question: Review Exercise 10, Question 5 (i): Construct triangle ABC according to the given measures and write the steps of construction in each case. (i) mAB = 8 cm, mAC = 6 cm, m∠A = 65°

Answer: This is a two-sides-and-included-angle (SAS) construction. Steps: (1) Draw AB = 8 cm. (2) Construct a 65° angle at A (in practice with a protractor). (3) On the other arm of the angle take AC = 6 cm from A to get point C. (4) Join C to B. The resulting △ABC is the required triangle.

Question: Review Exercise 10, Question 5 (ii): Construct triangle ABC

according to the given measures and write the steps of construction in each case. (ii) m∠A = 45°, m∠C = 45°, mAC = 7.5 cm

Answer: This is a one-side-and-two-angles (ASA) construction. Steps: (1) Draw AC = 7.5 cm. (2) Construct a 45° angle at A (by bisecting 90°). (3) Construct a 45° angle at C. (4) Extend the arms of both angles; where they meet is point B. The resulting △ABC is the required triangle.

Question: Review Exercise 10, Question 5 (iii): Construct triangle ABC according to the given measures and write the steps of construction in each case. (iii) mAB = 5 cm, mAC = 7 cm, m∠B = 90°

Answer: This is a two-sides-and-a-non-included-angle construction. Steps: (1) Draw AB = 5 cm. (2) Construct a 90° angle at B (with a compass). (3) Open the compass to AC = 7 cm and, with A as centre, draw an arc cutting the 90° ray at point C. (4) Join C to A. The resulting △ABC is the required triangle.

Question: Review Exercise 10, Question 6: Draw a line segment and write its name.

Answer: Draw a straight line segment with a scale, for example 6 cm long, and mark points at its two ends and name them, for example P and Q. This is the line segment PQ.

Question: Review Exercise 10, Question 7: Mark (✓) on the correct perpendicular bisector. Figures (i), (ii) and (iii) are given.

Answer: Mark (✓) on figure (ii), because in it the line cuts AB at its midpoint at a 90° angle, i.e. it is the true perpendicular bisector. In figures (i) and (iii) the lines are slanted and do not cut the segment perpendicularly at its middle. ✓ = (ii).

Review Exercise 10, Question 8 (i): Choose the correct answer. (i) Which shows the line segment AB:

(a) ↔AB

✓ (b) ¯AB

(c) →AB

(d) →BA

Review Exercise 10, Question 8 (ii): Choose the correct answer. (ii) The perpendicular bisector divides a line segment into _____ equal parts.

✓ (a) two

(b) three

(c) four

(d) five

Review Exercise 10, Question 8 (iii): Choose the correct answer. (iii) The

perpendicular bisector of PQ passes through its _____.

(a) end point

✓ (b) midpoint

(c) point P

(d) point Q

Review Exercise 10, Question 8 (iv): Choose the correct answer. (iv) An angle is measured with _____.

(a) scale

(b) compass

✓ (c) protractor

(d) set square

✗ Review Exercise 10, Question 9 (i): Write (T) for true and (F) for false for the following statements. (i) A line has a fixed length.

✗ Review Exercise 10, Question 9 (ii): Write (T) for true and (F) for false for the following statements. (ii) A ray has two ends.

✗ Review Exercise 10, Question 9 (iii): Write (T) for true and (F) for false for the following statements. (iii) Bisecting a 30° angle does not give a 15° angle.

✓ Review Exercise 10, Question 9 (iv): Write (T) for true and (F) for false for the following statements. (iv) A triangle has 6 parts.

✓ Review Exercise 10, Question 9 (v): Write (T) for true and (F) for false for the following statements. (v) A triangle cannot be constructed when the sum of two sides is less than the length of the third side.

11. Perimeter & Area

Question: Exercise 11.1, Question 1 (a): Find the perimeter and area of each of the following squares. Square ABCD, each side 5 cm.

Answer: Perimeter of square = 4 × (one side) = 4 × 5 cm = 20 cm. Area of square = side × side = 5 cm × 5 cm = 25 cm².

Question: Exercise 11.1, Question 1 (b): Find the perimeter and area of each of the following squares. Square EFGH, each side 3.5 cm.

Answer: Perimeter of square = 4 × (one side) = 4 × 3.5 cm = 14 cm. Area of square = side × side = 3.5 cm × 3.5 cm = 12.25 cm².

Question: Exercise 11.1, Question 1 (c): Find the perimeter and area of each of the following squares. Figure POIN: outer square 6 cm × 6 cm, with a 2 cm × 3 cm rectangle cut out from the bottom side (IJ = 2 cm, MN = 2 cm, KJ = 3 cm, LM = 3 cm, KL = 2 cm).

Answer: Perimeter = sum of all sides around the figure = 6 + 6 + 2 + 3 + 2 + 3 + 2 + 6 = 30 cm. Area = area of big square − area of cut-out rectangle = (6 × 6) − (2 × 3) = 36 − 6 = 30 cm².

Question: Exercise 11.1, Question 2 (a): Find the perimeter and area of each of the following rectangles. Rectangle OTSR, length 5 cm, breadth 3 cm.

Answer: Perimeter of rectangle = 2(length + breadth) = 2(5 + 3) = 2(8) = 16 cm. Area of rectangle = length × breadth = 5 × 3 = 15 cm².

Question: Exercise 11.1, Question 2 (b): Find the perimeter and area of each of the following rectangles. Rectangle ZYXW, length 6 cm, breadth 2.5 cm.

Answer: Perimeter of rectangle = 2(length + breadth) = 2(6 + 2.5) = 2(8.5) = 17 cm. Area of rectangle = length × breadth = 6 × 2.5 = 15 cm².

Question: Exercise 11.1, Question 2 (c): Find the perimeter and area of each of the following rectangles. Rectangle AEIO, length 4.8 cm, breadth 2.7 cm.

Answer: Perimeter of rectangle = 2(length + breadth) = 2(4.8 + 2.7) = 2(7.5) = 15 cm. Area of rectangle = length × breadth = 4.8 × 2.7 = 12.96 cm².

Question: Exercise 11.1, Question 2 (d): Find the perimeter and area of each of the following rectangles. Rectangle ONML, length 10 cm, breadth 2 cm.

Answer: Perimeter of rectangle = 2(length + breadth) = 2(10 + 2) = 2(12) = 24 cm. Area of rectangle = length × breadth = 10 × 2 = 20 cm².

Question: Exercise 11.1, Question 3 (i): Find the perimeter and area of the following. (i) A rectangle whose sides are shown as 5 cm, 2.5 cm, 5 cm, 2.5 cm.

Answer: This is a rectangle of length 5 cm and breadth 2.5 cm. Perimeter = 5 + 2.5 + 5 + 2.5 = 15 cm (or 2(5 + 2.5) = 15 cm). Area = 5 × 2.5 = 12.5 cm².

Question: Exercise 11.1, Question 3 (ii): Find the perimeter and area of the following. (ii) A square each of whose sides is shown as 5 cm.

Answer: Perimeter of square = 4 × 5 = 20 cm. Area of square = 5 × 5 = 25 cm².

Question: Exercise 11.1, Question 4 (i): Find the side of a square whose perimeter is: (i) 48 cm.

Answer: Side of square = perimeter ÷ 4 = 48 ÷ 4 = 12 cm.

Question: Exercise 11.1, Question 4 (ii): Find the side of a square whose perimeter is: (ii) 50 cm.

Answer: Side of square = perimeter ÷ 4 = 50 ÷ 4 = 12.5 cm.

Question: Exercise 11.1, Question 5: Find the length of a rectangle whose perimeter is 100 m and breadth is 200 m.

Answer: From the numbers as printed in the book: perimeter of rectangle = 2(length + breadth), so 2(length + 200) = 100, i.e. length + 200 = 50, i.e. length = 50 − 200 = −150 m. A length can never be negative, so no valid answer follows from the printed numbers — there is a printing error in the printed numbers (the breadth alone is greater than half the perimeter).

Question: Exercise 11.1, Question 6: Find the area of a rectangle whose two adjacent sides are shown as 6.2 cm and 8.5 cm.

Answer: Area of rectangle = length × breadth = 6.2 × 8.5 = 52.7 cm².

Question: Exercise 11.2, Question 1 (i): Find the area of the shaded part in the following figures. (i) Outer rectangle 25 cm × 20 cm, path 5 cm wide.

Answer: Area of the path = area of big rectangle − area of inner rectangle. Inner rectangle length = 25 − 5 − 5 = 15 cm, breadth = 20 − 5 − 5 = 10 cm. Area of shaded part = (25 × 20) − (15 × 10) = 500 − 150 = 350 cm².

Question: Exercise 11.2, Question 1 (ii): Find the area of the shaded part in the following figures. (ii) Outer rectangle 35 cm × 25 cm, path 10 cm wide.

Answer: Inner rectangle length = 35 − 10 − 10 = 15 cm, breadth = 25 − 10 − 10 = 5 cm. Area of shaded part = (35 × 25) − (15 × 5) = 875 − 75 = 800 cm².

Question: Exercise 11.2, Question 1 (iii): Find the area of the shaded part in the following figures. (iii) Outer rectangle 50.5 cm × 40.5 cm, inner rectangle 45.5 cm × 35.5 cm.

Answer: Area of shaded part = area of outer rectangle − area of inner rectangle = (50.5 × 40.5) − (45.5 × 35.5) = 2045.25 − 1615.25 = 430 cm².

Question: Exercise 11.2, Question 2: A rectangular field is 35 m long and 30 m wide. A 5 m wide path is made inside around it. Find the area of the path.

Answer: Area of the field = 35 × 30 = 1050 m². Inner field (without path) length = 35 − 5 − 5 = 25 m, breadth = 30 − 5 − 5 = 20 m, area = 25 × 20 = 500 m². Area of the path = 1050 − 500 = 550 m².

Question: Exercise 11.2, Question 3: A 5 m wide path is made outside around a rectangular field. The sides of the field are shown as 20 m and 10 m. Find the area of the path.

Answer: Area of the field = 20 × 10 = 200 m². Outer measurement (field + path): length = 20 + 5 + 5 = 30 m, breadth = 10 + 5 + 5 = 20 m, area = 30 × 20 = 600 m². Area of the path = 600 − 200 = 400 m².

Question: Exercise 11.3, Question 1: Find the cost of fencing around a square-shaped green field of side 250 m at Rs. 30 per metre.

Answer: Perimeter of square field = 4 × side = 4 × 250 = 1000 m. Cost of fencing = perimeter × rate per metre = 1000 × 30 = Rs. 30,000.

Question: Exercise 11.3, Question 2: Find the cost of fencing around a rectangular green field 250 m long and 150 m wide at Rs. 30 per metre.

Answer: Perimeter of rectangular field = 2(length + breadth) = 2(250 + 150) = 2(400) = 800 m. Cost of fencing = 800 × 30 = Rs. 24,000.

Question: Exercise 11.3, Question 3: Basma and Amima go for a morning exercise run every day. Basma runs around a field 210 m long and 100 m wide. Amima runs around a field 150 m long and 100 m wide. Whose distance is greater?

Answer: Perimeter of Basma's field = 2(210 + 100) = 2(310) = 620 m. Perimeter of Amima's field = 2(150 + 100) = 2(250) = 500 m. 620 > 500, so Basma's distance is greater.

Question: Exercise 11.3, Question 4: Find the length of a piece of land whose breadth is 15 m and area is 615 m².

Answer: Area of rectangle = length × breadth, so length = area ÷ breadth = 615 ÷ 15 = 41 m.

Question: Exercise 11.3, Question 5: A playground is 35.5 m long and 30.5 m wide. Find the cost of laying grass in the field at Rs. 150 per m².

Answer: Area of the field = 35.5 × 30.5 = 1082.75 m². Cost of laying grass = 1082.75 × 150 = Rs. 162,412.50.

Question: Exercise 11.3, Question 6: A path 120 m long and 8 m wide is to be made. Find the cost of paving this path at Rs. 200 per m².

Answer: Area of the path = length × breadth = 120 × 8 = 960 m². Cost of paving = 960 × 200 = Rs. 192,000.

Question: Exercise 11.3, Question 7: Find the area of the floor of a room which is covered with 650 bricks. The area of each brick is 0.5 m².

Answer: Area of the floor = number of bricks × area of each brick = 650 × 0.5 = 325 m².

Question: Exercise 11.4, Question 1 (a): Identify the altitude and its measure in the following figures. (a) Triangle ABC, in which AE is perpendicular to BC, AE = 4 cm.

Answer: Altitude = AE = 4 cm, its base = BC. AE is the perpendicular drawn from vertex A to the base BC.

Question: Exercise 11.4, Question 1 (b): Identify the altitude and its measure in the following figures. (b) Parallelogram LMON, in which OP is perpendicular to LM, OP = 4.5 cm.

Answer: Altitude = OP = 4.5 cm, its base = LM. OP is the perpendicular drawn from vertex O to the base LM.

Question: Exercise 11.4, Question 1 (c): Identify the altitude and its measure in the following figures. (c) Trapezium PQRS, in which SZ is perpendicular to PR, SZ = 4 cm.

Answer: Altitude = SZ = 4 cm, its base = PR. SZ is the perpendicular drawn from vertex S to the base PR.

Question: Exercise 11.4, Question 1 (d): Identify the altitude and its measure in the following figures. (d) Triangle XYT, in which YT is perpendicular to XT, YT = 6 cm (YT is shown with a dashed line).

Answer: Altitude = YT = 6 cm, its base = XT. YT is the perpendicular drawn from vertex Y to the base XT.

Question: Exercise 11.4, Question 2 (i): Find the area of a parallelogram when its altitude and base are given. (i) Base = 10 cm, altitude = 6 cm.

Answer: Area of parallelogram = base × altitude = 10 × 6 = 60 cm².

Question: Exercise 11.4, Question 2 (ii): Find the area of a parallelogram when its altitude and base are given. (ii) Base = 15 cm, altitude = 8 cm.

Answer: Area of parallelogram = base × altitude = 15 × 8 = 120 cm².

Question: Exercise 11.4, Question 2 (iii): Find the area of a parallelogram when its altitude and base are given. (iii) Base = 4.5 cm, altitude = 5.5 cm.

Answer: Area of parallelogram = base × altitude = 4.5 × 5.5 = 24.75 cm².

Question: Exercise 11.4, Question 2 (iv): Find the area of a parallelogram when its altitude and base are given. (iv) Base = 6.5 cm, altitude = 4.7 cm.

Answer: Area of parallelogram = base × altitude = 6.5 × 4.7 = 30.55 cm².

Question: Exercise 11.4, Question 3: The area of a parallelogram is 192 cm² and its altitude is 12 cm. Find its base.

Answer: Area = base × altitude, so base = area ÷ altitude = 192 ÷ 12 = 16 cm.

Question: Exercise 11.4, Question 4: The base of a parallelogram is 15 m and its area is 180 m². Find its altitude.

Answer: Area = base × altitude, so altitude = area ÷ base = 180 ÷ 15 = 12 m.

Question: Exercise 11.4, Question 5: The altitude of a parallelogram is 1.5 m and its area is 4.50 m². Find its base.

Answer: Area = base × altitude, so base = area ÷ altitude = 4.50 ÷ 1.5 = 3 m.

Question: Exercise 11.5, Question 1 (i): Find the area of a trapezium when the parallel sides and altitude are given as follows. (i) The parallel sides are 5 cm and 6 cm, while the altitude is 4 cm.

Answer: Area of trapezium = ½ × altitude × (sum of parallel sides) = ½ × 4 × (5 + 6) = 2 × 11 = 22 cm².

Question: Exercise 11.5, Question 1 (ii): Find the area of a trapezium when the parallel sides and altitude are given as follows. (ii) The parallel sides are 3.5 cm and 4.5 cm, while the altitude is 2 cm.

Answer: Area of trapezium = ½ × altitude × (sum of parallel sides) = ½ × 2 × (3.5 + 4.5) = 1 × 8 = 8 cm².

Question: Exercise 11.5, Question 1 (iii): Find the area of a trapezium when the parallel sides and altitude are given as follows. (iii) The parallel sides are 4.5 cm and 5.5 cm, while the altitude is 2 cm.

Answer: Area of trapezium = ½ × altitude × (sum of parallel sides) = ½ × 2 × (4.5 + 5.5) = 1 × 10 = 10 cm².

Question: Exercise 11.5, Question 2: The parallel sides of a trapezium are 6 cm and 8 cm and its altitude is 5 cm. Find its area.

Answer: Area of trapezium = ½ × altitude × (sum of parallel sides) = ½ × 5 × (6 + 8) = ½ × 5 × 14 = 35 cm².

Question: Exercise 11.5, Question 3: If the parallel sides of a trapezium are 15 cm and 20 cm and its area is 140 cm², find its altitude.

Answer: Area = ½ × altitude × (sum of parallel sides), so 140 = ½ × altitude × (15 + 20), i.e. 140 = ½ × altitude × 35, i.e. altitude = (140 × 2) ÷ 35 = 280 ÷ 35 = 8 cm.

Question: Exercise 11.5, Question 4 (i): Find the area of a triangle when its base and altitude are given as follows. (i) Base = 8 cm, altitude = 6 cm.

Answer: Area of triangle = ½ × base × altitude = ½ × 8 × 6 = 24 cm².

Question: Exercise 11.5, Question 4 (ii): Find the area of a triangle when its base and altitude are given as follows. (ii) Base = 9 cm, altitude = 5 cm.

Answer: Area of triangle = ½ × base × altitude = ½ × 9 × 5 = 22.5 cm².

Question: Exercise 11.5, Question 4 (iii): Find the area of a triangle when its base and altitude are given as follows. (iii) Base = 7 cm, altitude = 2.5 cm.

Answer: Area of triangle = ½ × base × altitude = ½ × 7 × 2.5 = 8.75 cm².

Question: Exercise 11.5, Question 4 (iv): Find the area of a triangle when its base and altitude are given as follows. (iv) Base = 6.5 cm, altitude = 3.5 cm.

Answer: Area of triangle = ½ × base × altitude = ½ × 6.5 × 3.5 = 11.375 cm².

Question: Exercise 11.5, Question 5: Find the area of a triangle whose base is 6 cm and altitude is 5.5 cm.

Answer: Area of triangle = ½ × base × altitude = ½ × 6 × 5.5 = 16.5 cm².

Question: Exercise 11.5, Question 6: The area of a triangle with base 5 m is 40 m². Find its altitude.

Answer: Area = ½ × base × altitude, so 40 = ½ × 5 × altitude, i.e. altitude = (40 × 2) ÷ 5 = 80 ÷ 5 = 16 m.

Question: Exercise 11.5, Question 7: Find the area of the given parallelogram in the figure, if the area of the shaded triangle is 20 m².

Answer: In the figure the parallelogram is divided by a diagonal into two equal triangles, one of which is shaded. So area of parallelogram = 2 × area of shaded triangle = 2 × 20 = 40 m².

Question: Review Exercise 11, Question 1 (i): Find the perimeter and area of the following figures. (i) Square, each side 5.3 cm.

Answer: Perimeter of square = 4 × 5.3 = 21.2 cm. Area of square = 5.3 × 5.3 = 28.09 cm².

Question: Review Exercise 11, Question 1 (ii): Find the perimeter and area of the following figures. (ii) Rectangle, length 11 cm, breadth 8 cm.

Answer: Perimeter of rectangle = 2(11 + 8) = 2(19) = 38 cm. Area of rectangle = 11 × 8 = 88 cm².

Question: Review Exercise 11, Question 2 (i): Find the area of the following figures. (i) Triangle, base 6 cm, altitude 2.5 cm.

Answer: Area of triangle = ½ × base × altitude = ½ × 6 × 2.5 = 7.5 cm².

Question: Review Exercise 11, Question 2 (ii): Find the area of the

following figures. (ii) Parallelogram, base 8 cm, altitude 6 cm.

Answer: Area of parallelogram = base × altitude = 8 × 6 = 48 cm².

Question: Review Exercise 11, Question 2 (iii): Find the area of the following figures. (iii) Trapezium, parallel sides 3 cm and 10 cm, altitude 2.5 cm.

Answer: Area of trapezium = ½ × altitude × (sum of parallel sides) = ½ × 2.5 × (3 + 10) = ½ × 2.5 × 13 = 16.25 cm².

Question: Review Exercise 11, Question 3: Find the area of the shaded part in the given figure. Outer rectangle 40 m × 25 m, inner rectangle 36 m × 21 m, path 2 m wide.

Answer: Area of shaded part = area of outer rectangle − area of inner rectangle = (40 × 25) − (36 × 21) = 1000 − 756 = 244 m².

Question: Review Exercise 11, Question 4: A square tile has side 20 cm. How many such square tiles are needed to cover the floor of a square bathroom with side 3 m?

Answer: Area of bathroom floor = 3 m × 3 m = 9 m² = 90,000 cm². Area of one tile = 20 × 20 = 400 cm². Number of tiles needed = 90,000 ÷ 400 = 225 tiles.

Question: Review Exercise 11, Question 5: The area of a square of side 16 cm is the same as that of a rectangle of length 64 cm. What will be the breadth of the rectangle?

Answer: Area of square = 16 × 16 = 256 cm². Area of rectangle is also 256 cm², so breadth = area ÷ length = 256 ÷ 64 = 4 cm.

Question: Review Exercise 11, Question 6: The area of a triangle is 72 cm². If its altitude is 8 cm, find its base.

Answer: Area = ½ × base × altitude, so 72 = ½ × base × 8, i.e. base = (72 × 2) ÷ 8 = 144 ÷ 8 = 18 cm.

Question: Review Exercise 11, Question 7: The area of a triangle is 60 cm². If its base is 12 cm, find its altitude.

Answer: Area = ½ × base × altitude, so 60 = ½ × 12 × altitude, i.e. altitude = (60 × 2) ÷ 12 = 120 ÷ 12 = 10 cm.

Question: Review Exercise 11, Question 8: In a right triangle the sides forming the right angle are 12 cm and 9 cm. Find the area of this triangle.

Answer: In a right triangle the two sides forming the right angle serve as each other's base and altitude. Area = ½ × 12 × 9 = 54 cm².

Question: Review Exercise 11, Question 9: In △ABC, CD is perpendicular to AB, such that mAB = 18 cm and the length of CD = 7.5 cm. Find the area of △ABC.

Answer: Here base AB = 18 cm and altitude CD = 7.5 cm. Area of △ABC =

½ × base × altitude = ½ × 18 × 7.5 = 67.5 cm².

Question: Review Exercise 11, Question 10: The parallel sides of a trapezium are 9 cm and 7 cm, its altitude is 5 cm. Find its area.

Answer: Area of trapezium = ½ × altitude × (sum of parallel sides) = ½ × 5 × (9 + 7) = ½ × 5 × 16 = 40 cm².

Question: Review Exercise 11, Question 11: If the parallel sides of a trapezium are 20 cm and 30 cm and its area is 200 cm², find its altitude.

Answer: Area = ½ × altitude × (sum of parallel sides), so 200 = ½ × altitude × (20 + 30), i.e. 200 = ½ × altitude × 50, i.e. altitude = (200 × 2) ÷ 50 = 400 ÷ 50 = 8 cm.

Question: Review Exercise 11, Question 12: The area of a trapezium is 10 cm². If its altitude is 8 cm and one parallel side is 20 cm, find the length of the other side.

Answer: From the numbers as printed in the book: area = ½ × altitude × (sum of both parallel sides), so 10 = ½ × 8 × (20 + other side), i.e. 10 = 4 × (20 + other side), i.e. 20 + other side = 10 ÷ 4 = 2.5, i.e. other side = 2.5 − 20 = −17.5 cm. A side length can never be negative, so no valid answer follows from the printed numbers — there is a printing error in the printed numbers.

Question: Review Exercise 11, Question 13: Find the area of the shaded part in the figure alongside. Figure: a parallelogram-like figure, top side 6 cm, altitude 4 cm, the blue triangle on the left has base 2 cm.

Answer: The shaded (blue) part is a triangle whose base is 2 cm and altitude is 4 cm. Area of shaded part = ½ × base × altitude = ½ × 2 × 4 = 4 cm².

Review Exercise 11, Question 14 (i): Fill in the blank. (i) Area is generally measured in _____.

Answer: square units.

Review Exercise 11, Question 14 (ii): Fill in the blank. (ii) Perimeter is generally measured in _____.

Answer: units of length.

Review Exercise 11, Question 14 (iii): Fill in the blank. (iii) The area of a rectangle is the _____ of length and breadth.

Answer: product

Review Exercise 11, Question 14 (iv): Fill in the blank. (iv) The perimeter of a rectangle is the _____ of its 4 sides.

Answer: sum

Review Exercise 11, Question 14 (v): Fill in the blank. (v) Area of square = _____ × _____

Answer: side, side.

Review Exercise 11, Question 14 (vi): Fill in the blank. (vi) Perimeter of square = _____ × _____

Answer: 4, side.

Review Exercise 11, Question 14 (vii): Fill in the blank. (vii) The formula for finding the perimeter of a rectangle is _____.

Answer: 2(length + breadth).

Review Exercise 11, Question 14 (viii): Fill in the blank. (viii) The formula for finding the area of a parallelogram is _____.

Answer: base × altitude.

Review Exercise 11, Question 14 (ix): Fill in the blank. (ix) The formula for finding the area of a triangle is _____.

Answer: ½ (base × altitude).

Review Exercise 11, Question 14 (x): Fill in the blank. (x) The area of a trapezium is _____.

Answer: ½ × altitude × (sum of parallel sides).

12. Three-Dimensional Shapes

Practical Work (Unit 12), Question (1): Write the names of the following 3D shapes and fill in the blanks. (1) Shape: cylinder. Name of the shape = cylinder, number of circular faces = _____.

Answer: 2 (two circular faces, top and bottom).

Practical Work (Unit 12), Question (2): Write the names of the following 3D shapes and fill in the blanks. (2) Shape: cuboid (labelled ABCDEFG). Name of the shape = cuboid, DE shows _____, points A, B show _____.

Answer: an edge,, vertices.

Practical Work (Unit 12), Question (3): Write the names of the following 3D shapes and fill in the blanks. (3) Shape: cone (point A, round base below). Name of the shape = cone, point A shows _____, the dark shaded part shows _____.

Answer: the vertex (apex),, the curved surface.

Question: Exercise 12.1, Question 1 (i): Identify the things and write the names of their 3-D shapes which these things show. (i) Football.

Answer: The shape of a football is a sphere.

Question: Exercise 12.1, Question 1 (ii): Identify the things and write the names of their 3-D shapes which these things show. (ii) Ghee tin.

Answer: The shape of a ghee tin is a cylinder.

Question: Exercise 12.1, Question 1 (iii): Identify the things and write the names of their 3-D shapes which these things show. (iii) Birthday cap.

Answer: The shape of a birthday cap is a cone.

Question: Exercise 12.1, Question 1 (iv): Identify the things and write the names of their 3-D shapes which these things show. (iv) Die.

Answer: The shape of a die is a cube.

Question: Exercise 12.1, Question 1 (v): Identify the things and write the names of their 3-D shapes which these things show. (v) Milk carton (packet).

Answer: The shape of a milk carton (packet) is a cuboid.

Question: Exercise 12.1, Question 1 (vi): Identify the things and write the names of their 3-D shapes which these things show. (vi) Matchbox.

Answer: The shape of a matchbox is a cuboid.

Exercise 12.1, Question 2 (i): Fill in the blanks for the following figures. (i) Cuboid: (a) Number of edges = _____ (b) Number of faces = _____ (c) Number of vertices = _____

Answer: 12, 6, 8.

Exercise 12.1, Question 2 (ii): Fill in the blanks for the following figures. (ii) Cone: (a) Point A shows _____ (b) Number of circular faces = _____

Answer: the vertex (apex), 1 (one circular face at the bottom).

Exercise 12.1, Question 2 (iii): Fill in the blanks for the following figures. (iii) Cube: (a) AB shows _____ (b) Point A shows _____ (c) Number of faces = _____

Answer: an edge, a vertex, 6.

Practical Work (Unit 12, surface and volume), Question (i): Fill in the blanks below. (i) The volume of a cuboid is measured in _____, whose sides are 5 m, 6 m, 7 m.

Answer: cubic metres,

Practical Work (Unit 12, surface and volume), Question (ii): Fill in the blanks below. (ii) The volume of a cube is measured in _____, each of whose sides is 6 cm.

Answer: cubic cm,

Practical Work (Unit 12, surface and volume), Question (iii): Fill in the blanks below. (iii) The surface area of a cuboid is measured in _____, whose sides are 6 m, 7 m and 4 m.

Answer: square metres,

Practical Work (Unit 12, surface and volume), Question (iv): Fill in the blanks below. (iv) The surface area of a cube is measured in _____, each of whose sides is 5 cm long.

Answer: square cm,

Question: Exercise 12.2, Question 1 (i): Write the surface area and volume of each of the following figures. (i) Cuboid: length 6 cm, breadth 5 cm, height 8 cm.

Answer: Surface area = 2(length × breadth + breadth × height + height × length) = 2(6 × 5 + 5 × 8 + 8 × 6) = 2(30 + 40 + 48) = 2(118) = 236 cm². Volume = length × breadth × height = 6 × 5 × 8 = 240 cm³.

Question: Exercise 12.2, Question 1 (ii): Write the surface area and volume of each of the following figures. (ii) Cuboid: length 3 mm, breadth 4 mm, height 5 mm.

Answer: Surface area = 2(3 × 4 + 4 × 5 + 5 × 3) = 2(12 + 20 + 15) = 2(47) = 94 mm². Volume = 3 × 4 × 5 = 60 mm³.

Question: Exercise 12.2, Question 2 (i): Find the surface area and volume of each of the following. (i) Cube, each side 2 cm.

Answer: Surface area = 6 × (side × side) = 6 × (2 × 2) = 24 cm². Volume = side × side × side = 2 × 2 × 2 = 8 cm³.

Question: Exercise 12.2, Question 2 (ii): Find the surface area and volume of each of the following. (ii) Cuboid: length 6.5 cm, breadth 2.5 cm, height 3 cm.

Answer: Surface area = 2(6.5 × 2.5 + 2.5 × 3 + 3 × 6.5) = 2(16.25 + 7.5 + 19.5) = 2(43.25) = 86.5 cm². Volume = 6.5 × 2.5 × 3 = 48.75 cm³.

Question: Exercise 12.2, Question 2 (iii): Find the surface area and volume of each of the following. (iii) Cube, each side 3.5 cm.

Answer: Surface area = 6 × (3.5 × 3.5) = 6 × 12.25 = 73.5 cm². Volume = 3.5 × 3.5 × 3.5 = 42.875 cm³.

Question: Exercise 12.2, Question 2 (iv): Find the surface area and volume of each of the following. (iv) Cuboid: length 2 cm, breadth 1.5 cm, height 6 cm.

Answer: Surface area = 2(2 × 1.5 + 1.5 × 6 + 6 × 2) = 2(3 + 9 + 12) = 2(24) = 48 cm². Volume = 2 × 1.5 × 6 = 18 cm³.

Question: Exercise 12.2, Question 3: Find the volume of each of the following cuboids, in which each side measures: (i) length 5 cm (ii) breadth 6.2 cm (iii) height 2.5 cm.

Answer: This is a single cuboid whose three sides are given: length = 5 cm, breadth = 6.2 cm, height = 2.5 cm. Volume = length × breadth × height = 5 × 6.2 × 2.5 = 77.5 cm³.

Question: Exercise 12.2, Question 4 (i): Find the volume of each of the following cuboids, in which: (i) length = 9 cm, breadth = 6 cm, height = 4 cm.

Answer: Volume = length × breadth × height = 9 × 6 × 4 = 216 cm³.

Question: Exercise 12.2, Question 4 (ii): Find the volume of each of the following cuboids, in which: (ii) length = 5 cm, breadth = 3.5 cm, height = 4 cm.

Answer: Volume = length × breadth × height = 5 × 3.5 × 4 = 70 cm³.

Question: Exercise 12.2, Question 4 (iii): Find the volume of each of the following cuboids, in which: (iii) length = 3.5 cm, breadth = 2 cm, height = 2.5 cm.

Answer: Volume = length × breadth × height = 3.5 × 2 × 2.5 = 17.5 cm³.

Question: Exercise 12.3, Question 1: Find the total cost of polishing a cupboard at Rs. 500 per m², if its sides are 3 m, 2 m and 1 m.

Answer: Surface area of the cupboard (cuboid) = 2(3 × 2 + 2 × 1 + 1 × 3)

= 2(6 + 2 + 3) = 2(11) = 22 m². Total cost of polishing = 22 × 500 = Rs. 11,000.

Question: Exercise 12.3, Question 2: A tank is 5 m long, 4 m wide and 2.5 m high. Find the capacity of the tank.

Answer: Capacity of the tank = its volume = length × breadth × height = 5 × 4 × 2.5 = 50 m³.

Question: Exercise 12.3, Question 3: The sides of a brick are 25 cm × 10 cm × 4.5 cm. How much space will 1000 such bricks occupy?

Answer: Volume of one brick = 25 × 10 × 4.5 = 1125 cm³. Volume of 1000 bricks = 1000 × 1125 = 1,125,000 cm³ = 1.125 m³. So 1000 bricks will occupy 1,125,000 cm³ (1.125 m³) of space.

Question: Exercise 12.3, Question 4: Find the cost of painting a cube-like tin box at Rs. 50 per m², each of whose sides is 2.5 m long.

Answer: Surface area of the cube = 6 × (side × side) = 6 × (2.5 × 2.5) = 6 × 6.25 = 37.5 m². Cost of painting = 37.5 × 50 = Rs. 1875.

Question: Exercise 12.3, Question 5: Find the capacity of a cube-like oil tank, each of whose sides is 5 m long.

Answer: Capacity of the tank = its volume = side × side × side = 5 × 5 × 5 = 125 m³.

Question: Exercise 12.3, Question 6: How many small boxes of 5 cm × 10 cm × 15 cm are needed to fill a big iron box of 10 cm × 20 cm × 30 cm?

Answer: Volume of big box = 10 × 20 × 30 = 6000 cm³. Volume of small box = 5 × 10 × 15 = 750 cm³. Number of small boxes needed = 6000 ÷ 750 = 8 boxes.

Question: Exercise 12.3, Question 7: If the price of a cake of 5 cm × 5 cm × 10 cm is Rs. 100, tell what the price of a cake of 15 cm × 15 cm × 20 cm will be?

Answer: Volume of first cake = 5 × 5 × 10 = 250 cm³. Volume of second cake = 15 × 15 × 20 = 4500 cm³. Ratio of volumes = 4500 ÷ 250 = 18. The price increases in proportion to the volume, so the price of the second cake = 18 × 100 = Rs. 1800.

Question: Exercise 12.3, Question 8: Find the capacity of a water tank of 5.5 m × 3.4 m × 3.5 m.

Answer: Capacity of the tank = its volume = 5.5 × 3.4 × 3.5 = 65.45 m³ (5.5 × 3.4 = 18.7, 18.7 × 3.5 = 65.45).

Question: Exercise 12.3, Question 9: The sides of a tea box measure 11/2 cm × 21/4 cm × 20/3 cm. How much space will 100 such boxes occupy?

Answer: Volume of one box = (11/2) × (21/4) × (20/3) = 5.5 × 5.25 × (20/3) = 28.875 × (20/3) = 577.5 ÷ 3 = 192.5 cm³. Volume of 100 boxes = 100 × 192.5 = 19,250 cm³. So 100 boxes will occupy 19,250 cm³ of

space.

Question: Exercise 12.3, Question 10: Find the cost of painting a cube at Rs. 15.5 per cubic metre, each of whose sides is 5.2 m long.

Answer: The book's printed words "per cubic metre" are a printing error for the painting rate, because paint goes on the surface and its rate is per square metre. Surface area of the cube = 6 × (5.2 × 5.2) = 6 × 27.04 = 162.24 m². Cost of painting = 162.24 × 15.5 = Rs. 2514.72.

Review Exercise 12, Question 1 (i): Fill in the blank. (i) A cube has _____ vertices.

Answer: 8

Review Exercise 12, Question 1 (ii): Fill in the blank. (ii) A cone has _____ circular faces.

Answer: 1

Review Exercise 12, Question 1 (iii): Fill in the blank. (iii) A cuboid has _____ faces.

Answer: 6

Review Exercise 12, Question 1 (iv): Fill in the blank. (iv) A cylinder has _____ circular faces.

Answer: 2

Review Exercise 12, Question 1 (v): Fill in the blank. (v) A cube has _____ edges.

Answer: 12

Question: Review Exercise 12, Question 2: Give any three examples of 3-D shapes from daily life.

Answer: Three examples of 3-D shapes from daily life: (1) football — sphere, (2) matchbox — cuboid, (3) a glass of water or a tin — cylinder.

Question: Review Exercise 12, Question 3 (i): Find the surface area and volume of the following figures. (i) Cube, each side 2.5 cm.

Answer: Surface area = 6 × (2.5 × 2.5) = 6 × 6.25 = 37.5 cm². Volume = 2.5 × 2.5 × 2.5 = 15.625 cm³.

Question: Review Exercise 12, Question 3 (ii): Find the surface area and volume of the following figures. (ii) Cuboid: length 3.5 cm, breadth 2 cm, height 1.5 cm.

Answer: Surface area = 2(3.5 × 2 + 2 × 1.5 + 1.5 × 3.5) = 2(7 + 3 + 5.25) = 2(15.25) = 30.5 cm². Volume = 3.5 × 2 × 1.5 = 10.5 cm³.

Question: Review Exercise 12, Question 4: Find the volume of a cube, each of whose sides is 2.5 cm long.

Answer: Volume of cube = side × side × side = 2.5 × 2.5 × 2.5 = 15.625 cm³.

Question: Review Exercise 12, Question 5: Find the cost of the tin of a cuboid at Rs. 100 per m², whose sides are 5 m, 6 m and 7 m.

Answer: Surface area of the cuboid = 2(5 × 6 + 6 × 7 + 7 × 5) = 2(30 + 42 + 35) = 2(107) = 214 m². Cost of the tin = 214 × 100 = Rs. 21,400.

Question: Review Exercise 12, Question 6: How much water will a tank hold whose sides are 6.5 m, 2.7 m and 10 m?

Answer: Capacity of the tank = its volume = 6.5 × 2.7 × 10 = 175.5 m³. So the tank will hold 175.5 m³ of water.

13. Data Handling

Question: Practical Work (Unit 13), Question 1: Ask each student of the class for their birth month. Then count them and complete the following table: Month | Tally marks | Number of students (January, February, March, April, May, June, July, August, September, October, November, December).

Answer: Method: ask each student for their birth month. For each student, put one tally mark in the row of their birth month. After every fourth mark, draw the fifth mark diagonally across them — this makes groups of five marks. Finally, count the tally marks in each month's row and write the number in the number-of-students column. The actual numbers are not given in the book — they depend on the class actually surveyed.

Question: Exercise 13.1, Question 1: Define data.

Answer: Data is a set of information and facts. The numerical facts we get in daily life are called data.

Question: Exercise 13.1, Question 2: State the difference between grouped data and ungrouped data.

Answer: Ungrouped data is data shown individually, in its original (raw) form. Grouped data is data arranged into groups/classes (such as by tally marks or intervals).

Question: Exercise 13.1, Question 3 (i): Which of the following tables shows grouped data? (i) Students' marks in the exam: Ali 581, Sana 786, Sara 678, Adil 725, Amir 788, Asma 580, Huma 690, Arif 780, Babar 599, Anas 509, Anil 619, Umair 560.

Answer: This is ungrouped data — each student's marks are shown individually, in their original form.

Question: Exercise 13.1, Question 3 (ii): Which of the following tables shows grouped data? (ii) Raza's book-store table: price interval 512–611 → 8 books, 612–711 → 5 books, 712–811 → 4 books, 812–911 → 9 books.

Answer: This is grouped data — the number of books is shown in price intervals (groups). ✓

Question: Exercise 13.1, Question 3 (iii): Which of the following tables shows grouped data? (iii) Sindh Museum visitors table: interval 150–199 → 3 weeks, 200–249 → 4 weeks, 250–299 → 1 week, 300–349 → 2 weeks, 350–399 → 3 weeks, 400–499 → 2 weeks.

Answer: This is grouped data — the number of visitors is shown in intervals (groups). ✓

Question: Exercise 13.1, Question 3 (iv): Which of the following tables shows grouped data? (iv) Cricket team scores in a one-day match: 62, 41,

15, 59, 22, 10, 8, 2, 43, 7, 21 (11 players).

Answer: This is ungrouped data — each player's score is shown individually, in its original form.

Question: Practical Work (Unit 13, bar graph), Question (1): Looking at the games bar graph, write sentences from the information in it. (1) Example given in the book: "Of all games, cricket is played by the most students. That is why cricket's bar is the longest of all bars. The length of that bar is 86 small squares." Graph data: Hockey 36, Cricket 43, Badminton 16, Football 20, Volleyball 8, Basketball 12, Boxing 1.

Answer: The book's given example sentence is correct: cricket is played by the most students (43), so cricket's bar is the longest of all bars, and its length is 86 small squares.

Question: Practical Work (Unit 13, bar graph), Question (2): Looking at the games bar graph, write sentences from the information in it. (1) Given: "Of all games, cricket is played by the most students. That is why cricket's bar is the longest of all bars. The length of that bar is 86 small squares." Graph data: Hockey 36, Cricket 43, Badminton 16, Football 20, Volleyball 8, Basketball 12, Boxing 1.

Answer: Sample sentence (2): "Boxing is played by the fewest students — only 1 student plays boxing, so boxing's bar is the shortest of all bars." (Students write their own sentences; this is a correct sentence from the book's numbers.)

Question: Practical Work (Unit 13, bar graph), Question (3): Looking at the games bar graph, write sentences from the information in it. Graph data: Hockey 36, Cricket 43, Badminton 16, Football 20, Volleyball 8, Basketball 12, Boxing 1.

Answer: Sample sentence (3): "Hockey is played by 36 students, which is more than football (20) but fewer than cricket (43)." (Students write their own sentences; this is a correct sentence from the book's numbers.)

Question: Practical Work (Unit 13, pie graph), Question 1: Look at the pie graph of 600 Hajj pilgrims (men, women, children). Who is the greatest in number?

Answer: Men. (The answer is given in the book's box: "Men")

Question: Practical Work (Unit 13, pie graph), Question 2: Look at the pie graph of 600 Hajj pilgrims (men, women, children). Who is the least in number?

Answer: Children — their sector is the smallest.

Question: Practical Work (Unit 13, pie graph), Question 3: Look at the pie graph of 600 Hajj pilgrims (men, women, children). Are the children equal in number to the women travellers?

Answer: No — the children's sector is smaller than the women's sector, so the children are not equal to the women.

Question: Practical Work (Unit 13, pie graph), Question 4: Look at the pie graph of 600 Hajj pilgrims (men, women, children). Are the male travellers more than the female travellers?

Answer: Yes — the men's sector is the largest, so the male travellers are more than the female travellers.

Question: Exercise 13.2, Question 1: Draw a vertical bar graph of a shopkeeper's monthly electric fan sales: January 70, February 50, March 35, April 60, May 90, June 95.

Answer: Construction method: on a large squared copy, mark the months (January to June) at equal intervals on the X-axis. Mark the number of electric fans on the Y-axis with a suitable scale (e.g. 1 square = 10 fans). For each month, draw a vertical rectangular bar of equal width, whose height equals that month's sales. Data: January 70, February 50, March 35, April 60, May 90, June 95. Graph title: "Monthly electric fan sales".

Question: Exercise 13.2, Question 2: Draw a vertical bar graph of Akbar's monthly household expenses: food 6000, education 5000, rent 4500, telephone 1500, electricity 2000, gas 300 (rupees).

Answer: Construction method: mark the expense types (food, education, rent, telephone, electricity, gas) at equal intervals on the X-axis. Mark rupees on the Y-axis with a suitable scale (e.g. 1 square = Rs. 500). For each type, draw a vertical bar of equal width, whose height equals that expense. Data: food 6000, education 5000, rent 4500, telephone 1500, electricity 2000, gas 300. Graph title: "Akbar's monthly household expenses".

Question: Exercise 13.2, Question 3: Draw a horizontal bar graph of the variety-show tickets sold in different cities: Hyderabad 10,000, Sukkur 6,000, Larkano 3,000, Dado 9,000.

Answer: Construction method: mark the cities (Hyderabad, Sukkur, Larkano, Dado) at equal intervals on the Y-axis. Mark the number of tickets on the X-axis with a suitable scale (e.g. 1 square = 1000 tickets). For each city, draw a horizontal bar of equal width, whose length equals that city's tickets. Data: Hyderabad 10,000, Sukkur 6,000, Larkano 3,000, Dado 9,000. Graph title: "Variety-show tickets sold in different cities".

Question: Exercise 13.2, Question 4: Draw a horizontal bar graph of Amjad's annual exam result: Islamiyat 70, English 60, Urdu 65, Math 90, Social Studies 45, Science 55, Sindhi 80.

Answer: Construction method: mark the subjects (Islamiyat, English, Urdu, Math, Social Studies, Science, Sindhi) at equal intervals on the Y-axis. Mark the marks on the X-axis with a suitable scale (e.g. 1 square = 5 marks). For each subject, draw a horizontal bar of equal width, whose length equals that subject's marks. Data: Islamiyat 70, English 60, Urdu 65, Math 90, Social Studies 45, Science 55, Sindhi 80. Graph title: "Amjad's annual exam result".

Question: Exercise 13.2, Question 5 (i): Looking at the pie graph of Class VI students playing different games, answer the following questions. (i) Which game is played by the fewest students?

Answer: Tennis — in the pie graph, tennis's sector is the smallest. (No numbers are printed on the graph, so the answer is read from the sizes of the sectors.)

Question: Exercise 13.2, Question 5 (ii): Looking at the pie graph of Class VI students playing different games, answer the following questions. (ii) Which game is played by the most students?

Answer: Cricket — in the pie graph, cricket's sector is the largest. (No numbers are printed on the graph, so the answer is read from the sizes of the sectors.)

Question: Exercise 13.2, Question 5 (iii): Looking at the pie graph of Class VI students playing different games, answer the following questions. (iii) In which two games did an equal number of students take part?

Answer: Football and tennis — in the pie graph their two sectors appear equal in size. (No numbers are printed on the graph, so the answer is read from the sizes of the sectors.)

Question: Exercise 13.2, Question 6 (i): Looking at the pie graph of Aftab's marks in different subjects, answer the following questions. (i) In which subject did Aftab get the most marks, and how many?

Answer: Math — in the pie graph, Math's sector is the largest. The exact number of marks is not printed on the graph, so that number cannot be read from the graph.

Question: Exercise 13.2, Question 6 (ii): Looking at the pie graph of Aftab's marks in different subjects, answer the following questions. (ii) In which subject did he get the fewest marks, and how many?

Answer: English — in the pie graph, English's sector is the smallest. The exact number of marks is not printed on the graph, so that number cannot be read from the graph.

Question: Exercise 13.2, Question 6 (iii): Looking at the pie graph of Aftab's marks in different subjects, answer the following questions. (iii) In which subject did he get more marks, Science or Math?

Answer: In Math — in the pie graph, Math's sector is slightly larger than Science's sector. (The exact marks are not printed on the graph.)

Question: Exercise 13.2, Question 7 (i): Looking at the pie graph of attendance of different classes, answer the following questions. (i) Which class has the highest attendance?

Answer: Class II — in the pie graph, Class II's sector is the largest. (No numbers are printed on the graph, so the answer is read from the sizes of the sectors.)

Question: Exercise 13.2, Question 7 (ii): Looking at the pie graph of attendance of different classes, answer the following questions. (ii) Which class has the lowest attendance?

Answer: Class I — in the pie graph, Class I's sector is the smallest. (No

numbers are printed on the graph, so the answer is read from the sizes of the sectors.)

Question: Exercise 13.2, Question 7 (iii): Looking at the pie graph of attendance of different classes, answer the following questions. (iii) In which classes is attendance equal?

Answer: Classes IV and V — in the pie graph their two sectors appear equal in size. (No numbers are printed on the graph, so the answer is read from the sizes of the sectors.)

Question: Review Exercise 13, Question 1 (i): Vertical bar graph of Saira's exam marks: Sindhi 50, Urdu 50, English 60, Science 70, Social Studies 40, Islamiyat 78, Math 80. (i) In which subject did Saira get the most marks, and how many?

Answer: In Math, 80 marks. (In the bar graph, Math's bar is the longest.)

Question: Review Exercise 13, Question 1 (ii): Vertical bar graph of Saira's exam marks: Sindhi 50, Urdu 50, English 60, Science 70, Social Studies 40, Islamiyat 78, Math 80. (ii) In which subject did Saira get the fewest marks, and how many?

Answer: In Social Studies, 40 marks. (In the bar graph, Social Studies' bar is the shortest.)

Question: Review Exercise 13, Question 1 (iii): Vertical bar graph of Saira's exam marks: Sindhi 50, Urdu 50, English 60, Science 70, Social Studies 40, Islamiyat 78, Math 80. (iii) In which subjects did Saira get equal marks, and how many?

Answer: In Sindhi and Urdu, 50 marks in each. (In the bar graph, the Sindhi and Urdu bars are of equal height.)

Question: Review Exercise 13, Question 1 (iv): Vertical bar graph of Saira's exam marks: Sindhi 50, Urdu 50, English 60, Science 70, Social Studies 40, Islamiyat 78, Math 80. (iv) Write Saira's marks in different subjects in a chart.

Answer: Subject | Marks — Sindhi 50, Urdu 50, English 60, Science 70, Social Studies 40, Islamiyat 78, Math 80.

✓ Review Exercise 13, Question 2 (i): Identify true and false. (i) From a pictograph, numerical data can be obtained by counting the pictures.

✓ Review Exercise 13, Question 2 (ii): Identify true and false. (ii) In a bar graph, numerical data is shown in the form of rectangular bars whose width is the same.

✗ Review Exercise 13, Question 2 (iii): Identify true and false. (iii) Arranging data in the same order is called ungrouped data.

Review Exercise 13, Question 2 (iv): Identify true and false. (iv) A rectangular shape is called a pie graph.

Review Exercise 13, Question 3 (i): Fill in the blank. (i) Showing numerical data in the form of pictures is called _____ graph.

Answer: a pictograph.

Review Exercise 13, Question 3 (ii): Fill in the blank. (ii) Showing data with _____ graphs is much easier and more useful than with a pictograph.

Answer: bar

Review Exercise 13, Question 3 (iii): Fill in the blank. (iii) In a bar graph the bars are in _____ form or _____ form.

Answer: vertical, horizontal

Review Exercise 13, Question 3 (iv): Fill in the blank. (iv) _____ is defined as a set of information and facts.

Answer: Data

Review Exercise 13, Question 3 (v): Fill in the blank. (v) The numerical facts that we get from daily life are called _____.

Answer: data.

14. Glossary